brGDGT-based NPP reconstructions, herbivore density estimations and carrying capacity of secondary consumers

R codes for the main and supplementary figures and tables available in:

'Increased top-down controls on terrestrial productivity across the Pleistocene-Holocene transition'

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Analysis overview

The analyses comprise brGDGT-NPP calibration, model validation, NPP reconstruction at Padul, Summed Probability Distribution of archaeological sites, herbivore biomass estimation and consumer carrying-capacity calculations. Code and its output are presented together, with separate sections for each main or supplementary figure. Run the chunks in document order.

Figure and table index

Output	Analysis	Section
Figure 1; Table S1	NPP and brGDGT abundances; association and regression statistics	brGDGT–NPP relationships
Figure 2	Environmental Spearman correlation matrix	Environmental correlations
Figure 3A	Zero-order and partial Spearman correlations	Partial correlations
Figure 3B	Multivariate redundancy analysis	RDA
Figure 3C	Constrained/unconstrained eigenvalue ratios	Eigenvalue comparison
Figure 4; Table 1	RF validation and residual distributions	RF model performance
Figure 5	Validation against independent NPP measurements	Independent validation
Figure 6B–C	NGRIP and pollen context	Palaeoenvironmental context
Figure 6D–E	Potential NPP, brGDGT-derived NPP and NPP deficit	NPP reconstruction
Figure 6F	Summed probability distributions and charcoal	SPD and charcoal
Figure 7A–C; Table S4	NPP and deficit distributions by culture	Cultural comparisons
Figure 8A–E	Archaeological indicators, charcoal and NPP deficit	Archaeological associations
Figure 9	Herbivore biomass, human carrying capacity and food-web summaries	Consumer analyses

Output	Analysis	Section
Figure S2	Compound–NPP correlations by sample type	Sample-type correlations
Figure S3; Table S2	Q–Q plots and Shapiro–Wilk tests	Normality
Figure S5	RF transfer between sample types	Cross-environment prediction
Figure S6	Independent validation without sample type as a predictor	Sample-type sensitivity
Figure S7	Environmental novelty relative to calibration support	Environmental novelty
Figure S8	Regional palaeoclimate records	Regional palaeoclimate
Figure S9	Padul predictions under different sample-type assignments	Padul sample-type sensitivity
Figure S10	NPP, pollen and pollen accumulation rates	Pollen associations
Figure S11	Charcoal and NPP associations	Charcoal associations
Figure S12	NPP deficit using alternative climate records	Climate-source sensitivity
Figure S13; Table S5	Sensitivity to temporal offsets	Temporal-offset sensitivity
Figure S15	Simulated stationary survival curves	Survival curves
Table S3	ANOVA and Tukey comparisons between cultures	Between-culture tests
Tables S6–S8	Biomass summaries by period and body-size class	Biomass summaries

Figures S1 and S4 were produced outside R. Figure 6A and the map component of Figure 5 were not generated in R.

Data and software

Place this document in the analysis directory containing `Dataset_1.xlsx`, `Dataset_2.xlsx`, `Prey_preferences.xlsx` and the model RDS files. The model filenames and the `WithoutSampleType/` subdirectory are retained. Set `retrain_models: true` to run the model-training chunks; otherwise, the saved models are loaded. The environmental-novelty analysis runs when all six input rasters are available in the same directory. The NPP uncertainty analysis uses the MAT and MAP RMSE columns in `Padul_climate`.

Knit the document to HTML in RStudio, or render it from R:

```
rmarkdown::render("brGDGT_NPP_reproducible_updated.Rmd", output_format = "html_document")

knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE,
  error = FALSE, cache = FALSE, fig.width = 10, fig.height = 6,
  fig.align = "center", out.width = "100%", dpi = 150)
document_dir <- dirname(normalizePath(knitr::current_input(), mustWork = TRUE))
data_path <- params$data_dir
if (!grepl("^(/|[A-Za-z]:[/\\\\]|~)", data_path))
  data_path <- file.path(document_dir, data_path)
data_path <- normalizePath(path.expand(data_path), mustWork = TRUE)
knitr::opts_knit$set(root.dir = data_path)
retrain_models <- isTRUE(params$retrain_models)
without_sampletype_files <- file.path("WithoutSampleType", c(
  "model_brGDGT-NPP_cv1.rds", "spatial_cv_50km_models1.rds",
  "spatial_cv_100km_models1.rds", "spatial_cv_500km_models1.rds",
  "spatial_cv_1000km_models1.rds"))
run_without_sampletype <- isTRUE(params$include_without_sampletype) &&
  all(file.exists(file.path(data_path, without_sampletype_files)))
raster_files <- c("Elevation.tif", "rbulk_reproj.tif", "rclay_reproj.tif",
  "rph_reproj.tif", "wc2.1_5m_bio_1.tif", "wc2.1_5m_bio_12.tif")
run_rasters <- isTRUE(params$include_rasters) &&
  all(file.exists(file.path(data_path, raster_files)))

# --- --- --- --- --- --- --- --- --- --- --- --- --- ---
#                            NOTE:
# spatial_cv_50km_models1.rds, spatial_cv_100km_models1.rds,
# spatial_cv_500km_models1.rds and spatial_cv_1000km_models1.rds cannot be stored
# in GitHub because they are >25 Mb, so they are in https://zenodo.org/records/23495441
# Likewise the rasters needed for the environmental novelty analyses are available in https://zenodo.org/records/23495441
# --- --- --- --- --- --- --- --- --- --- --- --- --- ---
```

Packages for analyses

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(scales)
library(openxlsx)
library(truncnorm)
library(vegan)
library(ggrepel)
library(grid)
library(compositions)
library(splines)
library(purrr)
library(patchwork)
library(ggpubr)
library(gridExtra)
library(analogue)
library(corrplot)
library(tidyverse)
library(RColorBrewer)
library(cowplot)
library(caret)
library(ggExtra)
library(randomForest)
library(reshape2)
library(rcarbon)
library(mgcv)
library(sf)
library(blockCV)
library(spdep)
library(ppcor)
library(terra)
library(RANN)
library(maps)
```

Input checks and prediction utilities

```
require_columns <- function(df, columns, worksheet) {
  missing <- setdiff(columns, names(df))
  if (length(missing)) stop("Missing columns in ", worksheet, ": ",
    paste(missing, collapse = ", "), call. = FALSE)
}
required_workbooks <- c("Dataset_1.xlsx", "Dataset_2.xlsx", "Prey_preferences
.xlsx")
missing_workbooks <- required_workbooks[!file.exists(required_workbooks)]
if (length(missing_workbooks)) stop("Missing input files: ",
  paste(missing_workbooks, collapse = ", "), call. = FALSE)
if (!retrain_models) {
  model_files <- c("model_brGDGT-NPP_all1.rds", "model_brGDGT-NPP_cv1.rds",
```

```
    "spatial_cv_50km_models1.rds", "spatial_cv_100km_models1.rds",
    "spatial_cv_500km_models1.rds", "spatial_cv_1000km_models1.rds")
  missing_models <- model_files[!file.exists(model_files)]
  if (length(missing_models)) stop("Missing model files: ",
    paste(missing_models, collapse = ", "), call. = FALSE)
}
required_sheets_1 <- c("Dataset_R", "Measured_NPP_brGDGTs", "Regional_Paleocl
imate",
  "NGRIP", "Padul_GDGTs", "Padul_climate", "Pollen", "SPD")
stopifnot(all(required_sheets_1 %in% openxlsx::getSheetNames("Dataset_1.xlsx"
)))
regional_climate_check <- openxlsx::read.xlsx("Dataset_1.xlsx", sheet = "Regi
onal_Paleoclimate")
require_columns(regional_climate_check, c("Age_P1", "MAT_P1", "MAP_P1", "Age_
GCM",
  "MAT_GCM", "MAP_GCM", "Age_CHELSA", "MAT_Chelsa", "MAP_Chelsa",
  "Age_SST", "SST", "Age_P2", "Camuerta.et.al..2022"), "Regional_Paleoclimate
")
climate_check <- openxlsx::read.xlsx("Dataset_1.xlsx", sheet = "Padul_climate
")
require_columns(climate_check, c("Age", "NPP_Miami", "Predicted_MAT", "Predic
ted_MAP",
  "MAT_RMSE", "RMSE_MAP"), "Padul_climate")

predict_fold_models <- function(models, newdata) {
  lapply(models, function(model) predict(model, newdata = newdata))
}
rf_design_matrix <- function(model, newdata) {
  design <- stats::model.matrix(stats::delete.response(model$terms),
    data = newdata, contrasts.arg = model$contrasts, xlev = model$xlevels)
  predictors <- names(model$finalModel$forest$ncat)
  if (is.null(predictors)) predictors <- rownames(model$finalModel$importance
)
  missing <- setdiff(predictors, colnames(design))
  if (length(missing)) stop("Missing RF predictors: ", paste(missing, collaps
e = ", "))
  design[, predictors, drop = FALSE]
}
```

Script 1: brGDGT-based NPP analyses

Calibration data and brGDGT indices

Read the calibration data, sum tetra-, penta- and hexamethylated compounds, and calculate the CBT and MBT indices.

```
data <- read.xlsx("Dataset_1.xlsx", sheet = "Dataset_R")
```

```
tetramethylated <- c("fIa", "fIb", "fIc")
pentamethylated <- c("fIIa", "fIIa_", "fIIb", "fIIb_", "fIIc", "fIIc_")
hexamethylated  <- c("fIIIa", "fIIIa_", "fIIIb", "fIIIb_", "fIIIc", "fIIIc_")

data$Tetramethylated <- rowSums(data[, tetramethylated], na.rm = TRUE)
data$Pentamethylated <- rowSums(data[, pentamethylated], na.rm = TRUE)
data$Hexamethylated  <- rowSums(data[, hexamethylated], na.rm = TRUE)

data$CBT <- log10((data$fIc + data$fIIa_+data$fIIb_+data$fIIc_+data$fIIIa_+da
ta$fIIIb_+data$fIIIc_) /
                    (data$fIa + data$fIIa + data$fIIIa))

all_compounds <- c("fIa", "fIb", "fIc",
                   "fIIa", "fIIb", "fIIc",
                   "fIIIa")

data$MBT <- (data$fIa + data$fIb + data$fIc) / rowSums(data[, all_compounds],
na.rm = TRUE)

require_columns(data, c(tetramethylated, pentamethylated, hexamethylated, "NP
P", "Sampletype"), "Dataset_R")
```

Figure 1: NPP and brGDGT abundances

Plot NPP against individual compounds and methylation groups, with points coloured by sample type. Compare linear, quadratic and GAM fits and calculate Spearman association statistics.

```
tet_cols  <- intersect(tetramethylated, names(data))
penta_cols <- intersect(pentamethylated, names(data))
hexa_cols <- intersect(hexamethylated, names(data))

for(col in c(tet_cols, penta_cols, hexa_cols)){
  data[[col]] <- as.numeric(as.character(data[[col]]))
}

data2 <- data %>%
  mutate(
    tetramethylated = rowSums(across(all_of(tet_cols)), na.rm = TRUE),
    pentamethylated = rowSums(across(all_of(penta_cols)), na.rm = TRUE),
    hexamethylated  = rowSums(across(all_of(hexa_cols)), na.rm = TRUE)
  )

all_cols <- c("tetramethylated", tetramethylated,
              "pentamethylated", pentamethylated,
              "hexamethylated", hexamethylated)

data_long <- data2 %>%
```

```r
  pivot_longer(cols = all_of(all_cols), names_to = "Fraction", values_to = "V
alue") %>%
  mutate(
    Group = case_when(
      Fraction %in% c("tetramethylated", tetramethylated) ~ "Tetra",
      Fraction %in% c("pentamethylated", pentamethylated) ~ "Penta",
      Fraction %in% c("hexamethylated", hexamethylated)  ~ "Hexa"
    ),
    is_sum = Fraction %in% c("tetramethylated","pentamethylated","hexamethyla
ted"),
    Fraction_plot = gsub("_", "'", Fraction)
  )

data_long$Sampletype <- factor(data_long$Sampletype)

sample_colors <- brewer.pal(n = length(levels(data_long$Sampletype)), "Set1")

compute_all_stats <- function(datf) {
  if(nrow(datf) < 3)
    return(list(
      n = nrow(datf),
      spearman_rho = NA, spearman_p = NA,
      linear_r2 = NA, linear_p = NA,
      quad_r2 = NA, quad_p = NA,
      gam_dev_expl = NA, gam_p = NA, gam_edf = NA
    ))

  results <- list(
    n = nrow(datf),
    spearman_rho = NA, spearman_p = NA,
    linear_r2 = NA, linear_p = NA,
    quad_r2 = NA, quad_p = NA,
    gam_dev_expl = NA, gam_p = NA, gam_edf = NA
  )

  try({
    sp <- suppressWarnings(cor.test(datf$NPP, datf$Value, method = "spearman"
))
    results$spearman_rho <- sp$estimate
    results$spearman_p <- sp$p.value
  }, silent = TRUE)

  try({
    lm_fit <- lm(Value ~ NPP, data = datf)
    lm_summary <- summary(lm_fit)
    results$linear_r2 <- lm_summary$r.squared
    results$linear_p <- ifelse(nrow(coef(lm_summary)) >= 2,
                               coef(lm_summary)[2, 4], NA)
  }, silent = TRUE)
```

```r
  try({
    quad_fit <- lm(Value ~ NPP + I(NPP^2), data = datf)
    quad_summary <- summary(quad_fit)
    results$quad_r2 <- quad_summary$r.squared
    results$quad_p <- ifelse(nrow(coef(quad_summary)) >= 3,
                             coef(quad_summary)[3, 4], NA)
  }, silent = TRUE)

  try({
    k_val <- min(8, nrow(datf) - 1)
    if(k_val >= 3) {
      gam_fit <- gam(Value ~ s(NPP, k = k_val), data = datf, method = "REML")
      gam_summary <- summary(gam_fit)
      results$gam_dev_expl <- gam_summary$dev.expl
      results$gam_p <- ifelse(!is.null(gam_summary$s.table) && nrow(gam_summa
ry$s.table) > 0,
                              gam_summary$s.table[1, "p-value"], NA)
      results$gam_edf <- ifelse(!is.null(gam_summary$s.table) && nrow(gam_sum
mary$s.table) > 0,
                                gam_summary$s.table[1, "edf"], NA)
    }
  }, silent = TRUE)

  return(results)
}

fractions <- unique(data_long$Fraction)
stats_all <- list()

for(frac in fractions) {
  datf <- data_long %>%
    filter(Fraction == frac) %>%
    drop_na(NPP, Value)

  stats_all[[frac]] <- compute_all_stats(datf)
}

format_p <- function(p) {
  if(is.na(p)) return("NA")
  if(p < 0.001) return("<0.001")
  return(sprintf("%.3f", p))
}

make_comparison_plot <- function(frac) {

  datf <- data_long %>%
    filter(Fraction == frac) %>%
    drop_na(NPP, Value)
```

```r
stats <- stats_all[[frac]]

if (is.null(stats) || length(stats) == 0) {
  p <- ggplot() +
    annotate("text", x = 0.5, y = 0.5, label = "No data") +
    theme_void()
  return(p)
}

p <- ggplot(datf, aes(x = NPP, y = Value, color = Sampletype)) +
  geom_point(size = 2, alpha = 0.8) +
  labs(
    title = gsub("_", "'", frac),
    x = expression("NPP (g C m"^-2*" yr"^-1*")"),
    y = "Percentage (%)"
  ) +
  scale_color_manual(values = sample_colors, drop = FALSE) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none",
    axis.text = element_text(size = 10),
    axis.title = element_text(size = 11)
  )

pred_data <- datf %>%
  arrange(NPP)

if (!is.na(stats$linear_r2)) {
  try({
    lm_fit <- lm(Value ~ NPP, data = datf)
    pred_data$linear <- predict(lm_fit, newdata = pred_data)

    p <- p + geom_line(
      data = pred_data,
      aes(x = NPP, y = linear),
      color = "black",
      linewidth = 1.5,
      linetype = "dashed",
      alpha = 0.9
    )
  }, silent = TRUE)
}

if (!is.na(stats$quad_r2)) {
  try({
    quad_fit <- lm(Value ~ NPP + I(NPP^2), data = datf)
    pred_data$quadratic <- predict(quad_fit, newdata = pred_data)
```

```
    p <- p + geom_line(
      data = pred_data,
      aes(x = NPP, y = quadratic),
      color = "black",
      linewidth = 1.5,
      linetype = "dotted"
    )
  }, silent = TRUE)
}

if (!is.na(stats$gam_dev_expl) && stats$n >= 8) {
  try({
    k_val <- min(8, nrow(datf) - 1)

    if (k_val >= 3) {
      gam_fit <- gam(Value ~ s(NPP, k = k_val),
                     data = datf,
                     method = "REML")

      pred_data$gam <- predict(gam_fit, newdata = pred_data)

      p <- p + geom_line(
        data = pred_data,
        aes(x = NPP, y = gam),
        color = "black",
        linewidth = 1.5,
        linetype = "solid",
        alpha = 0.6
      )
    }
  }, silent = TRUE)
}

stats_text <- paste0("n = ", stats$n)

if (!is.na(stats$spearman_rho)) {
  stats_text <- paste0(
    stats_text, "\n",
    "Spearman: ρ = ", sprintf("%.2f", stats$spearman_rho),
    ", p = ", format_p(stats$spearman_p)
  )
}

if (!is.na(stats$linear_r2)) {
  stats_text <- paste0(
    stats_text, "\n",
    "Linear: r² = ", sprintf("%.2f", stats$linear_r2),
    ", p = ", format_p(stats$linear_p)
```

```
    )
  }

  if (!is.na(stats$quad_r2)) {
    stats_text <- paste0(
      stats_text, "\n",
      "Quadratic: r² = ", sprintf("%.2f", stats$quad_r2),
      ", p = ", format_p(stats$quad_p)
    )
  }

  if (!is.na(stats$gam_dev_expl)) {
    stats_text <- paste0(
      stats_text, "\n",
      "GAM: r² = ", sprintf("%.2f", stats$gam_dev_expl),
      ", p = ", format_p(stats$gam_p),
      ", EDF = ",
      ifelse(!is.na(stats$gam_edf),
             sprintf("%.1f", stats$gam_edf),
             "NA")
    )
  }

  return(p)
}

plots_tetra <- lapply(c("tetramethylated", tetramethylated), make_comparison_
plot)
plots_penta <- lapply(c("pentamethylated", pentamethylated), make_comparison_
plot)
plots_hexa  <- lapply(c("hexamethylated", hexamethylated), make_comparison_pl
ot)
grid.arrange(grobs = plots_tetra, ncol = 4)
```

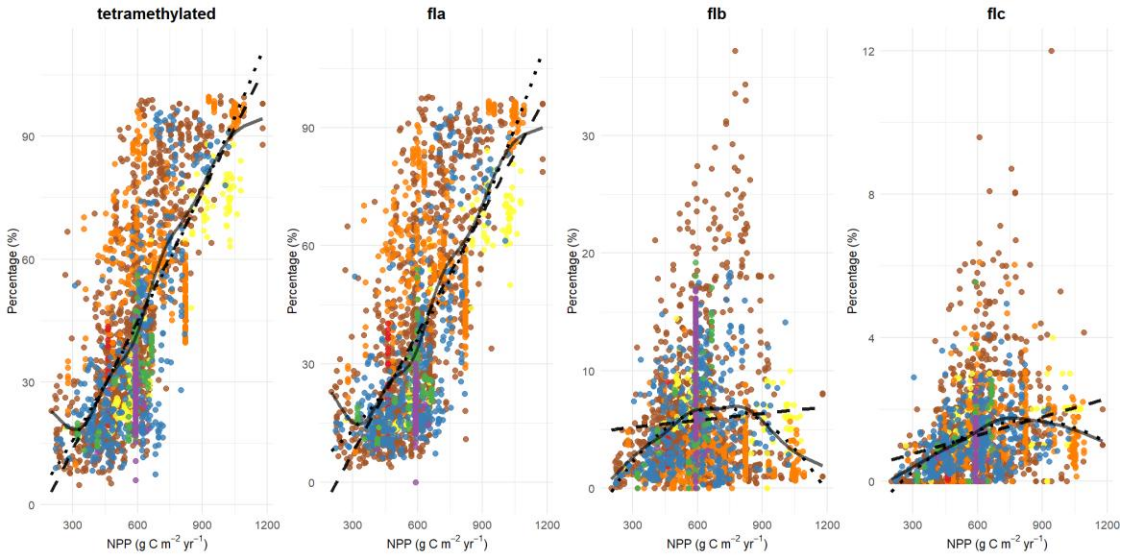


Figure 1. Relationships between NPP and brGDGT abundances.

```
grid.arrange(grobs = plots_penta, ncol = 4)
```

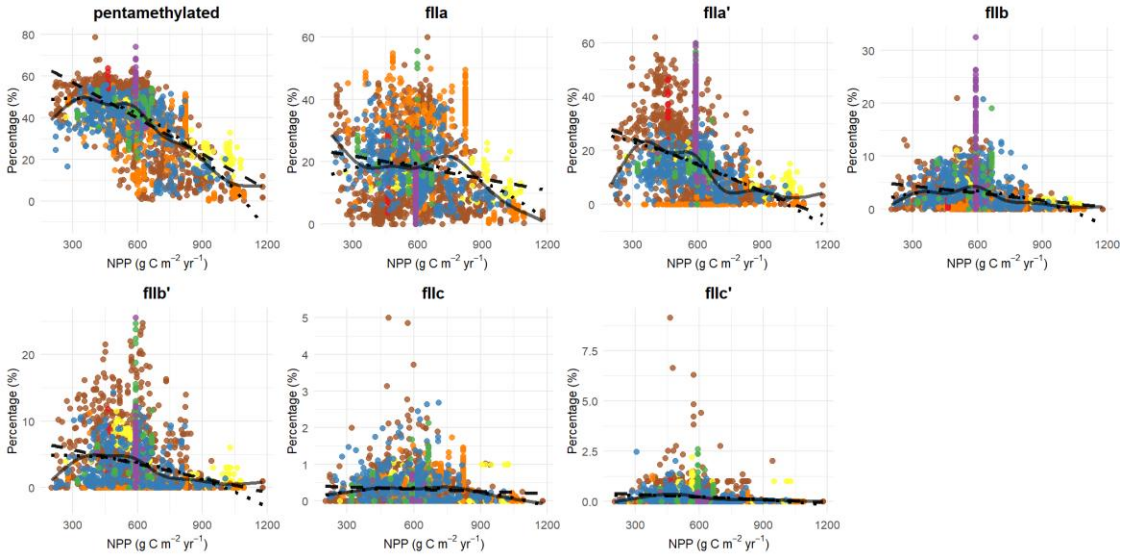


Figure 1. Relationships between NPP and brGDGT abundances.

```
grid.arrange(grobs = plots_hexa, ncol = 4)
```

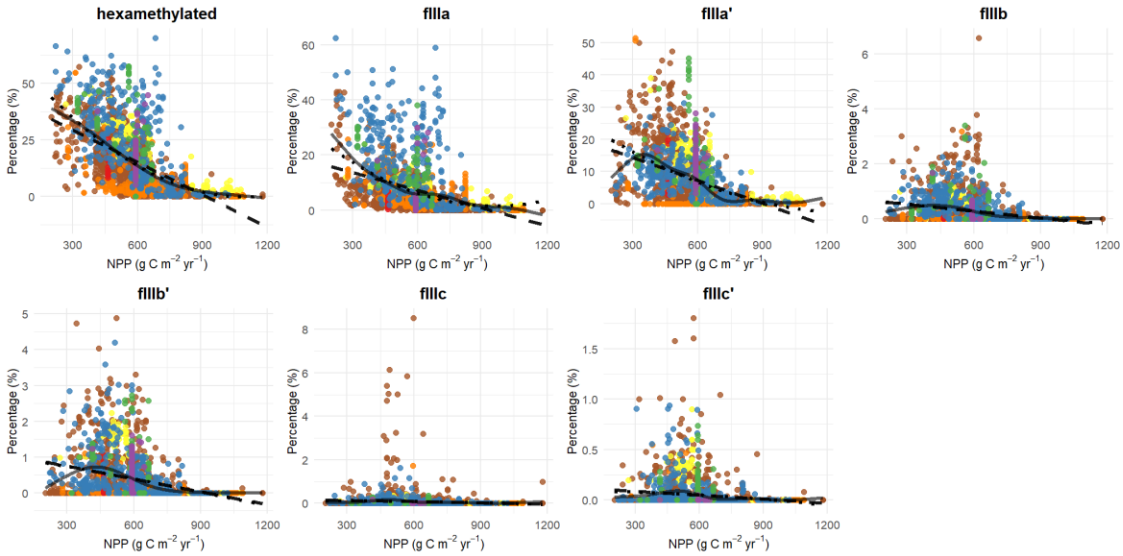


Figure 1. Relationships between NPP and brGDGT abundances.

Table S1: Compound-specific statistics

Report sample sizes, Spearman coefficients and p-values, linear and quadratic r^2, and the GAM explained-deviance statistic.

```
summary_table <- data.frame(
  Fraction = fractions,
  n = sapply(stats_all, function(x) ifelse(is.null(x$n), NA, x$n)),
  Spearman_rho = sapply(stats_all, function(x) ifelse(is.null(x$spearman_rho)
, NA, x$spearman_rho)),
  Spearman_p = sapply(stats_all, function(x) ifelse(is.null(x$spearman_p), NA
, x$spearman_p)),
  Linear_R2 = sapply(stats_all, function(x) ifelse(is.null(x$linear_r2), NA,
x$linear_r2)),
  Linear_p = sapply(stats_all, function(x) ifelse(is.null(x$linear_p), NA, x$
linear_p)),
  Quadratic_R2 = sapply(stats_all, function(x) ifelse(is.null(x$quad_r2), NA,
x$quad_r2)),
  GAM_R2 = sapply(stats_all, function(x) ifelse(is.null(x$gam_dev_expl), NA,
x$gam_dev_expl))
)

knitr::kable(summary_table, digits = 3, caption = "Table S1. Compound-specifi
c correlation and regression statistics.")
```

Table S1. Compound-specific correlation and regression statistics.

	Fraction	n	Spearman_rho	Spearman_p	Linear_R2	Linear_p	Quadratic_R2	GAM_R2
tetramethylated	tetramethylated	22234	0.707	0.000	0.517	0.000	0.518	0.533
fla	fla	22234	0.663	0.000	0.460	0.000	0.469	0.483
flb	flb	22234	0.054	0.011	0.004	0.002	0.064	0.069
flc	flc	22234	0.331	0.000	0.062	0.000	0.092	0.098
pentamethylated	pentamethylated	22234	-0.546	0.000	0.386	0.000	0.429	0.453
flIa	flIa	22234	-0.120	0.000	0.029	0.000	0.048	0.086
flIa_	flIa_	22234	-0.496	0.000	0.188	0.000	0.190	0.251
flIb	flIb	22234	-0.312	0.000	0.045	0.000	0.072	0.111
flIb_	flIb_	22234	-0.368	0.000	0.084	0.000	0.092	0.124

	Fraction	n	Spearman_rho	Spearman_p	Linear_R2	Linear_p	Quadratic_R2	GAM_R2
fIIc	fIIc	2234	-0.067	0.001	0.006	0.000	0.023	0.031
fIIc_	fIIc_	2234	-0.279	0.000	0.026	0.000	0.028	0.052
hexamethylated	hexamethylated	2234	-0.693	0.000	0.375	0.000	0.405	0.409
fIIIa	fIIIa	2234	-0.514	0.000	0.193	0.000	0.227	0.244
fIIIa_	fIIIa_	2234	-0.587	0.000	0.247	0.000	0.255	0.298
fIIIb	fIIIb	2234	-0.397	0.000	0.083	0.000	0.083	0.105
fIIIb_	fIIIb_	2234	-0.411	0.000	0.107	0.000	0.107	0.150
fIIIc	fIIIc	2234	-0.168	0.000	0.006	0.000	0.007	0.020
fIIIc_	fIIIc_	2234	-0.235	0.000	0.027	0.000	0.028	0.055

```
knitr::kable(summary_table, digits = 4, caption = "Table S1. Compound-specific association and model-fit statistics.")
```

Table S1. Compound-specific association and model-fit statistics.

	Fraction	n	Spearman_rho	Spearman_p	Linear_R2	Linear_p	Quadratic_R2	GAM_R2
tetramethylated	tetramethylated	22234	0.7070	0.0000	0.5165	0.0000	0.5181	0.5326
fIa	fIa	22234	0.6631	0.0000	0.4597	0.0000	0.4687	0.4829
fIb	fIb	22234	0.0538	0.0110	0.0045	0.0015	0.0641	0.0691
fIc	fIc	22234	0.3305	0.0000	0.0615	0.0000	0.0922	0.0975
pentamethylated	pentamethylated	22234	-0.5460	0.0000	0.3859	0.0000	0.4289	0.4529
fIIa	fIIa	22234	-0.1196	0.0000	0.0293	0.0000	0.0479	0.0860
fIIa_	fIIa_	22234	-0.4965	0.0000	0.1878	0.0000	0.1897	0.2507
fIIb	fIIb	22234	-0.3122	0.0000	0.0449	0.0000	0.0723	0.1110
fIIb_	fIIb_	22234	-0.3681	0.0000	0.0844	0.0000	0.0920	0.1243

	Fraction	n	Spearman_rho	Spearman_p	Linear_R2	Linear_p	Quadratic_R2	GAM_R2
fIIc	fIIc	2234	-0.0672	0.0015	0.0062	0.0002	0.0233	0.0308
fIIc_	fIIc_	2234	-0.2792	0.0000	0.0258	0.0000	0.0277	0.0517
hexamethylated	hexamethylated	2234	-0.6933	0.0000	0.3752	0.0000	0.4050	0.4088
fIIIa	fIIIa	2234	-0.5140	0.0000	0.1925	0.0000	0.2273	0.2443
fIIIa_	fIIIa_	2234	-0.5869	0.0000	0.2466	0.0000	0.2548	0.2977
fIIIb	fIIIb	2234	-0.3967	0.0000	0.0827	0.0000	0.0828	0.1053
fIIIb_	fIIIb_	2234	-0.4112	0.0000	0.1071	0.0000	0.1071	0.1498
fIIIc	fIIIc	2234	-0.1683	0.0000	0.0064	0.0001	0.0066	0.0204
fIIIc_	fIIIc_	2234	-0.2353	0.0000	0.0272	0.0000	0.0278	0.0546

Figure S3 and Table S2: Normality diagnostics

Apply Shapiro–Wilk tests to the analysis variables and compare their observed quantiles with theoretical normal quantiles.

```
vars <- c(
  "NPP",
  "Tetramethylated", "Pentamethylated", "Hexamethylated",
  "MBT", "CBT", "PET",
  "BIO01", "BIO12",
  "pH", "Bulk", "Clay", "CEC",
  "Nitrogen", "Phosphorus",
  "SOC_0-5", "SOC_5-15",
  "Elevation"
)

labels <- c(
  BIO01 = "MAT",
  BIO12 = "MAP",
  MBT = "MBT'5ME",
  CBT = "CBT'"
)

shapiro_table <- purrr::map_dfr(vars, function(v) {

  x <- data[[v]]
  x <- x[is.finite(x)]

  test <- shapiro.test(x)

  tibble(
    Variable = ifelse(v %in% names(labels), labels[v], v),
    W = round(unname(test$statistic), 3),
    `p-value` = ifelse(
      test$p.value < 0.001,
      "<0.001",
      sprintf("%.3f", test$p.value)
    )
  )
})

knitr::kable(shapiro_table, caption = "Table S2. Shapiro-Wilk normality tests
.")
```

Table S2. Shapiro-Wilk normality tests.

Variable	W	p-value
NPP	0.967	<0.001

Variable	W	p-value
Tetramethylated	0.943	<0.001
Pentamethylated	0.923	<0.001
Hexamethylated	0.840	<0.001
MBT’5ME	0.975	<0.001
CBT’	0.963	<0.001
PET	0.935	<0.001
MAT	0.972	<0.001
MAP	0.916	<0.001
pH	0.967	<0.001
Bulk	0.983	<0.001
Clay	0.955	<0.001
CEC	0.972	<0.001
Nitrogen	0.879	<0.001
Phosphorus	0.917	<0.001
SOC_0-5	0.814	<0.001
SOC_5-15	0.651	<0.001
Elevation	0.799	<0.001

```
knitr::kable(shapiro_table, caption = "Table S2. Shapiro-Wilk tests.")
```

Table S2. Shapiro-Wilk tests.

Variable	W	p-value
NPP	0.967	<0.001
Tetramethylated	0.943	<0.001
Pentamethylated	0.923	<0.001
Hexamethylated	0.840	<0.001
MBT’5ME	0.975	<0.001
CBT’	0.963	<0.001
PET	0.935	<0.001
MAT	0.972	<0.001
MAP	0.916	<0.001
pH	0.967	<0.001
Bulk	0.983	<0.001
Clay	0.955	<0.001
CEC	0.972	<0.001
Nitrogen	0.879	<0.001

Variable	W	p-value
Phosphorus	0.917	<0.001
SOC_0-5	0.814	<0.001
SOC_5-15	0.651	<0.001
Elevation	0.799	<0.001

```r
qq_data <- data %>%
  dplyr::select(all_of(vars)) %>%
  rename(
    MAT = BIO01,
    MAP = BIO12,
    `MBT'5ME` = MBT,
    `CBT'` = CBT
  ) %>%
  pivot_longer(
    everything(),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  filter(is.finite(Value))

ggplot(qq_data, aes(sample = Value)) +
  stat_qq(size = 1.3) +
  stat_qq_line(linewidth = 0.7) +
  facet_wrap(~ Variable, scales = "free", ncol = 4) +
  labs(
    x = "Theoretical quantiles",
    y = "Observed quantiles"
  ) +
  theme_minimal() +
  theme(
    panel.grid.minor = element_blank(),
    strip.text = element_text(size = 11)
  )
```

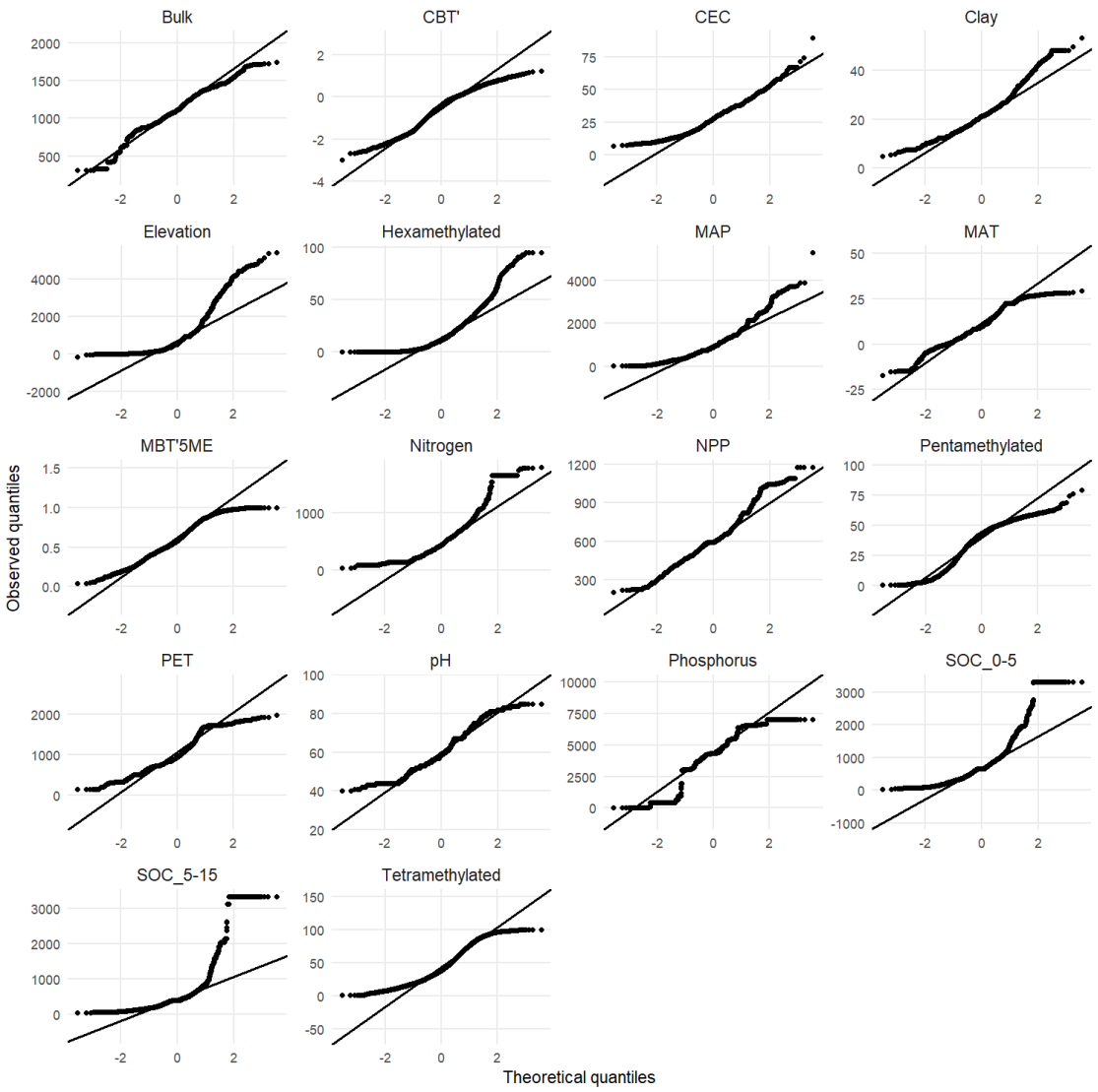


Figure S3. Normal Q-Q plots of the calibration variables.

Figure 2: Environmental correlation matrix

Calculate pairwise Spearman correlations between NPP, brGDGT groups, brGDGT indices and environmental variables.

```
selected_vars <- c("NPP","Tetramethylated", "Pentamethylated", "Hexamethylate
d", "MBT" , "CBT",
                   "PET","BIO01", "BIO12", "pH",
                    "Bulk", "Clay", "CEC", "Nitrogen", "Phosphorus", "SOC_0-5
", "SOC_5-15",
                   "Elevation" )

cor_data <- data[, selected_vars]
cor_data <- as.data.frame(lapply(cor_data, as.numeric))
```

```r
cor_matrix <- cor(cor_data, use = "pairwise.complete.obs", method = "spearman
")

colnames(cor_matrix)[colnames(cor_matrix) == "NPP"] <- "NPP"
colnames(cor_matrix)[colnames(cor_matrix) == "BIO01"] <- "MAT"
colnames(cor_matrix)[colnames(cor_matrix) == "BIO12"] <- "MAP"

rownames(cor_matrix) <- colnames(cor_matrix)

corrplot(
  cor_matrix,
  method = "square",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 1.1,
  number.cex = 0.85,
  cl.pos = "r",

  col = colorRampPalette(c(
    "firebrick3",
    "mistyrose",
    "white",
    "lightblue",
    "steelblue4"
  ))(200)
)
```

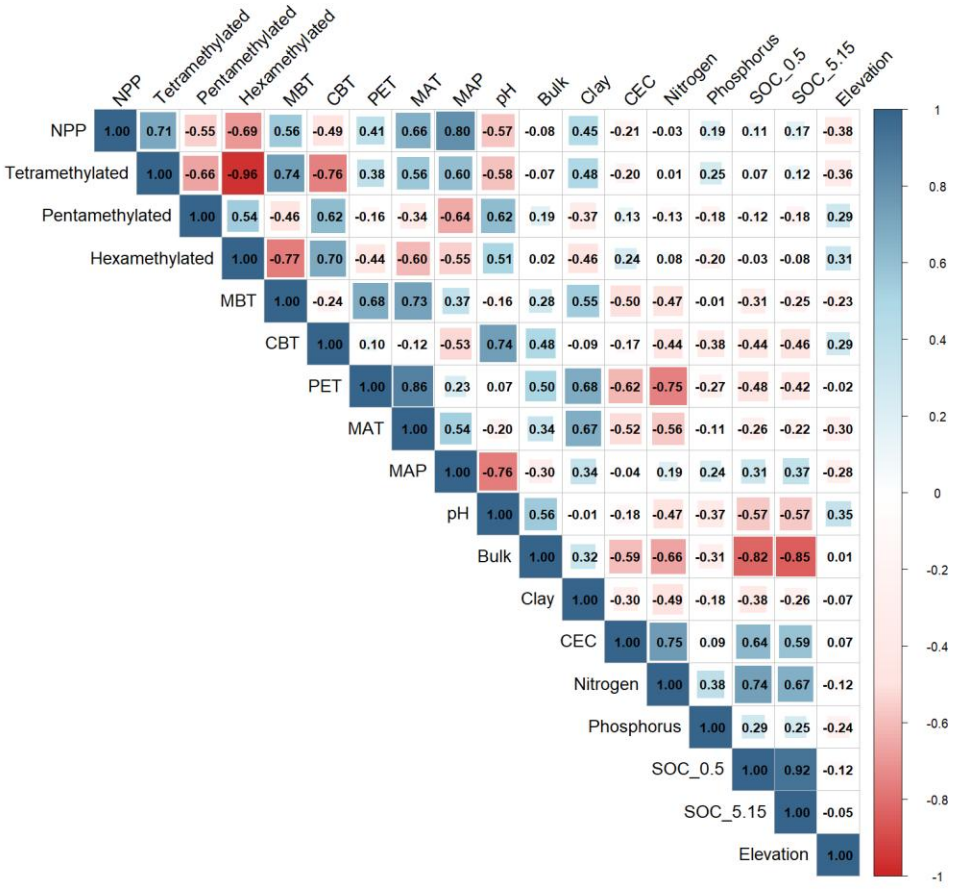


Figure 2. Spearman correlation matrix.

Figure 3A: Zero-order and partial correlations

Compare zero-order NPP–brGDGT correlations with partial Spearman correlations adjusted separately for SOC, pH, clay, elevation, MAT and MAP. Apply Benjamini–Hochberg correction to the p-values displayed in the panels.

```
br_vars <- c(
  "Tetramethylated",
  "Pentamethylated",
  "Hexamethylated"
)

partial_spearman <- function(var, controls) {

  df <- data %>%
    dplyr::select(all_of(c("NPP", var, controls))) %>%
    na.omit()

  partial <- ppcor::pcor.test(
```

```
    x = df$NPP,
    y = df[[var]],
    z = df[, controls, drop = FALSE],
    method = "spearman"
  )

  data.frame(
    Variable = var,
    rho = partial$estimate,
    p = partial$p.value
  )
}

zero_order <- bind_rows(
  lapply(br_vars, function(var) {

    df <- data %>%
      dplyr::select(NPP, all_of(var)) %>%
      na.omit()

    x <- cor.test(
      df$NPP,
      df[[var]],
      method = "spearman",
      exact = FALSE
    )

    data.frame(
      Variable = var,
      rho = unname(x$estimate),
      p = x$p.value
    )
  })
)

partial_SOC <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = c("SOC_0-5", "SOC_5-15")
  )
)

partial_pH <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = "pH"
  )
```

```
)

partial_Clay <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = "Clay"
  )
)

partial_Elevation <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = "Elevation"
  )
)

partial_MAT <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = "BIO01"
  )
)

partial_MAP <- bind_rows(
  lapply(
    br_vars,
    partial_spearman,
    controls = "BIO12"
  )
)

plot_df <- bind_rows(

  zero_order %>%
    mutate(Adjustment = "Zero-order"),

  partial_SOC %>%
    mutate(Adjustment = "SOC"),

  partial_pH %>%
    mutate(Adjustment = "pH"),

  partial_Clay %>%
    mutate(Adjustment = "Clay"),
```

```r
  partial_Elevation %>%
    mutate(Adjustment = "Elevation"),

  partial_MAT %>%
    mutate(Adjustment = "MAT"),

  partial_MAP %>%
    mutate(Adjustment = "MAP")
)

plot_df <- plot_df %>%
  mutate(
    p_adj = p.adjust(p, method = "BH"),

    sig = case_when(
      p_adj < 0.001 ~ "***",
      p_adj < 0.01  ~ "**",
      p_adj < 0.05  ~ "*",
      TRUE          ~ ""
    )
  )

plot_df$Adjustment <- factor(
  plot_df$Adjustment,
  levels = c(
    "Zero-order",
    "SOC",
    "pH",
    "Clay",
    "Elevation",
    "MAT",
    "MAP"
  )
)

plot_df$Variable <- factor(
  plot_df$Variable,
  levels = c(
    "Tetramethylated",
    "Hexamethylated",
    "Pentamethylated"
  )
)

partial <- ggplot(
  plot_df,
  aes(
    x = Adjustment,
    y = rho
```

```
  )
) +

  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    linewidth = 0.4
  ) +

  geom_segment(
    aes(
      x = Adjustment,
      xend = Adjustment,
      y = 0,
      yend = rho
    ),
    linewidth = 0.5
  ) +

  geom_point(
    size = 3.2
  ) +

  geom_text(
    aes(
      label = sig,
      y = rho + ifelse(rho >= 0, 0.045, -0.045)
    ),
    size = 4.2
  ) +

  facet_wrap(
    ~ Variable,
    nrow = 1
  ) +

  scale_y_continuous(
    limits = c(-0.82, 0.82),
    breaks = seq(-0.8, 0.8, 0.4)
  ) +

  labs(
    x = NULL,
    y = expression("Spearman " * rho)
  ) +

  theme_classic(
    base_size = 12
  ) +
```

```
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    ),
    strip.text = element_text(
      face = "bold",
      size = 12
    ),
    panel.spacing = unit(1, "lines")
  )

partial
```

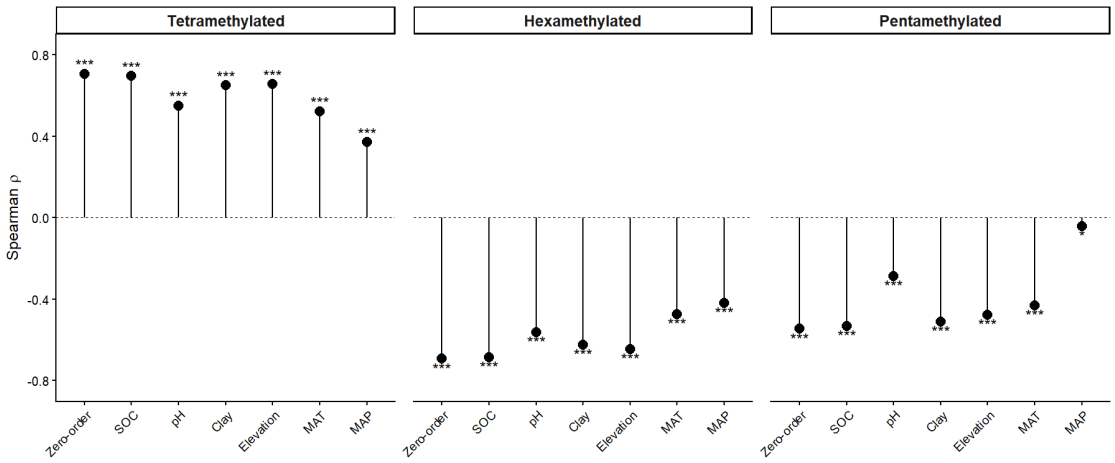


Figure 3A. Zero-order and partial Spearman correlations.

```
knitr::kable(plot_df, digits = 4, caption = "Coefficients and adjusted p-valu
es for Figure 3A.")
```

Coefficients and adjusted p-values for Figure 3A.

Variable	rho	p	Adjustment	p_adj	sig
Tetramethylated	0.7070	0.0000	Zero-order	0.0000	***
Pentamethylated	-0.5460	0.0000	Zero-order	0.0000	***
Hexamethylated	-0.6933	0.0000	Zero-order	0.0000	***
Tetramethylated	0.6957	0.0000	SOC	0.0000	***
Pentamethylated	-0.5335	0.0000	SOC	0.0000	***
Hexamethylated	-0.6859	0.0000	SOC	0.0000	***
Tetramethylated	0.5502	0.0000	pH	0.0000	***
Pentamethylated	-0.2880	0.0000	pH	0.0000	***
Hexamethylated	-0.5622	0.0000	pH	0.0000	***

Variable	rho	p	Adjustment	p_adj	sig
Tetramethylated	0.6490	0.0000	Clay	0.0000	***
Pentamethylated	-0.5102	0.0000	Clay	0.0000	***
Hexamethylated	-0.6241	0.0000	Clay	0.0000	***
Tetramethylated	0.6551	0.0000	Elevation	0.0000	***
Pentamethylated	-0.4773	0.0000	Elevation	0.0000	***
Hexamethylated	-0.6467	0.0000	Elevation	0.0000	***
Tetramethylated	0.5230	0.0000	MAT	0.0000	***
Pentamethylated	-0.4311	0.0000	MAT	0.0000	***
Hexamethylated	-0.4743	0.0000	MAT	0.0000	***
Tetramethylated	0.3721	0.0000	MAP	0.0000	***
Pentamethylated	-0.0429	0.0425	MAP	0.0425	*
Hexamethylated	-0.4199	0.0000	MAP	0.0000	***

Figure 3B: Redundancy analysis

Fit a multivariate RDA of the 15 brGDGT compounds using NPP, MAT, MAP, pH, SOC and clay as explanatory variables. Calculate adjusted r^2 and global and marginal permutation tests, and plot sample, compound and predictor scores.

```
brgdgt_vars <- c(
  "fIa", "fIb", "fIc",
  "fIIa", "fIIa_", "fIIb", "fIIb_", "fIIc", "fIIc_",
  "fIIIa", "fIIIa_", "fIIIb", "fIIIb_", "fIIIc", "fIIIc_"
)

rda_data <- data %>%
  dplyr::select(
    all_of(brgdgt_vars),
    NPP,
    BIO01,
    BIO12,
    `SOC_0-5`,
    `SOC_5-15`,
    pH,
    Clay,
    Elevation
  ) %>%
  rename(
    MAT = BIO01,
    MAP = BIO12,
    SOC05 = `SOC_0-5`,
    SOC515 = `SOC_5-15`
  ) %>%
```

```r
  filter(complete.cases(.))

br_clr <- as.matrix(
  rda_data[, brgdgt_vars]
)

m_full <- vegan::rda(
  br_clr ~
    NPP +
    MAT +
    MAP +
    pH +
    SOC05 +
    SOC515 +
    Clay,
  data = rda_data
)

vegan::RsquareAdj(m_full)
```

```
## $r.squared
## [1] 0.5479516
##
## $adj.r.squared
## [1] 0.5462522
```

```r
set.seed(123)
anova(
  m_full,
  permutations = 999
)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = br_clr ~ NPP + MAT + MAP + pH + SOC05 + SOC515 + Clay
, data = rda_data)
##            Df Variance      F Pr(>F)
## Model       7   654.54 322.43  0.001 ***
## Residual 1862   539.98
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
set.seed(123)
anova(
  m_full,
  by = "margin",
  permutations = 999
)
```

```
## Permutation test for rda under reduced model
## Marginal effects of terms
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = br_clr ~ NPP + MAT + MAP + pH + SOC05 + SOC515 + Clay
, data = rda_data)
##            Df Variance       F Pr(>F)
## NPP         1    15.06  51.942  0.001 ***
## MAT         1    34.13 117.691  0.001 ***
## MAP         1     3.68  12.678  0.001 ***
## pH          1    95.86 330.553  0.001 ***
## SOC05       1     8.29  28.589  0.001 ***
## SOC515      1     3.51  12.098  0.001 ***
## Clay        1     3.10  10.693  0.001 ***
## Residual 1862   539.98
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
mod <- m_full

scaling_use <- 2

site_scores <- as.data.frame(
  scores(mod, display = "sites", choices = 1:2, scaling = scaling_use)
)

s_scores <- as.data.frame(
  scores(mod, display = "species", choices = 1:2, scaling = scaling_use)
)

env_scores <- as.data.frame(
  scores(mod, display = "bp", choices = 1:2, scaling = scaling_use)
)

colnames(site_scores)[1:2] <- c("Axis1", "Axis2")
colnames(s_scores)[1:2] <- c("Axis1", "Axis2")
colnames(env_scores)[1:2] <- c("Axis1", "Axis2")

s_scores$Compound <- rownames(s_scores)
env_scores$label <- rownames(env_scores)

site_range_x <- diff(range(site_scores$Axis1, na.rm = TRUE))
site_range_y <- diff(range(site_scores$Axis2, na.rm = TRUE))

env_range_x <- diff(range(env_scores$Axis1, na.rm = TRUE))
env_range_y <- diff(range(env_scores$Axis2, na.rm = TRUE))

sp_range_x <- diff(range(s_scores$Axis1, na.rm = TRUE))
```

```r
sp_range_y <- diff(range(s_scores$Axis2, na.rm = TRUE))

env_mult <- 0.85 * min(site_range_x / env_range_x,
                       site_range_y / env_range_y)

species_mult <- 0.35 * min(site_range_x / sp_range_x,
                           site_range_y / sp_range_y)

env_scores_plot <- env_scores %>%
  mutate(
    Axis1_plot = Axis1 * env_mult,
    Axis2_plot = Axis2 * env_mult
  )

s_scores_plot <- s_scores %>%
  mutate(
    Axis1_plot = Axis1 * species_mult,
    Axis2_plot = Axis2 * species_mult
  )

if (!is.null(mod$CCA) && length(mod$CCA$eig) >= 2) {
  axis_percent <- mod$CCA$eig[1:2] / sum(mod$CCA$eig) * 100

  x_lab <- paste0(
    "RDA1 (", round(axis_percent[1], 1), "% of constrained variance)"
  )

  y_lab <- paste0(
    "RDA2 (", round(axis_percent[2], 1), "% of constrained variance)"
  )
} else {
  x_lab <- "RDA1"
  y_lab <- "RDA2"
}

p_rda <- ggplot() +

  geom_hline(
    yintercept = 0,
    linewidth = 0.3,
    linetype = "dashed",
    colour = "grey60"
  ) +

  geom_vline(
    xintercept = 0,
    linewidth = 0.3,
    linetype = "dashed",
```

```
    colour = "grey60"
  ) +

  geom_point(
    data = site_scores,
    aes(x = Axis1, y = Axis2),
    size = 1.0,
    alpha = 0.22,
    shape = 21,
    fill = "grey75",
    colour = "grey40",
    stroke = 0.2
  ) +

  geom_segment(
    data = env_scores_plot,
    aes(
      x = 0, y = 0,
      xend = Axis1_plot,
      yend = Axis2_plot
    ),
    arrow = arrow(length = unit(0.18, "cm"), type = "closed"),
    linewidth = 0.8,
    colour = "black"
  ) +

  geom_text_repel(
    data = env_scores_plot,
    aes(
      x = Axis1_plot,
      y = Axis2_plot,
      label = label
    ),
    size = 3.5,
    colour = "black",
    segment.color = NA,
    max.overlaps = Inf
  ) +

  geom_segment(
    data = s_scores_plot,
    aes(
      x = 0, y = 0,
      xend = Axis1_plot,
      yend = Axis2_plot
    ),
    arrow = arrow(length = unit(0.08, "cm")),
    linewidth = 0.3,
    alpha = 0.55,
```

```
    colour = "red"
  ) +

  geom_text_repel(
    data = s_scores_plot,
    aes(
      x = Axis1_plot,
      y = Axis2_plot,
      label = Compound
    ),
    size = 2.4,
    colour = "blue",
    segment.color = NA,
    max.overlaps = Inf
  ) +

  labs(
    x = x_lab,
    y = y_lab
  ) +

  coord_equal() +

  theme_classic(base_size = 13) +

  theme(
    axis.title = element_text(size = 13),
    axis.text = element_text(size = 11),
    plot.margin = ggplot2::margin(
      t = 10, r = 20, b = 10, l = 10
    )
  )

p_rda
```

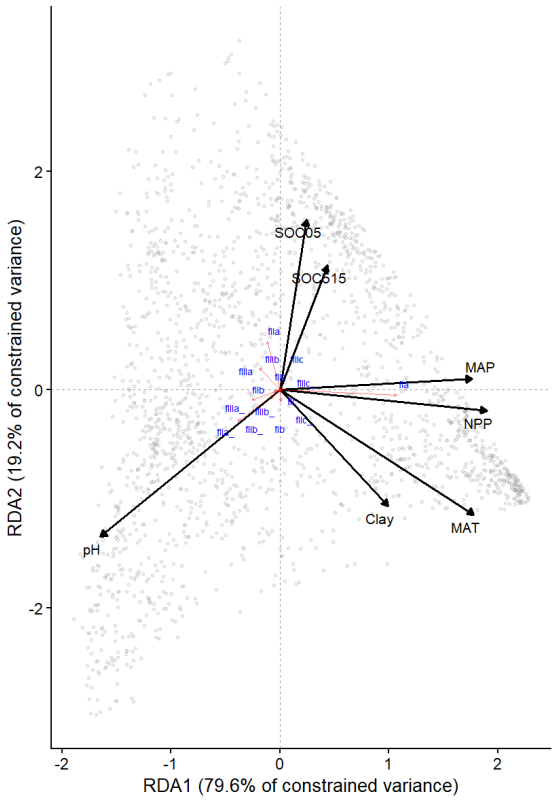


Figure 3B. Redundancy analysis of brGDGT composition.

Figure 3C: RDA eigenvalue ratios

Fit single-predictor RDAs to individual compounds and grouped compounds. Compare the first constrained eigenvalue with the first unconstrained eigenvalue and calculate adjusted r^2 for each predictor.

```
br_comp <- as.matrix(
  rda_data[, brgdgt_vars]
)

env_vars <- c("MAT", "NPP", "MAP", "pH",
              "Clay", "SOC515", "SOC05")

groups <- list(
  Tetra = c("fIa", "fIb", "fIc"),
  Penta = c("fIIa", "fIIa_", "fIIb", "fIIb_", "fIIc", "fIIc_"),
  Hexa  = c("fIIIa", "fIIIa_", "fIIIb", "fIIIb_", "fIIIc", "fIIIc_")
)

br_raw <- as.matrix(rda_data[, brgdgt_vars])

br_groups <- sapply(groups, function(x) {
  rowSums(rda_data[, x, drop = FALSE])
```

```
})

fit_rda <- function(Y, var) {

  mod <- rda(Y, rda_data[, var])

  lambda1 <- eigenvals(mod, model = "constrained")[1]
  lambda2 <- eigenvals(mod, model = "unconstrained")[1]

  tibble(
    Variable = var,
    Ratio = unname(lambda1 / lambda2),
    R2_adj = RsquareAdj(mod)$adj.r.squared
  )
}

results <- imap_dfr(
  list("Individual brGDGTs" = br_raw,
       "Grouped brGDGTs" = br_groups),
  function(Y, name) {
    map_dfr(env_vars, ~ fit_rda(Y, .x)) %>%
      mutate(Composition = name)
  }
)

results <- results %>%
  mutate(
    Variable = recode(
      Variable,
      SOC515 = "SOC 5–15 cm",
      SOC05 = "SOC 0–5 cm"
    ),
    Variable = factor(
      Variable,
      levels = rev(c(
        "MAT", "NPP", "MAP", "pH",
        "Clay", "SOC 5–15 cm",
        "SOC 0–5 cm"
      ))
    )
  )

knitr::kable(results, digits = 4, caption = "RDA eigenvalue ratios and adjust
ed R-squared.")
```

RDA eigenvalue ratios and adjusted R-squared.

Variable	Ratio	R2_adj	Composition
MAT	0.7255	0.3266	Individual brGDGTs

Variable	Ratio	R2_adj	Composition
NPP	0.9122	0.3392	Individual brGDGTs
MAP	0.7117	0.2930	Individual brGDGTs
pH	0.6298	0.2953	Individual brGDGTs
Clay	0.1927	0.1190	Individual brGDGTs
SOC 5–15 cm	0.0696	0.0471	Individual brGDGTs
SOC 0–5 cm	0.0876	0.0605	Individual brGDGTs
MAT	1.1852	0.4907	Grouped brGDGTs
NPP	1.1118	0.4762	Grouped brGDGTs
MAP	0.7633	0.3929	Grouped brGDGTs
pH	0.4236	0.2696	Grouped brGDGTs
Clay	0.2588	0.1857	Grouped brGDGTs
SOC 5–15 cm	0.0096	0.0081	Grouped brGDGTs
SOC 0–5 cm	0.0002	-0.0004	Grouped brGDGTs

```r
ggplot(results, aes(x = Ratio, y = Variable, fill = Composition)) +
  geom_col(
    position = position_dodge(width = 0.72),
    width = 0.28,
    color = NA
  ) +
  geom_text(
    aes(label = ifelse(Ratio == 0, "0", sprintf("%.2f", Ratio))),
    position = position_dodge(width = 0.72),
    hjust = -0.15,
    size = 3
  ) +
  geom_vline(
    xintercept = 1,
    linetype = "dashed",
    linewidth = 0.5,
    color = "#2E4DA7"
  ) +
  scale_fill_manual(
    values = c(
      "Grouped brGDGTs" = "#7A0000",
      "Individual brGDGTs" = "grey55"
    )
  ) +
  scale_x_continuous(
    limits = c(0, 1.35),
    breaks = c(0.0, 0.5, 1.0),
    expand = expansion(mult = c(0, 0.06))
  ) +
  labs(
```

```
    x = expression(lambda[1] / lambda[2]),
    y = NULL
  ) +
  theme_classic(base_size = 11) +
  theme(
    legend.position = "none",
    axis.line = element_line(color = "black", linewidth = 0.6),
    axis.ticks = element_line(color = "black", linewidth = 0.5),
    axis.text.y = element_text(color = "black"),
    axis.text.x = element_text(color = "black"),
    panel.background = element_rect(fill = "grey92", color = NA),
    plot.background = element_rect(fill = "grey92", color = NA)
  )
```

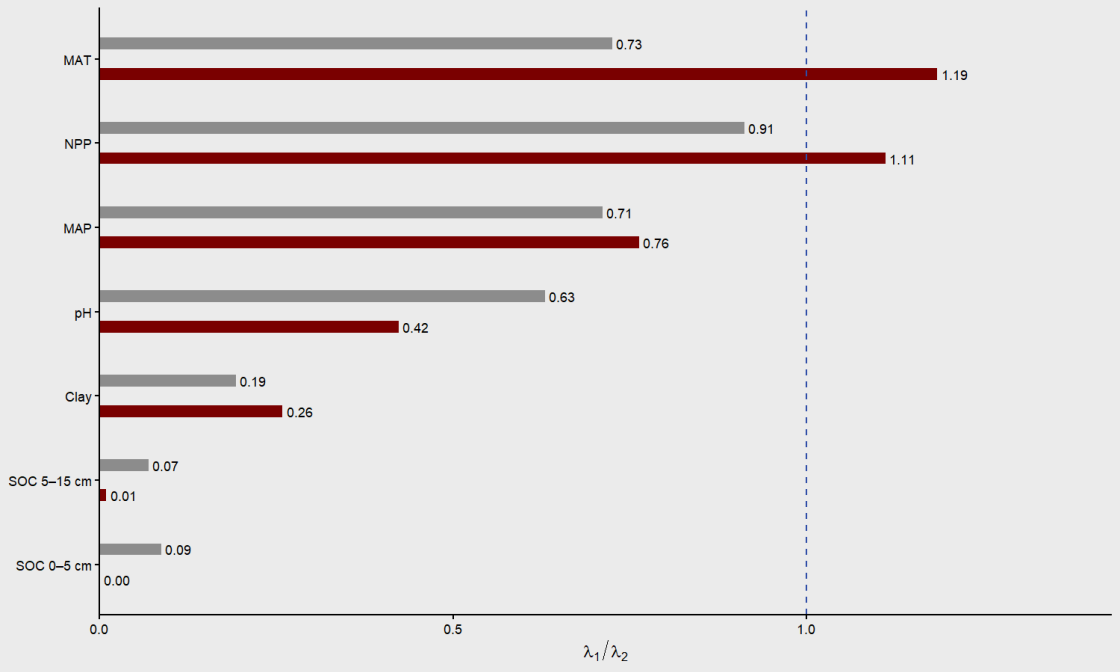


Figure 3C. Constrained-to-unconstrained eigenvalue ratios.

Random Forest predictors

Transform the 15 brGDGT compounds (CLR). Combine aquatic suspended-particle categories in `Sampletype_fixed` and retain NPP and sample coordinates.

```
data <- read.xlsx("Dataset_1.xlsx", sheet = "Dataset_R")
colnames(data)
```

```
##  [1] "No"             "Sample.name"    "Samplegroup"    "Sampletype"
##  [5] "Sample.subtype" "Region"         "Site"           "Latitude"
##  [9] "Longitude"      "pH"             "fIa"            "fIb"
## [13] "fIc"            "fIIa"           "fIIa_"          "fIIb"
## [17] "fIIb_"          "fIIc"           "fIIc_"          "fIIIa"
## [21] "fIIIa_"         "fIIIb"          "fIIIb_"         "fIIIc"
```

```
## [25] "fIIIc_"         "Reference"      "BIO01"          "BIO12"
## [29] "NPP"            "Elevation"      "PET"            "Bulk"
## [33] "Clay"           "CEC"            "Nitrogen"       "Phosphorus"
## [37] "SOC_0-5"        "SOC_5-15"
```

```
vars <- c(tetramethylated, pentamethylated, hexamethylated)
require_columns(data, vars, "Dataset_R")

vars_extra <- c("Sampletype", "NPP", "Latitude", "Longitude")

comp_data <- data[, vars]

comp_data <- comp_data / 100

comp_data <- comp_data + 1e-6

clr_data <- clr(comp_data)

clr_df <- as.data.frame(clr_data)
colnames(clr_df) <- paste0("clr_", colnames(clr_df))

model_data <- data.frame(clr_data, data[, vars_extra])

model_data <- model_data[complete.cases(model_data), ]

model_data <- model_data %>%
  mutate(
    Sampletype_fixed = case_when(

      Sampletype %in% c("Lacustrine SPM",
                        "Low DO Lacustrine SPM",
                        "Riverine Sediment and SPM",
                        "Lacustrine Meso/Microcosm") ~ "Aquatic_SPM_combined"
,
      TRUE ~ as.character(Sampletype)
    ),
    Sampletype_fixed = as.factor(Sampletype_fixed)
  )

formula <- as.formula(paste("NPP ~", paste(c(colnames(clr_data), "Sampletype_
fixed"), collapse = " + ")))

formula
```

```
## NPP ~ fIa + fIb + fIc + fIIa + fIIa_ + fIIb + fIIb_ + fIIc +
##     fIIc_ + fIIIa + fIIIa_ + fIIIb + fIIIb_ + fIIIc + fIIIc_ +
##     Sampletype_fixed
```

Figure S2: Correlations by sample type

Calculate compound–NPP Spearman correlations separately for soil, peat and lacustrine sediment samples using CLR-transformed values.

```
unique(model_data$Sampletype)
```

```
## [1] "Soil"                      "Riverine Sediment and SPM"
## [3] "Peat"                      "Lacustrine Meso/Microcosm"
## [5] "Lacustrine Sediment"       "Lacustrine SPM"
## [7] "Low DO Lacustrine SPM"
```

```
gdgt <- colnames(clr_data)

cors_long <- model_data %>%
  filter(Sampletype_fixed %in% c("Soil", "Peat", "Lacustrine Sediment")) %>%
  group_by(Sampletype_fixed) %>%
  summarise(across(all_of(gdgt),
                   ~cor(.x, NPP, method = "spearman",
                        use = "complete.obs"))) %>%
  pivot_longer(-Sampletype_fixed,
               names_to = "brGDGT",
               values_to = "rho")

ggplot(cors_long,
       aes(x = brGDGT, y = rho,
           group = Sampletype_fixed,
           colour = Sampletype_fixed)) +
  geom_line() +
  geom_point(size = 2) +
  geom_hline(yintercept = 0, linetype = 2) +
  theme_classic() +
  labs(x = NULL,
       y = "Spearman rho with NPP",
       colour = "Sample type")
```



Figure S2. Compound-NPP correlations by sample type.

Figure S5: Cross-environment RF prediction

Train a compound-based RF on each sample type and predict the other types. Calculate correlation coefficient and RMSE for each training–test combination and display the transfer results as a matrix.

```
types <- c("Soil",
           "Peat",
           "Lacustrine Sediment",
           "Aquatic_SPM_combined")

results <- expand.grid(
  Train = types,
  Test = types,
  stringsAsFactors = FALSE
) %>%
  filter(Train != Test)

results <- results %>%
  rowwise() %>%
  mutate(
    metrics = list({

      train <- model_data %>% filter(Sampletype_fixed == Train)
      test  <- model_data %>% filter(Sampletype_fixed == Test)

      rf <- randomForest(
```

```
        x = train[, gdgt],
        y = train$NPP
      )

      pred <- predict(rf, test[, gdgt])

      data.frame(
        r = cor(pred, test$NPP),
        RMSE = sqrt(mean((pred - test$NPP)^2))
      )
    })
  ) %>%
  unnest(metrics)

results
```

```
## # A tibble: 12 × 4
##    Train                Test                    r  RMSE
##    <chr>                <chr>               <dbl> <dbl>
##  1 Peat                 Soil                0.567  139.
##  2 Lacustrine Sediment  Soil                0.636  134.
##  3 Aquatic_SPM_combined Soil                0.569  147.
##  4 Soil                 Peat                0.682  158.
##  5 Lacustrine Sediment  Peat                0.494  180.
##  6 Aquatic_SPM_combined Peat                0.472  205.
##  7 Soil                 Lacustrine Sediment 0.690  117.
##  8 Peat                 Lacustrine Sediment 0.484  150.
##  9 Aquatic_SPM_combined Lacustrine Sediment 0.694  114.
## 10 Soil                 Aquatic_SPM_combined 0.756  140.
## 11 Peat                 Aquatic_SPM_combined 0.516  144.
## 12 Lacustrine Sediment  Aquatic_SPM_combined 0.649  124.
```

```
results$label <- paste0(
  "r = ", sprintf("%.2f", results$r),
  "\nRMSE = ", round(results$RMSE)
)

results$Test <- gsub("Aquatic_SPM_combined", "Aquatic SPM", results$Test)
results$Train <- gsub("Aquatic_SPM_combined", "Aquatic SPM", results$Train)

knitr::kable(results, digits = 3, caption = "Cross-environment prediction met
rics.")
```

Cross-environment prediction metrics.

Train	Test	r	RMSE	label
Peat	Soil	0.567	138.980	r = 0.57 RMSE = 139

Train	Test	r	RMSE	label
Lacustrine Sediment	Soil	0.636	133.747	r = 0.64
RMSE = 134				
Aquatic SPM	Soil	0.569	146.915	r = 0.57
RMSE = 147				
Soil	Peat	0.682	157.906	r = 0.68
RMSE = 158				
Lacustrine Sediment	Peat	0.494	179.808	r = 0.49
RMSE = 180				
Aquatic SPM	Peat	0.472	205.485	r = 0.47
RMSE = 205				
Soil	Lacustrine Sediment	0.690	117.356	r = 0.69
RMSE = 117				
Peat	Lacustrine Sediment	0.484	150.257	r = 0.48
RMSE = 150				
Aquatic SPM	Lacustrine Sediment	0.694	113.852	r = 0.69
RMSE = 114				
Soil	Aquatic SPM	0.756	139.778	r = 0.76
RMSE = 140				
Peat	Aquatic SPM	0.516	144.393	r = 0.52
RMSE = 144				
Lacustrine Sediment	Aquatic SPM	0.649	124.455	r = 0.65
RMSE = 124				

```
ggplot(results, aes(x = Test, y = Train, fill = r)) +
  geom_tile(color = "white") +
  geom_text(aes(label = label), size = 3.5) +
  scale_fill_gradient2(
    low = "white",
```

```
    mid = "lightblue",
    high = "steelblue4",
    midpoint = 0.5,
    limits = c(0, 0.8)
  ) +
  theme_classic() +
  labs(
    x = "Test sample type",
    y = "Training sample type",
    fill = "Correlation (r)"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

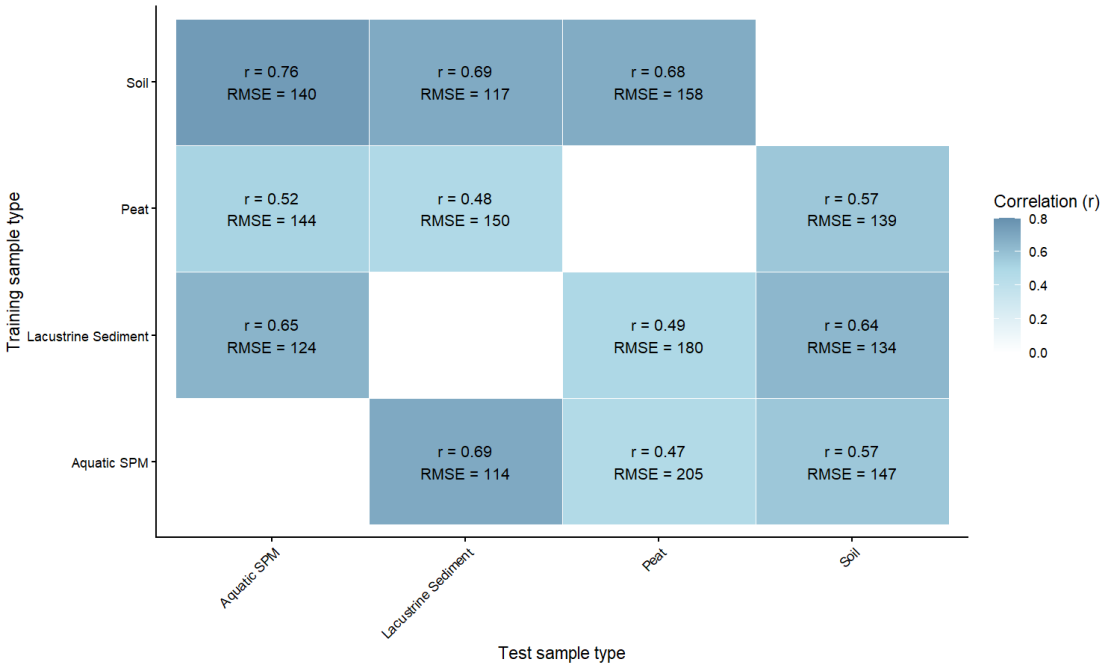


Figure S5. Cross-environment RF prediction performance.

RF fitting and cross-validation

Fit a 1000-tree RF to the complete calibration data. For random five-fold cross-validation, select `mtry` with `caret`. For nested spatial validation, use five outer and five inner folds with block sizes of 50, 100, 500 and 1000 km. Load the saved model objects for prediction and figure generation.

```
model_all_data <- train(
  formula,
  data = model_data,
  method = "rf",
  trControl = trainControl(method = "none"),
```

```r
  importance = TRUE,
  ntree = 1000,
  tuneGrid = data.frame(mtry = 11)
)

print(model_all_data)

predictions_all <- predict(model_all_data, model_data)
r2_all <- cor(model_data$NPP, predictions_all)^2
r2_all

saveRDS(model_all_data, file = "model_brGDGT-NPP_all1.rds")

model_all_data <- readRDS("model_brGDGT-NPP_all1.rds")

set.seed(2026)

train_control <- trainControl(
  method = "cv",
  number = 5,
  savePredictions = "final",
  returnResamp = "final"
)

model_cv <- train(
  formula,
  data = model_data,
  method = "rf",
  trControl = train_control,
  importance = TRUE,
  ntree = 1000,
  tuneLength = 3
)

print(model_cv)

saveRDS(model_cv, file = "model_brGDGT-NPP_cv1.rds")
model_cv <- readRDS("model_brGDGT-NPP_cv1.rds")

nested_spatial_cv_complete <- function(model_data_sf, block_size_km, k_outer
= 5, k_inner = 5,
                                       save_models = FALSE, model_name = NULL
) {

  results <- data.frame()
  model_df <- st_drop_geometry(model_data_sf)

  all_fold_predictions <- list()
  all_final_models <- list()
```

```
  outer_blocks <- cv_spatial(model_data_sf, k = k_outer, size = block_size_km
)

  for(outer_fold in 1:k_outer) {
    cat(sprintf("\n--- Outer Fold %d/%d ---\n", outer_fold, k_outer))

    train_idx <- outer_blocks$folds_list[[outer_fold]][[1]]
    test_idx <- outer_blocks$folds_list[[outer_fold]][[2]]

    train_df <- model_df[train_idx, ]
    test_df <- model_df[test_idx, ]

    train_sf <- model_data_sf[train_idx, ]
    inner_blocks <- cv_spatial(train_sf, k = k_inner, size = block_size_km)

    inner_train_ids <- lapply(inner_blocks$folds_list, function(x) x[[1]])
    inner_test_ids <- lapply(inner_blocks$folds_list, function(x) x[[2]])

    tr_control_inner <- trainControl(
      method = "cv",
      index = inner_train_ids,
      indexOut = inner_test_ids
    )

    inner_model <- train(
      NPP ~ Sampletype_fixed + fIa + fIb + fIc + fIIa + fIIa_ + fIIb + fIIb_
+
        fIIc + fIIc_ + fIIIa + fIIIa_ + fIIIb + fIIIb_ + fIIIc + fIIIc_,
      data = train_df,
      method = "rf",
      trControl = tr_control_inner,
      tuneGrid = expand.grid(mtry = c(3, 7, 11, 15)),
      ntree = 1000
    )

    final_model <- train(
      NPP ~ Sampletype_fixed + fIa + fIb + fIc + fIIa + fIIa_ + fIIb + fIIb_
+
        fIIc + fIIc_ + fIIIa + fIIIa_ + fIIIb + fIIIb_ + fIIIc + fIIIc_,
      data = train_df,
      method = "rf",
      tuneGrid = inner_model$bestTune,
      trControl = trainControl(method = "none"),
      ntree = 1000
    )

    if(save_models) {
```

```r
    all_final_models[[outer_fold]] <- final_model
  }

  predictions <- predict(final_model, test_df)
  residuals <- test_df$NPP - predictions
  r = cor(test_df$NPP, predictions)
  r2 <- cor(test_df$NPP, predictions)^2
  rmse <- sqrt(mean((test_df$NPP - predictions)^2))
  mae <- mean(abs(test_df$NPP - predictions))

  all_fold_predictions[[outer_fold]] <- data.frame(
    Observed = test_df$NPP,
    Predicted = predictions,
    Fold = outer_fold,
    BlockSize_km = block_size_km/1000,
    Residuals = residuals,
    Sampletype_fixed = test_df$Sampletype_fixed,
    X = st_coordinates(model_data_sf[test_idx, ])[, 1],
    Y = st_coordinates(model_data_sf[test_idx, ])[, 2]
  )

  results <- rbind(results, data.frame(
    outer_fold = outer_fold,
    r = r,
    r2 = r2,
    rmse = rmse,
    mae = mae,
    best_mtry = inner_model$bestTune$mtry,
    n_train = nrow(train_df),
    n_test = nrow(test_df)
  ))

  cat(sprintf("  r² = %.3f, RMSE = %.1f, MAE = %.1f, mtry = %d\n",
              r2, rmse, mae, inner_model$bestTune$mtry))
}

cat(sprintf("\n=== Summary: r² = %.3f ± %.3f ===\n",
            mean(results$r2), sd(results$r2)))

all_predictions_df <- do.call(rbind, all_fold_predictions)

return_list <- list(
  results = results,
  predictions = all_predictions_df
)

if(save_models && length(all_final_models) > 0) {
  return_list$models <- all_final_models
```

```r
    if(!is.null(model_name)) {
      saveRDS(return_list, file = paste0(model_name, ".rds"))
      cat(sprintf("Models saved: %s.rds\n", model_name))
    }
  }

  return(return_list)
}

model_data_sf_fixed <- st_as_sf(
  model_data,
  coords = c("Longitude", "Latitude"),
  crs = 4326
)
result_caret_50km_complete <- nested_spatial_cv_complete(
  model_data_sf = model_data_sf_fixed,
  block_size_km = 50000,
  save_models = TRUE,
  model_name = "spatial_cv_50km_models1"
)

result_caret_100km_complete <- nested_spatial_cv_complete(
  model_data_sf = model_data_sf_fixed,
  block_size_km = 100000,
  save_models = TRUE,
  model_name = "spatial_cv_100km_models1"
)

result_caret_500km_complete <- nested_spatial_cv_complete(
  model_data_sf = model_data_sf_fixed,
  block_size_km = 500000,
  save_models = TRUE,
  model_name = "spatial_cv_500km_models1"
)

result_caret_1000km_complete <- nested_spatial_cv_complete(
  model_data_sf = model_data_sf_fixed,
  block_size_km = 1000000,
  save_models = TRUE,
  model_name = "spatial_cv_1000km_models1"
)

model_all_data <- readRDS("model_brGDGT-NPP_all1.rds")
model_cv <- readRDS("model_brGDGT-NPP_cv1.rds")

load_spatial_cv_models <- function(block_size_km) {
  file_name <- paste0("spatial_cv_", block_size_km, "km_models1.rds")
  if (file.exists(file_name)) {
    cat(sprintf("Loading %d km models...\n", block_size_km))
    return(readRDS(file_name))
```

```
  } else {
    cat(sprintf("File %s not found\n", file_name))
    return(NULL)
  }
}

model_spatial_cv_50<-load_spatial_cv_models(50)

## Loading 50 km models...

model_spatial_cv_100<-load_spatial_cv_models(100)

## Loading 100 km models...

model_spatial_cv_500<-load_spatial_cv_models(500)

## Loading 500 km models...

model_spatial_cv_1000<-load_spatial_cv_models(1000)

## Loading 1000 km models...
```

Table 1: Validation statistics

Select the best-tuned random-CV predictions and summarise the spatial prediction metrics by sample type and across samples.

```
# Random cross-validation
best_preds <- model_cv$pred %>%
  filter(mtry == model_cv$bestTune$mtry) %>%
  mutate(
    Sampletype_fixed = as.character(
      model_data$Sampletype_fixed[rowIndex]
    ),
    Observed = obs,
    Predicted = pred
  )

get_stats <- function(df, validation, block) {

  df <- df %>%
    mutate(Sampletype_fixed = as.character(Sampletype_fixed))

  bind_rows(df, mutate(df, Sampletype_fixed = "All")) %>%
    group_by(Sampletype_fixed) %>%
    summarise(
      r = cor(Observed, Predicted),
      r2 = r^2,
      RMSE = sqrt(mean((Observed - Predicted)^2)),
      .groups = "drop"
    ) %>%
```

```
    pivot_longer(c(r, r2, RMSE),
                 names_to = "Summary",
                 values_to = "Value") %>%
    mutate(Validation = validation, `Block size` = block)
}

table1 <- bind_rows(
  get_stats(best_preds, "Random cross-validation", "-"),
  get_stats(model_spatial_cv_50$predictions, "Nested spatial cross-validation
", "50"),
  get_stats(model_spatial_cv_100$predictions, "Nested spatial cross-validatio
n", "100"),
  get_stats(model_spatial_cv_500$predictions, "Nested spatial cross-validatio
n", "500"),
  get_stats(model_spatial_cv_1000$predictions, "Nested spatial cross-validati
on", "1000")
) %>%
  mutate(
    Sampletype_fixed = recode(Sampletype_fixed,
      "Soil" = "S",
      "Lacustrine Sediment" = "LS",
      "Peat" = "P",
      "Aquatic_SPM_combined" = "SPM"
    ),
    Validation = factor(Validation, levels = c(
      "Random cross-validation",
      "Nested spatial cross-validation"
    )),
    `Block size` = factor(`Block size`,
                          levels = c("-", "50", "100", "500", "1000")),
    Summary = factor(Summary, levels = c("r", "r2", "RMSE")),
    Value = ifelse(Summary == "RMSE",
                   sprintf("%.1f", Value),
                   sprintf("%.2f", Value))
  ) %>%
  arrange(Validation, `Block size`, Summary) %>%
  pivot_wider(names_from = Sampletype_fixed, values_from = Value) %>%
  dplyr::select(Validation, `Block size`, Summary, S, LS, P, SPM, All)

types <- c("Soil", "Lacustrine Sediment",
           "Peat", "Aquatic_SPM_combined")

n <- sapply(types, function(x)
  sum(model_data$Sampletype_fixed == x))

names(table1)[4:8] <- paste0(
  c("S", "LS", "P", "SPM", "All"),
  " (n=", c(n, nrow(model_data)), ")"
)
```

```
knitr::kable(
  table1,
  caption = "Table 1. Random Forest model performance by sample type after random and nested spatial cross-validation."
)
```

Table 1. Random Forest model performance by sample type after random and nested spatial cross-validation.

Validation	Block size	Summary	S (n=1045)	LS (n=388)	P (n=383)	SPM (n=418)	All (n=2234)
Random cross-validation	-	r	0.81	0.81	0.87	0.92	0.86
Random cross-validation	-	r2	0.65	0.66	0.76	0.85	0.74
Random cross-validation	-	RMSE	93.3	91.4	99.5	64.8	89.5
Nested spatial cross-validation	50	r	0.75	0.78	0.73	0.82	0.78
Nested spatial cross-validation	50	r2	0.56	0.61	0.53	0.68	0.61
Nested spatial cross-validation	50	RMSE	104.3	98.1	138.9	96.9	108.7
Nested spatial cross-validation	100	r	0.75	0.77	0.75	0.80	0.78
Nested spatial cross-validation	100	r2	0.57	0.60	0.57	0.64	0.61
Nested spatial cross-validation	100	RMSE	103.4	100.2	133.7	102.1	108.4
Nested spatial cross-validation	500	r	0.72	0.71	0.70	0.73	0.73
Nested spatial cross-validation	500	r2	0.51	0.51	0.49	0.54	0.54
Nested spatial cross-validation	500	RMSE	109.4	112.6	145.3	119.3	118.7
Nested spatial cross-validation	1000	r	0.70	0.77	0.66	0.80	0.74
Nested spatial cross-validation	1000	r2	0.50	0.59	0.44	0.64	0.54

Validation	Block size	Summary	S (n=1045)	LS (n=388)	P (n=383)	SPM (n=418)	All (n=2234)
Nested spatial cross-validation	1000	RMSE	111.6	101.3	158.6	101.4	117.7

Figure 4: RF predictions and residuals

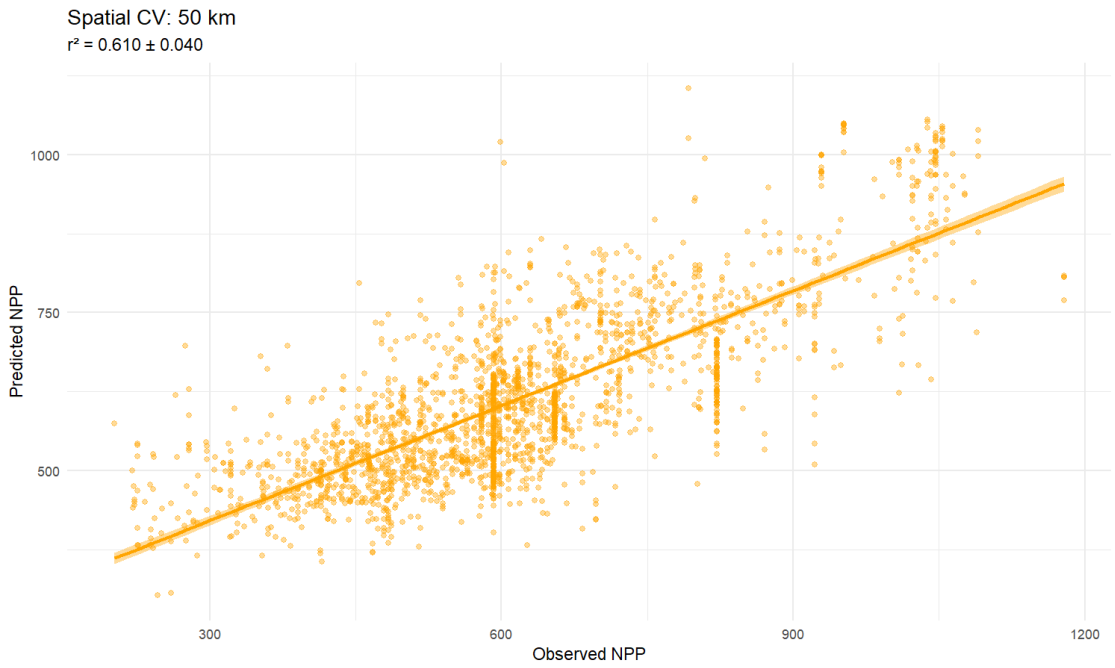
Compare observed and predicted NPP for in-sample, random-CV and spatial-CV predictions. Plot residuals against predictions, with marginal histograms.

```
obs_in <- model_data$NPP
pred_in <- predict(model_all_data, model_data)
r2_in <- round((cor(obs_in, pred_in)^2),2)

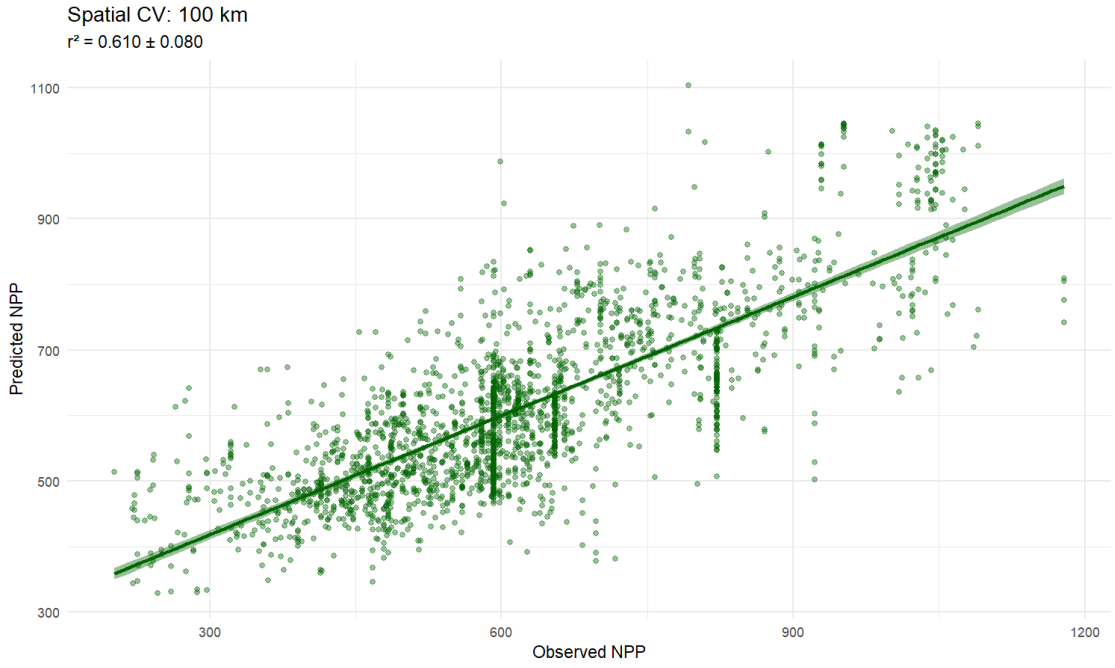
cv_predictions <- model_cv$pred
obs_cv <- cv_predictions$obs
pred_cv <- cv_predictions$pred
r2_cv <- round( (cor(obs_cv, pred_cv)^2), 2)

plot_spatial_cv <- function(result_obj, block_name, color) {
  if(!is.null(result_obj$predictions)) {
    p <- ggplot(result_obj$predictions, aes(x = Observed, y = Predicted)) +
      geom_point(color = color, alpha = 0.4, size = 1.5) +
      geom_smooth(method = "lm", se = TRUE,
                  color = color, fill = adjustcolor(color, alpha.f = 0.1)) +
      labs(title = paste("Spatial CV:", block_name),
           subtitle = sprintf("r² = %.3f ± %.3f",
                              mean( round((result_obj$results$r2), 2)  ),
                              sd(result_obj$results$r2)),
           x = "Observed NPP", y = "Predicted NPP") +
      theme_minimal()
    return(p)
  }
  return(NULL)
}

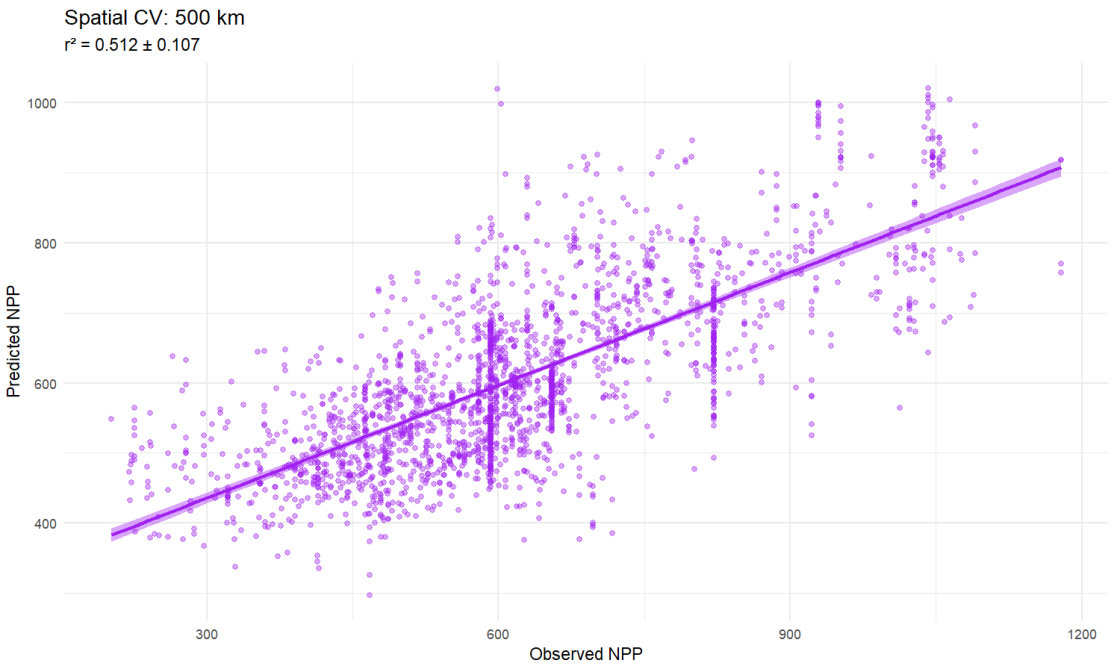
p50 <- plot_spatial_cv(model_spatial_cv_50, "50 km", "orange")
p100 <- plot_spatial_cv(model_spatial_cv_100, "100 km", "darkgreen")
p500 <- plot_spatial_cv(model_spatial_cv_500, "500 km", "purple")
p1000 <- plot_spatial_cv(model_spatial_cv_1000, "1000 km", "brown")
p50
```


Spatial CV: 50 km
r² = 0.610 ± 0.040
Predicted NPP
1000
750
500
300
600
900
1200
Observed NPP

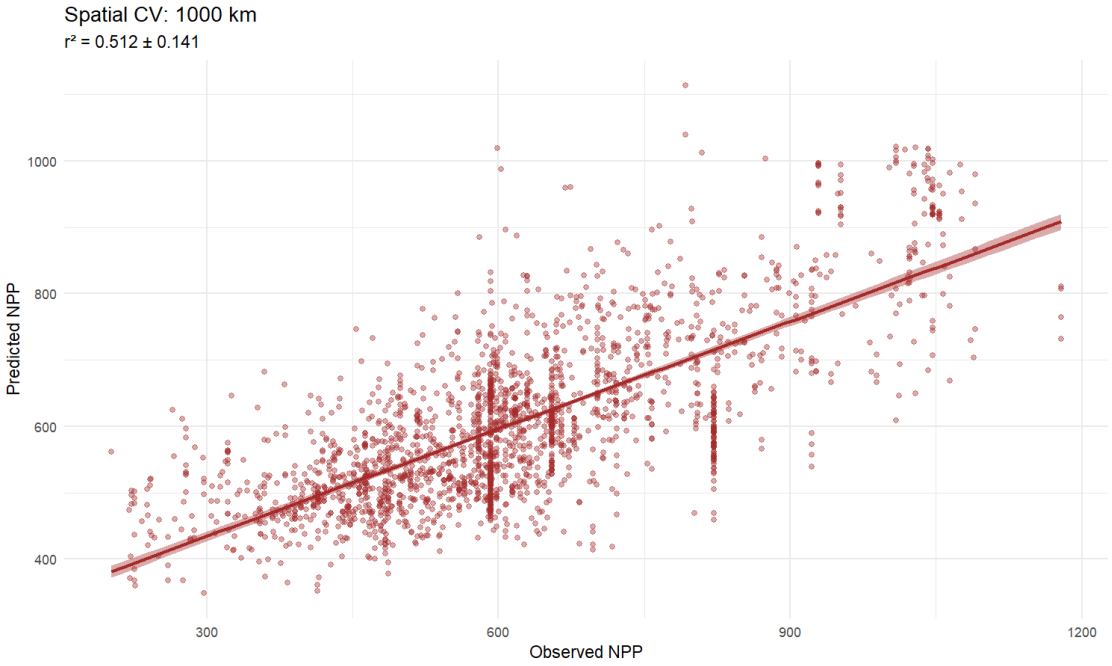
```
p100
```


Spatial CV: 100 km
r² = 0.610 ± 0.080
Predicted NPP
1100
900
700
500
300
600
900
1200
Observed NPP

```
p500
```




```
p1000
```




```
all_fracs <- colnames(model_data)[grepl("^clr_", colnames(model_data))]
newdata_in <- model_data

obs_in <- newdata_in$NPP
pred_in <- predict(model_cv, newdata = newdata_in)
r2_in <- round((1 - sum((obs_in - pred_in)^2) / sum((obs_in - mean(obs_in))^2
)), 2)
```

```r
cv_pred <- model_cv$pred
cv_pred <- subset(cv_pred, mtry == model_cv$bestTune$mtry)
obs_cv <- cv_pred$obs

pred_cv <- cv_pred$pred
r2_cv <- 1 - sum((obs_cv - pred_cv)^2) / sum((obs_cv - mean(obs_cv))^2)

df_cv <- data.frame(Observed = obs_cv, Predicted = pred_cv, Type = "Cross-val
idated")
df_in <- data.frame(Observed = obs_in, Predicted = pred_in, Type = "In-sample
")

p1 <- ggplot(df_in, aes(x = Observed, y = Predicted)) +
  geom_point(color = "darkblue", alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", se = TRUE,
              color = "darkblue", fill = "lightblue", alpha = 0.3) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray")
+
  labs(title = "Random Forest: In-sample Predictions",
       subtitle = "Model trained and tested on all data",
       x = "Observed NPP",
       y = "Predicted NPP") +
  annotate("text", x = min(obs_in), y = max(pred_in),
           label = paste("r² =", round(r2_in, 2)),
           hjust = 0, vjust = 1.5, color = "black", size = 5, fontface = "bol
d") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(size = 11, color = "gray40"))



p2 <- ggplot(df_cv, aes(x = Observed, y = Predicted)) +
  geom_point(color = "darkred", alpha = 0.5, size = 2) +
  geom_smooth(method = "lm", se = TRUE,
              color = "darkred", fill = "pink", alpha = 0.2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray")
+
  labs(title = "Random Forest: Random Cross-Validation",
       subtitle = "5-fold CV assuming independent samples",
       x = "Observed NPP",
       y = "Predicted NPP") +
  annotate("text", x = min(obs_cv), y = max(pred_cv),
           label = paste("r² =", round(r2_cv, 2)),
           hjust = 0, vjust = 1.5, color = "black", size = 5, fontface = "bol
d") +
  theme_minimal() +
```

```r
  theme(plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(size = 11, color = "gray40"))

calc_auto_tolerance <- function(residuals, percentiles = c(95, 75)) {

  abs_res <- abs(residuals)

  thresholds <- quantile(abs_res, probs = percentiles/100)

  return(thresholds)
}

auto_residual_plot <- function(predictions_df, title, color,
                               percentiles = c(95, 75)) {

  if(!"Residuals" %in% names(predictions_df)) {
    predictions_df$Residuals <- predictions_df$Observed - predictions_df$Pred
icted
  }

  tolerances <- calc_auto_tolerance(predictions_df$Residuals, percentiles)
  tol_95 <- tolerances[1]
  tol_75 <- tolerances[2]

  pct_95 <- mean(abs(predictions_df$Residuals) <= tol_95) * 100
  pct_75 <- mean(abs(predictions_df$Residuals) <= tol_75) * 100

  p <- ggplot(predictions_df, aes(x = Predicted, y = Residuals)) +

    annotate("rect", xmin = -Inf, xmax = Inf,
             ymin = -tol_95, ymax = tol_95,
             fill = "lightgreen", alpha = 0.05) +

    annotate("rect", xmin = -Inf, xmax = Inf,
             ymin = -tol_75, ymax = tol_75,
             fill = "green", alpha = 0.1) +

    geom_point(color = color, alpha = 0.4, size = 1.2) +

    geom_hline(yintercept = 0, linetype = "dashed", color = "black", size = 0
.8) +
    geom_hline(yintercept = c(-tol_95, tol_95),
               linetype = "dotted", color = "darkgreen", alpha = 0.5) +
    geom_hline(yintercept = c(-tol_75, tol_75),
               linetype = "dashed", color = "darkgreen", alpha = 0.7) +

    labs(title = paste("Residuals:", title),
         subtitle = sprintf("%.0f%% within ±%.0f | %.0f%% within ±%.0f",
```

```
                              pct_75, tol_75, pct_95, tol_95),
         x = "Predicted NPP", y = "Residuals (Obs - Pred)") +
    theme_minimal(base_size = 9) +
    theme(plot.title = element_text(face = "bold", size = 10),
          plot.subtitle = element_text(size = 8))+
    ylim(-500, 500)+
    scale_x_continuous(limits = c(300, 1300), breaks = seq(300, 1200, 300))

  p_with_marginal <- ggMarginal(p, type = "histogram", margins = "y",
                                groupColour = FALSE, groupFill = FALSE,
                                fill = color, alpha = 0.6)

  return(list(plot = p_with_marginal, tol_95 = tol_95, tol_75 = tol_75))
}

res_in <- obs_in - pred_in
auto_r1 <- auto_residual_plot(
  data.frame(Predicted = pred_in, Residuals = res_in, Observed = obs_in),
  "In-sample", "darkblue"
)

res_cv <- obs_cv - pred_cv

df_random <- data.frame(
  Predicted = pred_cv,
  Residuals = res_cv,
  Observed = obs_cv
) %>%
  dplyr::filter(if_all(everything(), is.finite))

clean_preds <- function(df) {
  df %>%
    dplyr::filter(
      if_all(where(is.numeric), is.finite)
    )
}

auto_r2 <- auto_residual_plot(
  df_random, "Random CV", "darkred"
)

auto_r50 <- auto_residual_plot(
  clean_preds(model_spatial_cv_50$predictions),
  "Spatial CV 50 km", "orange"
)

auto_r100 <- auto_residual_plot(
  clean_preds(model_spatial_cv_100$predictions),
  "Spatial CV 100 km", "darkgreen"
```

```
)

auto_r500 <- auto_residual_plot(
  clean_preds(model_spatial_cv_500$predictions),
  "Spatial CV 500 km", "purple"
)

auto_r1000 <- auto_residual_plot(
  clean_preds(model_spatial_cv_1000$predictions),
  "Spatial CV 1000 km", "brown"
)

grid.arrange(p1,auto_r1$plot, p2,auto_r2$plot, p50,auto_r50$plot, p100,auto_r
100$plot,
             p500,auto_r500$plot, p1000, auto_r1000$plot, ncol=4)
```

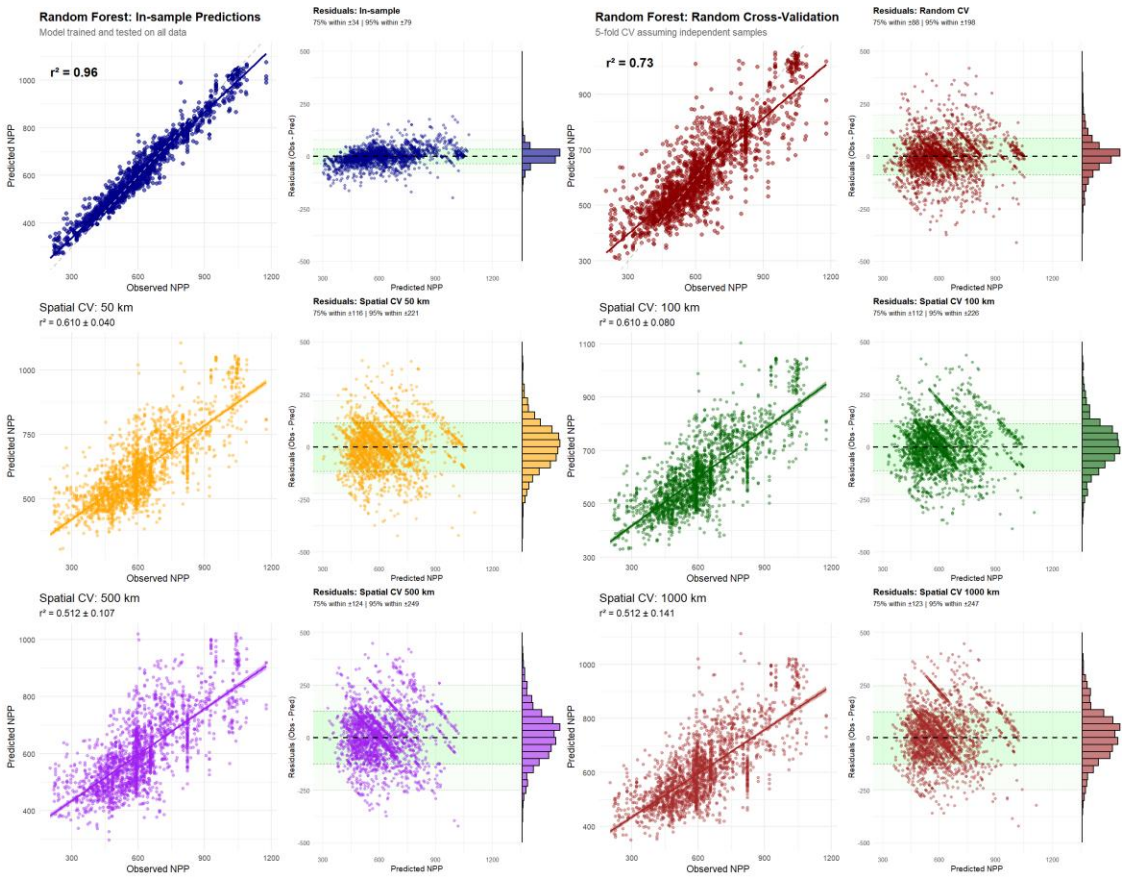


Figure 4. RF prediction and residual diagnostics.

Figure 5: Independent validation

Apply the calibration models to the independent brGDGT–NPP dataset. Average predictions across the saved fold models for each spatial block size. Compare measured NPP with RF and MOD17A3 predictions using Pearson r, regression r^2 and RMSE.

```r
m_NPP_df <- read.xlsx("Dataset_1.xlsx", sheet = "Measured_NPP_brGDGTs")

tetramethylated <- c("fIa", "fIb", "fIc")
pentamethylated <- c("fIIa", "fIIa_", "fIIb", "fIIb_", "fIIc", "fIIc_")
hexamethylated  <- c("fIIIa", "fIIIa_", "fIIIb", "fIIIb_", "fIIIc", "fIIIc_")
all_fracs <- c(tetramethylated, pentamethylated, hexamethylated)

frac_data <- m_NPP_df %>%
  dplyr::select(all_of(all_fracs)) %>%
  mutate(across(everything(), ~ .x / 100 + 1e-6))

clr_transformed <- clr(frac_data)
clr_df <- as.data.frame(clr_transformed)
colnames(clr_df) <- paste0( colnames(clr_df))

newdata <- bind_cols(
  Sampletype = factor(m_NPP_df$Sampletype, levels = levels(model_data$Samplet
ype)),
  clr_df
)
newdata$Sampletype_fixed<- m_NPP_df$Sampletype

model_all_data

## Random Forest
##
## 2234 samples
##   16 predictor
##
## No pre-processing
## Resampling: None

model_cv

## Random Forest
##
## 2234 samples
##   16 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 1788, 1788, 1787, 1787, 1786
## Resampling results across tuning parameters:
##
##   mtry  RMSE      Rsquared   MAE
##    2    92.27355  0.7240895  65.20059
##   10    89.43811  0.7355846  61.88159
##   18    89.74069  0.7327392  61.60257
##
```

```
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was mtry = 10.

predicted_npp <- predict(model_all_data, newdata = newdata)

predicted_npp

##        1        2        3        4        5        6        7        8
## 718.8542 663.9250 718.8542 663.9250 640.6080 657.2584 784.6382 784.6382
##        9       10       11       12       13       14       15       16
## 975.6845 995.6961 995.6961 506.5285 503.7429 539.4275 539.4275 550.2306
##       17       18       19       20       21       22
## 399.7631 577.1762 586.5285 497.7830 609.2471 608.3943

m_NPP_df$Predicted_NPP_all<- predicted_npp

predicted_npp <- predict(model_cv, newdata = newdata)

predicted_npp

##        1        2        3        4        5        6        7        8
## 718.1100 662.6445 718.1100 662.6445 639.5649 649.8875 774.0513 774.0513
##        9       10       11       12       13       14       15       16
## 958.6516 989.8266 989.8266 513.8235 503.1334 541.0209 541.0209 550.6475
##       17       18       19       20       21       22
## 401.8822 585.6194 594.5206 501.3053 615.6559 608.0725

m_NPP_df$Predicted_NPP_cv<- predicted_npp

predicted_npp <- predict_fold_models(model_spatial_cv_50$models, newdata = ne
wdata)

predicted_npp

## [[1]]
##        1        2        3        4        5        6        7        8
## 697.0242 664.2116 697.0242 664.2116 647.8388 656.5440 773.8274 773.8274
##        9       10       11       12       13       14       15       16
## 974.8697 985.6139 985.6139 587.5689 499.1027 550.6388 550.6388 553.2027
##       17       18       19       20       21       22
## 412.1153 567.4116 581.1558 508.2273 612.6556 611.9571
##
## [[2]]
##        1        2        3        4        5        6        7        8
## 738.5018 675.4197 738.5018 675.4197 651.9970 661.4730 784.2991 784.2991
##        9       10       11       12       13       14       15       16
## 766.2320 808.4305 808.4305 540.5915 522.6773 541.2069 541.2069 549.3454
##       17       18       19       20       21       22
## 405.4079 590.3449 607.8233 500.0608 595.1441 581.5315
##
```

```
## [[3]]
##         1         2         3         4         5         6         7
8
##  712.6620  677.6803  712.6620  677.6803  640.7988  648.1153  782.5487  782
.5487
##         9        10        11        12        13        14        15
16
##  928.1237 1098.7747 1098.7747  549.1767  528.1586  541.9174  541.9174  546
.5975
##        17        18        19        20        21        22
##  399.5546  593.0362  603.9782  599.3888  608.3479  611.7942
##
## [[4]]
##        1        2        3        4        5        6        7        8
## 692.8097 659.9397 692.8097 659.9397 659.4152 652.8041 766.0778 766.0778
##        9       10       11       12       13       14       15       16
## 932.0290 990.4826 990.4826 543.7714 510.8149 537.1843 537.1843 543.5411
##       17       18       19       20       21       22
## 545.3915 594.9394 595.9003 525.1254 584.9173 594.2050
##
## [[5]]
##        1        2        3        4        5        6        7        8
## 705.0863 662.2102 705.0863 662.2102 643.6996 654.8750 753.3249 753.3249
##        9       10       11       12       13       14       15       16
## 977.7128 998.6068 998.6068 533.6616 499.4293 546.2510 546.2510 546.8095
##       17       18       19       20       21       22
## 402.2528 578.3029 589.9975 492.6426 609.3332 603.8190
```

```
pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_50<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_100$models, newdata = n
ewdata)

predicted_npp
```

```
## [[1]]
##        1        2        3        4        5        6        7        8
## 720.4869 660.5911 720.4869 660.5911 637.3763 635.9329 802.0873 802.0873
##        9       10       11       12       13       14       15       16
## 735.3745 809.3328 809.3328 558.8410 511.6601 545.5636 545.5636 549.6130
##       17       18       19       20       21       22
## 420.7734 577.6370 586.7683 486.9968 609.3032 605.6291
##
## [[2]]
##        1        2        3        4        5        6        7        8
## 722.8922 665.0642 722.8922 665.0642 658.6524 660.1739 791.2696 791.2696
##        9       10       11       12       13       14       15       16
```

```
## 947.2172 994.6744 994.6744 511.7561 523.7598 549.2553 549.2553 548.2099
##       17       18       19       20       21       22
## 399.9182 566.4363 569.2462 505.7854 611.4450 607.6038
##
## [[3]]
##        1        2        3        4        5        6        7        8
## 710.8890 678.0339 710.8890 678.0339 667.5764 663.0942 772.5646 772.5646
##        9       10       11       12       13       14       15       16
## 983.6202 994.3430 994.3430 560.1809 498.4683 540.7549 540.7549 548.1551
##       17       18       19       20       21       22
## 414.0647 565.6677 578.5686 499.2273 611.4256 611.8265
##
## [[4]]
##         1         2         3         4         5         6         7
8
##  714.3440  684.9791  714.3440  684.9791  654.6419  666.5142  768.1442  768
.1442
##         9        10        11        12        13        14        15
16
##  927.4821 1096.0950 1096.0950  577.5840  523.0910  540.0841  540.0841  549
.6860
##        17        18        19        20        21        22
##  532.6009  580.0186  595.0184  601.0378  612.4793  613.8280
##
## [[5]]
##        1        2        3        4        5        6        7        8
## 715.3654 664.7142 715.3654 664.7142 638.5355 654.3530 753.2573 753.2573
##        9       10       11       12       13       14       15       16
## 960.7284 990.7266 990.7266 504.8033 506.6342 539.6533 539.6533 544.1114
##       17       18       19       20       21       22
## 380.6304 606.5324 610.4950 519.9510 572.5050 559.2838
```

```
pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_100<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_500$models, newdata = n
ewdata)

predicted_npp
```

```
## [[1]]
##        1        2        3        4        5        6        7        8
## 709.4317 677.6069 709.4317 677.6069 638.0529 637.9395 742.0107 742.0107
##        9       10       11       12       13       14       15       16
## 934.5082 990.7054 990.7054 511.2591 510.7228 541.9476 541.9476 547.0002
##       17       18       19       20       21       22
## 396.4933 597.4842 604.1608 524.8210 582.1714 569.4380
##
```

```
## [[2]]
##        1        2        3        4        5        6        7        8
## 712.8854 672.3279 712.8854 672.3279 650.2093 668.2880 748.7291 748.7291
##        9       10       11       12       13       14       15       16
## 982.1119 989.1534 989.1534 579.5897 506.5580 552.8445 552.8445 551.9358
##       17       18       19       20       21       22
## 420.9552 574.1103 584.1237 506.6627 606.6076 608.0755
##
## [[3]]
##        1        2        3        4        5        6        7        8
## 740.7402 654.8770 740.7402 654.8770 621.4404 657.3443 829.4394 829.4394
##        9       10       11       12       13       14       15       16
## 748.9321 918.1076 918.1076 520.7879 498.9649 537.9318 537.9318 541.6057
##       17       18       19       20       21       22
## 390.7587 596.0677 603.4265 591.5411 617.6351 615.6540
##
## [[4]]
##        1        2        3        4        5        6        7        8
## 700.2477 657.9585 700.2477 657.9585 659.5833 655.1565 789.3446 789.3446
##        9       10       11       12       13       14       15       16
## 966.5817 989.4345 989.4345 489.8282 510.4554 539.8348 539.8348 543.6417
##       17       18       19       20       21       22
## 556.1509 550.4198 555.7174 490.8734 602.5223 603.4663
##
## [[5]]
##        1        2        3        4        5        6        7        8
## 716.7398 675.1266 716.7398 675.1266 660.4988 671.9351 785.7921 785.7921
##        9       10       11       12       13       14       15       16
## 983.9652 980.5528 980.5528 544.8037 529.1303 543.8249 543.8249 550.4551
##       17       18       19       20       21       22
## 408.9026 593.0384 610.8564 497.8567 621.0044 614.0458
```

```
pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_500<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_1000$models, newdata =
newdata)

predicted_npp
```

```
## [[1]]
##         1         2         3         4         5         6         7
8
##  707.3737  645.6164  707.3737  645.6164  711.1231  705.5617  784.3973  784
.3973
##         9        10        11        12        13        14        15
16
##  935.2323 1109.5902 1109.5902  510.4812  509.4652  544.1447  544.1447  543
```

```
.3279
##        17        18        19        20        21        22
##  403.4093  569.2713  569.3386  505.0681  608.5577  603.7105
##
## [[2]]
##        1        2        3        4        5        6        7        8
## 709.0277 670.2648 709.0277 670.2648 641.4329 661.9580 777.7466 777.7466
##        9       10       11       12       13       14       15       16
## 953.8289 991.7464 991.7464 564.3670 513.1722 540.6649 540.6649 544.4027
##       17       18       19       20       21       22
## 552.2174 583.0045 596.6784 505.7555 614.0482 613.5168
##
## [[3]]
##        1        2        3        4        5        6        7        8
## 727.7828 678.7756 727.7828 678.7756 632.4100 637.5897 723.6703 723.6703
##        9       10       11       12       13       14       15       16
## 935.2986 987.3276 987.3276 576.1420 512.4131 542.4774 542.4774 546.4323
##       17       18       19       20       21       22
## 432.5855 615.0072 616.8269 641.1983 602.0905 605.3013
##
## [[4]]
##         1         2         3         4         5         6         7
8
##  716.6463  654.7535  716.6463  654.7535  640.0255  653.6655  783.2865  783
.2865
##         9        10        11        12        13        14        15
16
## 1009.5540  990.2368  990.2368  505.2148  528.8957  558.3848  558.3848  556
.2661
##        17        18        19        20        21        22
##  394.2514  611.4766  623.0252  489.3689  598.5281  567.6345
##
## [[5]]
##        1        2        3        4        5        6        7        8
## 699.4591 676.7171 699.4591 676.7171 643.5086 638.8163 752.4589 752.4589
##        9       10       11       12       13       14       15       16
## 726.8599 809.9289 809.9289 524.8292 504.4877 535.8468 535.8468 537.2566
##       17       18       19       20       21       22
## 391.4672 557.6583 566.9375 511.2587 594.6469 606.7651
```

```r
pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_1000<- pred_mean

data <- m_NPP_df

metrics <- function(obs, pred) {
  cor_val <- cor(obs, pred, use = "complete.obs")
  lm_model <- lm(obs ~ pred)
```

```
  r2_val <- summary(lm_model)$r.squared
  rmse_val <- sqrt(mean((obs - pred)^2, na.rm = TRUE))
  list(correlation = cor_val, R2 = r2_val, RMSE = rmse_val)
}

results <- tibble(
  model = c( "Predicted_NPP_all", "Predicted_NPP_cv", "Predicted_NPP_50", "Pr
edicted_NPP_100",
             "Predicted_NPP_500", "Predicted_NPP_1000",  "MOD17A3_NPP"),
  correlation = c(
    metrics(data$NPP, data$Predicted_NPP_all)$correlation,
    metrics(data$NPP, data$Predicted_NPP_cv)$correlation,
    metrics(data$NPP, data$Predicted_NPP_50)$correlation,
    metrics(data$NPP, data$Predicted_NPP_100)$correlation,
    metrics(data$NPP, data$Predicted_NPP_500)$correlation,
    metrics(data$NPP, data$Predicted_NPP_1000)$correlation,
    metrics(data$NPP, data$MOD17A3_NPP)$correlation
  ),
  R2 = c(
    metrics(data$NPP, data$Predicted_NPP_all)$R2,
    metrics(data$NPP, data$Predicted_NPP_cv)$R2,
    metrics(data$NPP, data$Predicted_NPP_50)$R2,
    metrics(data$NPP, data$Predicted_NPP_100)$R2,
    metrics(data$NPP, data$Predicted_NPP_500)$R2,
    metrics(data$NPP, data$Predicted_NPP_1000)$R2,
    metrics(data$NPP, data$MOD17A3_NPP)$R2
  ),
  RMSE = c(
    metrics(data$NPP, data$Predicted_NPP_all)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_cv)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_50)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_100)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_500)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_1000)$RMSE,
    metrics(data$NPP, data$MOD17A3_NPP)$RMSE
  )
)

print(results)
```

```
## # A tibble: 7 × 4
##   model              correlation    R2  RMSE
##   <chr>                    <dbl> <dbl> <dbl>
## 1 Predicted_NPP_all        0.886 0.785  209.
## 2 Predicted_NPP_cv         0.885 0.783  212.
## 3 Predicted_NPP_50         0.879 0.773  220.
## 4 Predicted_NPP_100        0.880 0.775  219.
## 5 Predicted_NPP_500        0.880 0.774  219.
## 6 Predicted_NPP_1000       0.877 0.769  221.
## 7 MOD17A3_NPP              0.891 0.795  292.
```

```
knitr::kable(results, digits = 3, caption = "Independent-validation statistics.")
```

Independent-validation statistics.

model	correlation	R2	RMSE
Predicted_NPP_all	0.886	0.785	208.630
Predicted_NPP_cv	0.885	0.783	212.073
Predicted_NPP_50	0.879	0.773	220.363
Predicted_NPP_100	0.880	0.775	219.208
Predicted_NPP_500	0.880	0.774	219.081
Predicted_NPP_1000	0.877	0.769	221.194
MOD17A3_NPP	0.891	0.795	291.670

```
plot_data <- data %>%
  dplyr::select(
    NPP,
    Predicted_NPP_all,
    Predicted_NPP_cv,
    Predicted_NPP_50,
    Predicted_NPP_100,
    Predicted_NPP_500,
    Predicted_NPP_1000,
    MOD17A3_NPP
  ) %>%
  pivot_longer(
    cols = -NPP,
    names_to = "model",
    values_to = "predicted"
  )

labels <- results %>%
  mutate(
    label = paste0(
      "r = ", round(correlation, 2), "\n",
      "r² = ", round(R2, 2), "\n",
      "RMSE = ", round(RMSE, 0)
    )
  )

plot_obs_pred <- function(obs, pred, model_name, stats) {

  df <- data.frame(
    Observed = obs,
    Predicted = pred
  )
  m <- stats %>% filter(model == model_name)
  subtitle_txt <- sprintf(
```

```
    "r = %.3f | r² = %.3f | RMSE = %.0f",
    m$correlation,
    m$R2,
    m$RMSE
  )

  ggplot(df, aes(x = Observed, y = Predicted)) +
    geom_point(color = "black", alpha = 0.4, size = 1.5) +
    geom_smooth(
      method = "lm",
      se = TRUE,
      color = "darkred",
      fill = adjustcolor("gray50")
    ) +
    labs(
      title = model_name,
      subtitle = subtitle_txt,
      x = "Observed NPP",
      y = "Predicted NPP"
    ) +
    theme_minimal()
}

p1 <- plot_obs_pred(data$NPP, data$Predicted_NPP_all,    "Predicted_NPP_all",
results)
p2 <- plot_obs_pred(data$NPP, data$Predicted_NPP_cv,     "Predicted_NPP_cv",
results)
p3 <- plot_obs_pred(data$NPP, data$Predicted_NPP_50,     "Predicted_NPP_50",
results)
p4 <- plot_obs_pred(data$NPP, data$Predicted_NPP_100,    "Predicted_NPP_100",
results)
p5 <- plot_obs_pred(data$NPP, data$Predicted_NPP_500,    "Predicted_NPP_500",
results)
p6 <- plot_obs_pred(data$NPP, data$Predicted_NPP_1000,   "Predicted_NPP_1000"
,   results)
p7 <- plot_obs_pred(data$NPP, data$MOD17A3_NPP,          "MOD17A3_NPP",
results)

grid.arrange(
  p7, nullGrob(), nullGrob(),
  p1, p2, p3,
  p4, p5, p6,
  ncol = 3
)
```

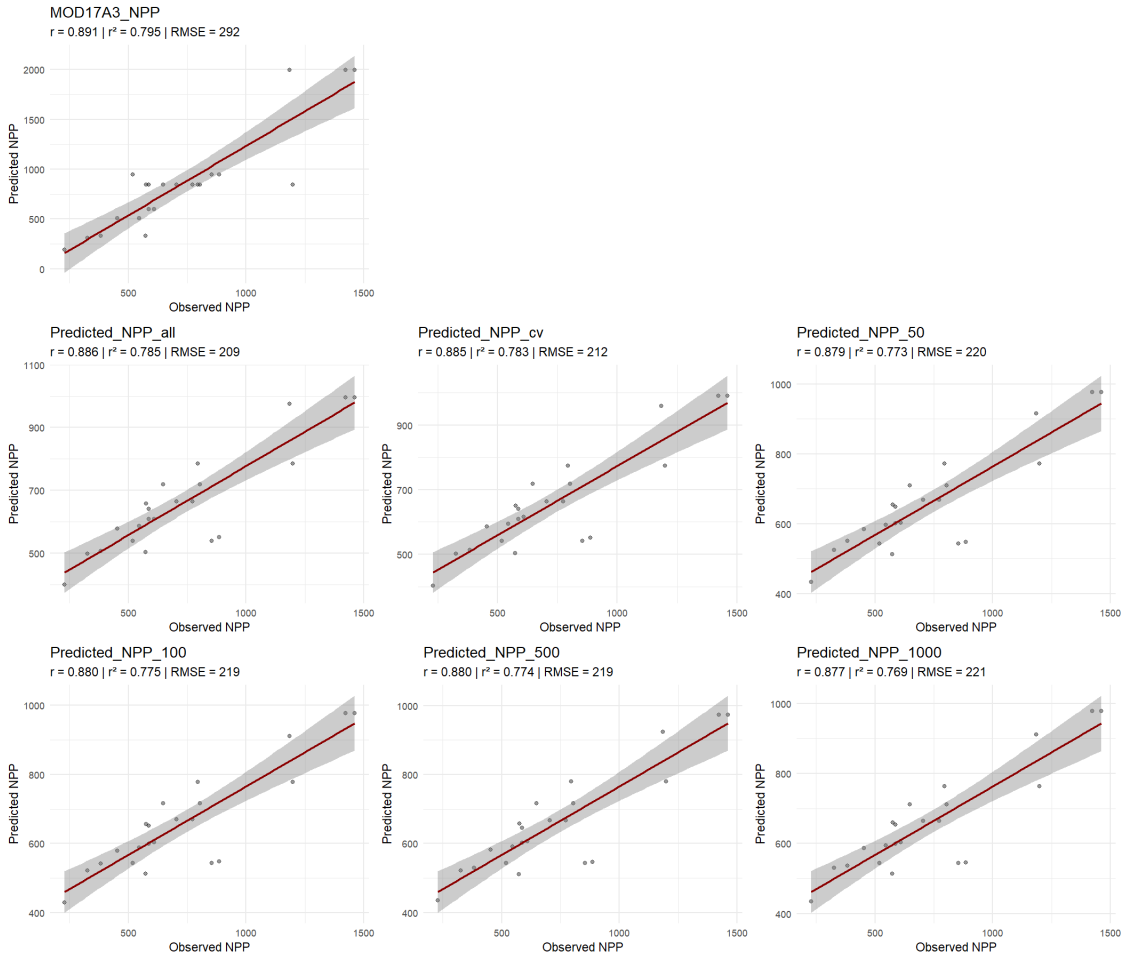


Figure 5. Independent NPP validation.

Figure S6: Validation without sample type

Load the random-CV and spatial models trained without sample type as a predictor and repeat the independent-validation calculations. Retain the complete-data model and MOD17A3 comparison shown in the validation workflow.

```
local({
formula <- as.formula(paste("NPP ~", paste(c(colnames(clr_data)), collapse =
" + ")))

formula

model_cv <- readRDS(paste0("WithoutSampleType/", "model_brGDGT-NPP_cv1.rds"))
model_spatial_cv_50<- readRDS(paste0("WithoutSampleType/", "spatial_cv_50km_m
odels1.rds"))
model_spatial_cv_100<- readRDS(paste0("WithoutSampleType/", "spatial_cv_100km
_models1.rds"))
model_spatial_cv_500<- readRDS(paste0("WithoutSampleType/", "spatial_cv_500km
_models1.rds"))
```

```r
model_spatial_cv_1000<- readRDS(paste0("WithoutSampleType/", "spatial_cv_1000
km_models1.rds"))
m_NPP_df <- read.xlsx("Dataset_1.xlsx", sheet = "Measured_NPP_brGDGTs")

tetramethylated <- c("fIa", "fIb", "fIc")
pentamethylated <- c("fIIa", "fIIa_", "fIIb", "fIIb_", "fIIc", "fIIc_")
hexamethylated  <- c("fIIIa", "fIIIa_", "fIIIb", "fIIIb_", "fIIIc", "fIIIc_")
all_fracs <- c(tetramethylated, pentamethylated, hexamethylated)

frac_data <- m_NPP_df %>%
  dplyr::select(all_of(all_fracs)) %>%
  mutate(across(everything(), ~ .x / 100 + 1e-6))

clr_transformed <- clr(frac_data)
clr_df <- as.data.frame(clr_transformed)
colnames(clr_df) <- paste0( colnames(clr_df))

newdata <- bind_cols(
  Sampletype = factor(m_NPP_df$Sampletype, levels = levels(model_data$Samplet
ype)),
  clr_df
)
newdata$Sampletype_fixed<- m_NPP_df$Sampletype

model_all_data
model_cv
model_spatial_cv_50
model_spatial_cv_100
model_spatial_cv_500

predicted_npp <- predict(model_all_data, newdata = newdata)

predicted_npp

m_NPP_df$Predicted_NPP_all<- predicted_npp

predicted_npp <- predict(model_cv, newdata = newdata)

predicted_npp

m_NPP_df$Predicted_NPP_cv<- predicted_npp

predicted_npp <- predict_fold_models(model_spatial_cv_50$models, newdata = ne
wdata)

predicted_npp

pred_matrix <- do.call(cbind, predicted_npp)
```

```r
pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_50<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_100$models, newdata = n
ewdata)


pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_100<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_500$models, newdata = n
ewdata)


pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_500<- pred_mean

predicted_npp <- predict_fold_models(model_spatial_cv_1000$models, newdata =
newdata)


pred_matrix <- do.call(cbind, predicted_npp)

pred_mean <- rowMeans(pred_matrix)
m_NPP_df$Predicted_NPP_1000<- pred_mean

data <- m_NPP_df

metrics <- function(obs, pred) {
  cor_val <- cor(obs, pred, use = "complete.obs")
  lm_model <- lm(obs ~ pred)
  r2_val <- summary(lm_model)$r.squared
  rmse_val <- sqrt(mean((obs - pred)^2, na.rm = TRUE))
  list(correlation = cor_val, R2 = r2_val, RMSE = rmse_val)
}

results <- tibble(
  model = c( "Predicted_NPP_all", "Predicted_NPP_cv", "Predicted_NPP_50", "Pr
edicted_NPP_100",
             "Predicted_NPP_500", "Predicted_NPP_1000",  "MOD17A3_NPP"),
  correlation = c(
    metrics(data$NPP, data$Predicted_NPP_all)$correlation,
    metrics(data$NPP, data$Predicted_NPP_cv)$correlation,
```

```
    metrics(data$NPP, data$Predicted_NPP_50)$correlation,
    metrics(data$NPP, data$Predicted_NPP_100)$correlation,
    metrics(data$NPP, data$Predicted_NPP_500)$correlation,
    metrics(data$NPP, data$Predicted_NPP_1000)$correlation,
    metrics(data$NPP, data$MOD17A3_NPP)$correlation
  ),
  R2 = c(
    metrics(data$NPP, data$Predicted_NPP_all)$R2,
    metrics(data$NPP, data$Predicted_NPP_cv)$R2,
    metrics(data$NPP, data$Predicted_NPP_50)$R2,
    metrics(data$NPP, data$Predicted_NPP_100)$R2,
    metrics(data$NPP, data$Predicted_NPP_500)$R2,
    metrics(data$NPP, data$Predicted_NPP_1000)$R2,
    metrics(data$NPP, data$MOD17A3_NPP)$R2
  ),
  RMSE = c(
    metrics(data$NPP, data$Predicted_NPP_all)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_cv)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_50)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_100)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_500)$RMSE,
    metrics(data$NPP, data$Predicted_NPP_1000)$RMSE,
    metrics(data$NPP, data$MOD17A3_NPP)$RMSE
  )
)


plot_data <- data %>%
  dplyr::select(
    NPP,
    Predicted_NPP_all,
    Predicted_NPP_cv,
    Predicted_NPP_50,
    Predicted_NPP_100,
    Predicted_NPP_500,
    Predicted_NPP_1000,
    MOD17A3_NPP
  ) %>%
  pivot_longer(
    cols = -NPP,
    names_to = "model",
    values_to = "predicted"
  )

labels <- results %>%
  mutate(
    label = paste0(
      "r = ", round(correlation, 2), "\n",
      "r² = ", round(R2, 2), "\n",
```

```
      "RMSE = ", round(RMSE, 0)
    )
  )

plot_obs_pred <- function(obs, pred, model_name, stats) {

  df <- data.frame(
    Observed = obs,
    Predicted = pred
  )
  m <- stats %>% filter(model == model_name)
  subtitle_txt <- sprintf(
    "r = %.3f | r² = %.3f | RMSE = %.0f",
    m$correlation,
    m$R2,
    m$RMSE
  )

  ggplot(df, aes(x = Observed, y = Predicted)) +
    geom_point(color = "black", alpha = 0.4, size = 1.5) +
    geom_smooth(
      method = "lm",
      se = TRUE,
      color = "darkred",
      fill = adjustcolor("gray50")
    ) +
    labs(
      title = model_name,
      subtitle = subtitle_txt,
      x = "Observed NPP",
      y = "Predicted NPP"
    ) +
    theme_minimal()
}

p1 <- plot_obs_pred(data$NPP, data$Predicted_NPP_all,    "Predicted_NPP_all",
results)
p2 <- plot_obs_pred(data$NPP, data$Predicted_NPP_cv,     "Predicted_NPP_cv",
results)
p3 <- plot_obs_pred(data$NPP, data$Predicted_NPP_50,     "Predicted_NPP_50",
results)
p4 <- plot_obs_pred(data$NPP, data$Predicted_NPP_100,    "Predicted_NPP_100",
results)
p5 <- plot_obs_pred(data$NPP, data$Predicted_NPP_500,    "Predicted_NPP_500",
results)
p6 <- plot_obs_pred(data$NPP, data$Predicted_NPP_1000,   "Predicted_NPP_1000"
,   results)
p7 <- plot_obs_pred(data$NPP, data$MOD17A3_NPP,          "MOD17A3_NPP",
results)
```

```
grid.arrange(
  p7, nullGrob(), nullGrob(),
  p1, p2, p3,
  p4, p5, p6,
  ncol = 3
)
})
```

The validation without sample type requires the model files in WithoutSampleType/.

Figure S7: Environmental novelty

Align MAT, MAP, pH, clay, bulk density and elevation rasters. Standardise environmental variables using the calibration mean and standard deviation, then calculate distance to the nearest calibration analogue. Express novelty relative to the 95th percentile of withheld-sample distances from five-fold cross-validation.

```
data <- read.xlsx("Dataset_1.xlsx", sheet = "Dataset_R")

r_elev <- rast("Elevation.tif")
r_bulk <- rast("rbulk_reproj.tif")
r_clay <- rast("rclay_reproj.tif")
r_ph   <- rast("rph_reproj.tif")
r_mat  <- rast("wc2.1_5m_bio_1.tif")
r_map  <- rast("wc2.1_5m_bio_12.tif")

template <- r_mat

r_map  <- project(r_map,  template)
r_ph   <- project(r_ph,   template)
r_clay <- project(r_clay, template)
r_bulk <- project(r_bulk, template)
r_elev <- project(r_elev, template)

env_rast <- c(
  r_mat,
  r_map,
  r_ph,
  r_clay,
  r_bulk,
  r_elev
)

names(env_rast) <- c(
  "BIO01",
  "BIO12",
  "pH",
  "Clay",
```

```
  "Bulk",
  "Elevation"
)

vars <- names(env_rast)

train <- data %>%
  dplyr::select(all_of(vars)) %>%
  drop_na()

rbind(
  Calibration_min  = sapply(train, min),
  Calibration_max  = sapply(train, max),
  Calibration_mean = sapply(train, mean),
  Raster_min = as.numeric(
    global(env_rast, "min", na.rm = TRUE)[,1]
  ),
  Raster_max = as.numeric(
    global(env_rast, "max", na.rm = TRUE)[,1]
  )
)
```

```
##                      BIO01     BIO12       pH     Clay      Bulk Elevation
## Calibration_min  -17.09492    16.000 40.00000  4.83517  315.2660 -153.0000
## Calibration_max   29.23092  5320.000 85.00000 53.44531 1738.5514 5439.0000
## Calibration_mean  10.27635  1108.304 59.81845 22.01405 1111.3623 1020.1791
## Raster_min       -54.73946     0.000 39.15727  0.00000  151.7619 -408.1406
## Raster_max        31.05112 11191.000 92.51916 74.34597 1819.8074 6369.0781
```

```
mu  <- sapply(train, mean)
sdv <- sapply(train, sd)

train_z <- scale(
  train,
  center = mu,
  scale = sdv
)

env_z <- env_rast

for(i in seq_along(vars)){
  env_z[[i]] <- (env_rast[[i]] - mu[i]) / sdv[i]
}

env_df <- as.data.frame(
  env_z,
  xy = TRUE,
  na.rm = TRUE
)
```

```r
nn <- nn2(
  data = train_z,
  query = as.matrix(env_df[, vars]),
  k = 1
)

env_df$Novelty <- nn$nn.dists[,1]

set.seed(123)

K <- 5

fold <- sample(
  rep(1:K, length.out = nrow(train_z))
)

cv_dist <- rep(NA_real_, nrow(train_z))

for(k in 1:K){

  ref <- train_z[
    fold != k,
    ,
    drop = FALSE
  ]

  test <- train_z[
    fold == k,
    ,
    drop = FALSE
  ]

  nn_cv <- nn2(
    data = ref,
    query = test,
    k = 1
  )

  cv_dist[fold == k] <- nn_cv$nn.dists[,1]
}

cv_quantiles <- quantile(
  cv_dist,
  probs = c(0.50, 0.75, 0.90, 0.95, 0.99),
  na.rm = TRUE
)

cv_quantiles
```

```
##       50%       75%       90%       95%       99%
## 0.0000000 0.2307706 0.4315039 0.5872833 0.9008890
```

```
q95 <- quantile(
  cv_dist,
  0.95,
  na.rm = TRUE
)

env_df$Novelty_relative <- env_df$Novelty / q95

env_df$Extrapolative <- env_df$Novelty_relative > 1

mean(env_df$Extrapolative, na.rm = TRUE) * 100
```

```
## [1] 48.58079
```

```
novelty_map <- rast(
  env_df[, c("x", "y", "Novelty_relative")],
  type = "xyz",
  crs = crs(env_rast)
)

map_df <- env_df %>%
  dplyr::select(
    x,
    y,
    Novelty_relative
  )

display_cap <- quantile(
  map_df$Novelty_relative,
  0.99,
  na.rm = TRUE
)

map_df$Novelty_plot <- pmin(
  map_df$Novelty_relative,
  display_cap
)

world <- map_data("world")

ggplot() +

  geom_raster(
    data = map_df,
    aes(
      x = x,
      y = y,
```

```r
    fill = Novelty_plot
  )
) +

geom_polygon(
  data = world,
  aes(
    x = long,
    y = lat,
    group = group
  ),
  fill = NA,
  colour = "grey25",
  linewidth = 0.15
) +

scale_fill_viridis_c(
  option = "magma",
  direction = -1,
  limits = c(0, display_cap),
  oob = squish,
  name = "Environmental novelty"
) +

coord_quickmap(
  xlim = c(-180, 180),
  ylim = c(-60, 85),
  expand = FALSE
) +

labs(
  x = NULL,
  y = NULL
) +

theme_void(base_size = 11) +

theme(
  legend.position = "right",
  legend.title = element_text(size = 9),
  legend.text = element_text(size = 8)
)
```

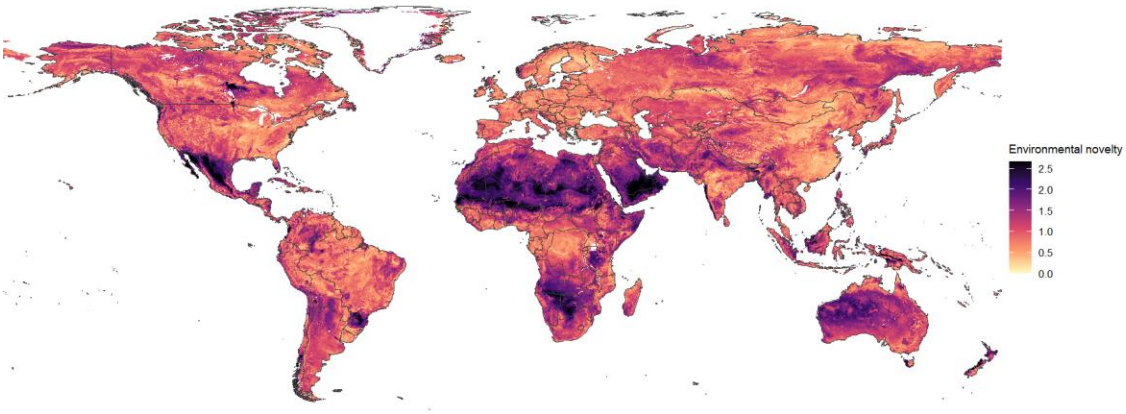


Figure S7. Environmental novelty relative to calibration support.

Figure S8: Regional palaeoclimate

Plot pollen-derived MAT and MAP, HadCM3B-M2.1 and CHELSA-TraCE21k climate series, Alboran Sea temperature and the precipitation reconstruction of Camuera et al. (2022). Mark the H2, LGM, H1, Younger Dryas and 8.2 ka intervals.

```
regional_climate <- read.xlsx("Dataset_1.xlsx", sheet = "Regional_Paleoclimat
e")

colnames(regional_climate)
```

```
##  [1] "Age_SST"              "SST"                  "Age_P1"
##  [4] "MAT_P1"               "MAP_P1"               "NPP_Miami"
##  [7] "Age_P2"               "Camuerta.et.al..2022" "Age_GCM"
## [10] "MAT_GCM"              "MAT_GCM_SD"           "MAP_GCM"
## [13] "MAP_GCM_SD"           "NPP_Miami_GCM"        "Age_CHELSA"
## [16] "MAT_Chelsa"           "MAP_Chelsa"           "NPP_Miami_CHELSA"
```

```
NGRIP_df <- read.xlsx("Dataset_1.xlsx", sheet = "NGRIP", detectDates = FALSE)
head(NGRIP_df)
```

```
##    Age   d18O
## 1 5.00 -34.76
## 2 5.00 -34.65
## 3 5.05 -34.65
## 4 5.05 -34.98
## 5 5.10 -34.98
## 6 5.10 -34.41
```

```
events <- data.frame(
  Event = c( "H2", "LGM", "H1",  "Younger Dryas", "8.2"),
```

```
  start = c( 24500, 21500, 17000,  12900,  8400),
  end   = c( 23000, 19000, 15000, 11700,  8100)
)

require_columns(regional_climate, c("Age_P1", "MAT_P1", "MAP_P1"), "Regional_
Paleoclimate")

a <- ggplot(regional_climate, aes(x = Age_P1, y = MAT_P1)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Temperature (MAT) from Pollen (This study)",
       x = "Age (cal BP)",
       y = "MAT (°C)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(0, 15)) +
  scale_x_reverse()

b <- ggplot(regional_climate, aes(x = Age_P1, y = MAP_P1)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Precipitation (MAP) from Pollen (This study)",
       x = "Age (cal BP)",
       y = "MAP (mm/yr)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(0, 800)) +
  scale_x_reverse()

c <- ggplot(regional_climate, aes(x = Age_GCM, y = MAT_GCM)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Temperature (MAT) from HadCM3B-M2.1 ",
       x = "Age (cal BP)",
       y = "MAT (°C)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(0, 15)) +
  scale_x_reverse()

d<- ggplot(regional_climate, aes(x = Age_GCM, y = MAP_GCM)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
```

```
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Precipitation (MAP) from HadCM3B-M2.1",
       x = "Age (cal BP)",
       y = "MAP (mm/yr)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim =c(200, 800)) +
  scale_x_reverse()

e<- ggplot(regional_climate, aes(x = Age_CHELSA, y = MAT_Chelsa)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Temperature (MAT) from CHELSA-TraCE21k ",
       x = "Age (cal BP)",
       y = "MAT (°C)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(0, 10)) +
  scale_x_reverse()

f<- ggplot(regional_climate, aes(x = Age_CHELSA, y = MAP_Chelsa)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Precipitation (MAP) from CHELSA-TraCE21k",
       x = "Age (cal BP)",
       y = "MAP (mm/yr)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(400, 900)) +
  scale_x_reverse()

g<- ggplot(regional_climate, aes(x = Age_SST, y = SST)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Temperature from Alboran Sea Temperature ",
       x = "Age (cal BP)",
       y = "MAT (°C)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(0, 22)) +
  scale_x_reverse()

h <- ggplot(regional_climate, aes(x = Age_P2, y = Camuerta.et.al..2022)) +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
```

```
Inf),
            fill = "grey80", alpha = 0.4, inherit.aes = F) +
  geom_line(color = "black", linewidth = 1) +
  labs(title = "Mean Annual Precipitation (MAP) from Camuera et al. 2022",
       x = "Age (cal BP)",
       y = "MAP (mm/yr)") +
  geom_point(color = "black", size = 2) +
  theme_minimal()+
  coord_cartesian(ylim = c(200, 800)) +
  scale_x_reverse()

grid.arrange(a, b, c, d, e, f, g, h, ncol = 2)
```

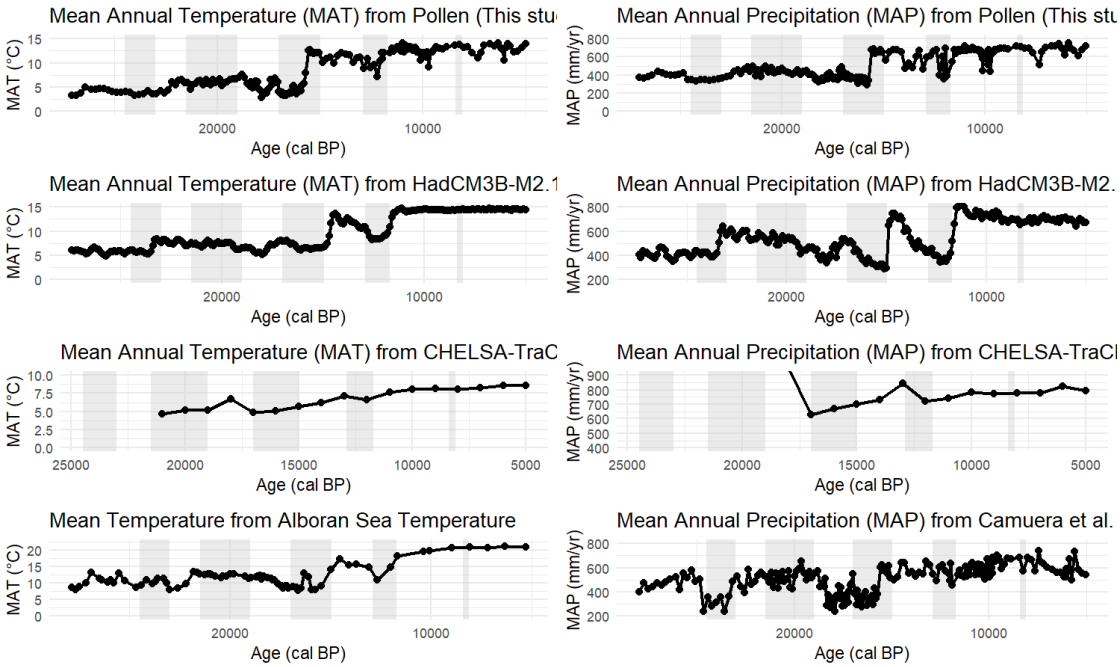


Figure S8G-H. MAT and MAP reconstructed from different proxies.

Padul brGDGT–NPP reconstruction

Apply the RF model to Padul brGDGT compositions, using a lacustrine sample-type assignment.

```
Padul_df <- read.xlsx("Dataset_1.xlsx", sheet = "Padul_GDGTs")

vars <- colnames(Padul_df)[4:18]

vars
```

```
##  [1] "fIIIa"   "fIIIa_" "fIIIb"   "fIIIb_" "fIIIc"   "fIIIc_" "fIIa"    "fIIa_
"
##  [9] "fIIb"    "fIIb_"  "fIIc"    "fIIc_"  "fIa"     "fIb"    "fIc"
```

```r
vars_extra <- c("Sampletype")

comp_new <- Padul_df[, vars]

clr_new <- clr(comp_new)
clr_new_df <- as.data.frame(clr_new)

newdata_model <- cbind(clr_new_df,
                       Sampletype = Padul_df$Sampletype,
                       Sampletype_fixed = "Lacustrine Sediment")

rf_model <- model_all_data$finalModel
model_vars <- rownames(rf_model$importance)

newdata_ready <- rf_design_matrix(model_all_data, newdata_model)

rf_preds_all <- predict(rf_model,
                        newdata = newdata_ready,
                        predict.all = TRUE)

ci_95_lower <- apply(rf_preds_all$individual, 1, quantile, probs = 0.025, na.
rm = TRUE)

ci_95_upper <- apply(rf_preds_all$individual, 1, quantile, probs = 0.975, na.
rm = TRUE)

results_complete <- data.frame(
  SampleID = 1:nrow(Padul_df),
  Age_calBP = Padul_df$`Age.(yr.cal.BP)`,
  NPP_pred_mean = rf_preds_all$aggregate,
  NPP_CI95_lower = ci_95_lower,
  NPP_CI95_upper = ci_95_upper,
  NPP_range = ci_95_upper - ci_95_lower
)

head(results_complete)
```

```
##   SampleID Age_calBP NPP_pred_mean NPP_CI95_lower NPP_CI95_upper NPP_range
## 1        1     -62.0      472.7762       333.7719       639.8841  306.1121
## 2        2      -1.0      452.8727       327.8324       627.7277  299.8954
## 3        3     123.0      466.9476       311.1520       639.7486  328.5966
## 4        4    5015.0      627.0802       476.7877       804.8959  328.1082
## 5        5    5203.5      643.9611       495.4219       812.3082  316.8863
## 6        6    5404.5      641.2053       501.8220       805.6728  303.8508
```

```r
tree_indices <- sample(1:ncol(rf_preds_all$individual), 1000)

tree_data <- data.frame(Age_calBP = Padul_df$`Age.(yr.cal.BP)`)
```

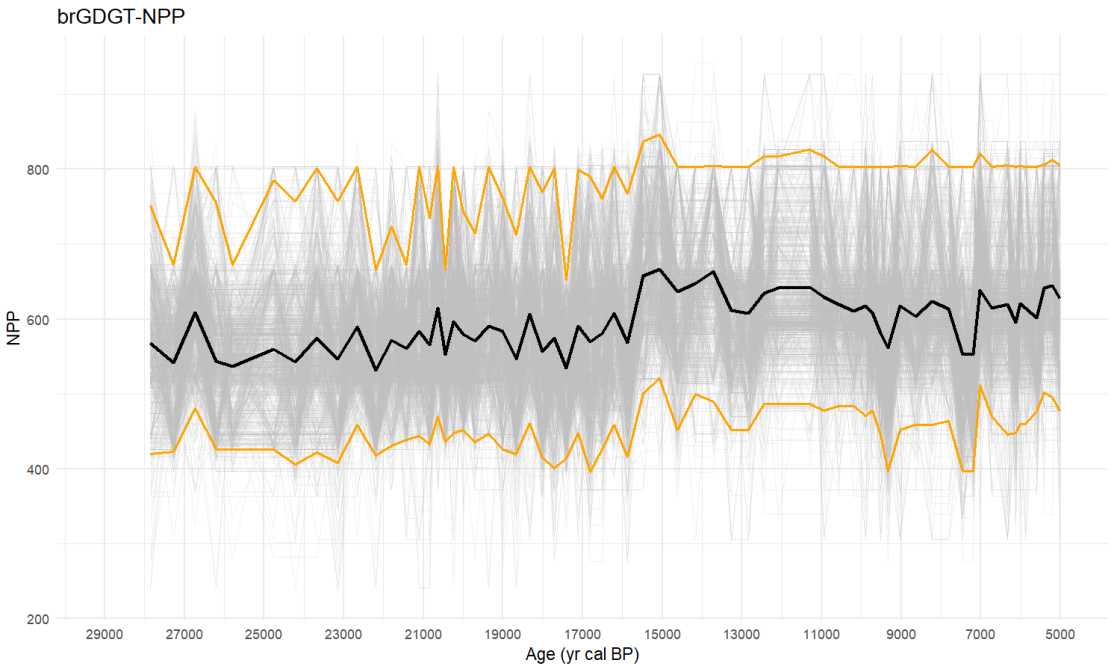
```r
for (i in 1:1000) {
  tree_data[[paste0("Tree_", i)]] <- rf_preds_all$individual[, tree_indices[i
]]
}

tree_long <- melt(tree_data,
                  id.vars = "Age_calBP",
                  variable.name = "Tree",
                  value.name = "NPP")

tree_long_main <- tree_long

Padul_NPP_plot_simple <- ggplot() +
  geom_line(data = tree_long, aes(x = Age_calBP, y = NPP, group = Tree),
            alpha = 0.2, color = "gray", linewidth = 0.3) +
  geom_line(data = results_complete, aes(x = Age_calBP, y = NPP_pred_mean),
            color = "black", linewidth = 1) +
  geom_line(data = results_complete, aes(x = Age_calBP, y = NPP_CI95_lower),
            color = "orange", linewidth = 0.8) +
  geom_line(data = results_complete, aes(x = Age_calBP, y = NPP_CI95_upper),
            color = "orange", linewidth = 0.8) +
  scale_x_reverse(limits = c(29000, 5000), breaks = seq(29000, 5000, by = -20
00)) +
  labs(x = "Age (yr cal BP)", y = "NPP", title = "brGDGT-NPP") +
  theme_minimal()

print(Padul_NPP_plot_simple)
```



Padul brGDGT-derived NPP: individual-tree predictions, mean and percentile bounds.

Figure S9: Sensitivity to sample-type assignment

Repeat Padul predictions for the sample-type categories represented in the calibration model and compare their mean NPP trajectories. Plot total organic carbon from the Padul core alongside the sample-type comparison.

```
local({
Padul_df <- read.xlsx("Dataset_1.xlsx", sheet = "Padul_GDGTs")

vars <- colnames(Padul_df)[4:18]
comp_new <- Padul_df[, vars]

clr_new <- clr(comp_new)
clr_new_df <- as.data.frame(clr_new)

sampletypes_fixed <- model_all_data$xlevels$Sampletype_fixed

results_list <- list()
tree_long_list <- list()

rf_model <- model_all_data$finalModel
model_vars <- rownames(rf_model$importance)

for (stype in sampletypes_fixed) {

  newdata_model <- cbind(
    clr_new_df,
    Sampletype = Padul_df$Sampletype,
    Sampletype_fixed = stype
  )

  newdata_ready <- rf_design_matrix(model_all_data, newdata_model)

  rf_preds_all <- predict(rf_model, newdata = newdata_ready, predict.all = TR
UE)

  ci_95_lower <- apply(rf_preds_all$individual, 1, quantile, probs = 0.025, n
a.rm = TRUE)
  ci_95_upper <- apply(rf_preds_all$individual, 1, quantile, probs = 0.975, n
a.rm = TRUE)

  results_complete <- data.frame(
    SampleID = 1:nrow(Padul_df),
    Age_calBP = Padul_df$`Age.(yr.cal.BP)`,
    NPP_pred_mean = rf_preds_all$aggregate,
    NPP_CI95_lower = ci_95_lower,
    NPP_CI95_upper = ci_95_upper,
    NPP_range = ci_95_upper - ci_95_lower,
```

```
    Sampletype_fixed = stype
  )

  results_list[[stype]] <- results_complete

  tree_indices <- sample(1:ncol(rf_preds_all$individual), 1000)
  tree_data <- data.frame(Age_calBP = Padul_df$`Age.(yr.cal.BP)`)

  for (i in 1:1000) {
    tree_data[[paste0("Tree_", i)]] <- rf_preds_all$individual[, tree_indices
[i]]
  }

  tree_data$Sampletype_fixed <- stype
  tree_long <- melt(tree_data, id.vars = c("Age_calBP", "Sampletype_fixed"),
                    variable.name = "Tree", value.name = "NPP")

  tree_long_list[[stype]] <- tree_long
}

all_results <- bind_rows(results_list)
all_tree_long <- bind_rows(tree_long_list)

NPP_plot_combined <- ggplot(data = all_results,
                            aes(x = Age_calBP, y = NPP_pred_mean, color = Sam
pletype_fixed)) +
  geom_line(linewidth = 0.8) +
  labs(x = "Age (yr cal BP)",
       y = "NPP (g/m^2/yr)",
       title = "brGDGT-NPP by Sample Type",
       color = "Sample Type") +
  theme_minimal() +
  theme(legend.position = "top",
        plot.title = element_text(face = "bold", hjust = 0.5))+
  scale_x_reverse(limits=c(27000, 5000))


TOC_df<- read.xlsx("Dataset_1.xlsx", sheet ="Padul_GDGTs")

TOC_df<- subset(TOC_df, TOC_df$`Age.(yr.cal.BP)`>4000)

TOC_plot<- ggplot(data=TOC_df,
                  aes(x=TOC_df$`Age.(yr.cal.BP)`, y=as.numeric(TOC))) +
  geom_point()+
  geom_line()+
  scale_x_reverse()+
  theme_minimal()+
  xlab("Age (kyr BP)")+
```

```
  ylab("Total Organic Content (%)")


plot_grid(
  NPP_plot_combined, TOC_plot,
  ncol = 1,
  align = "v"
)
})
```

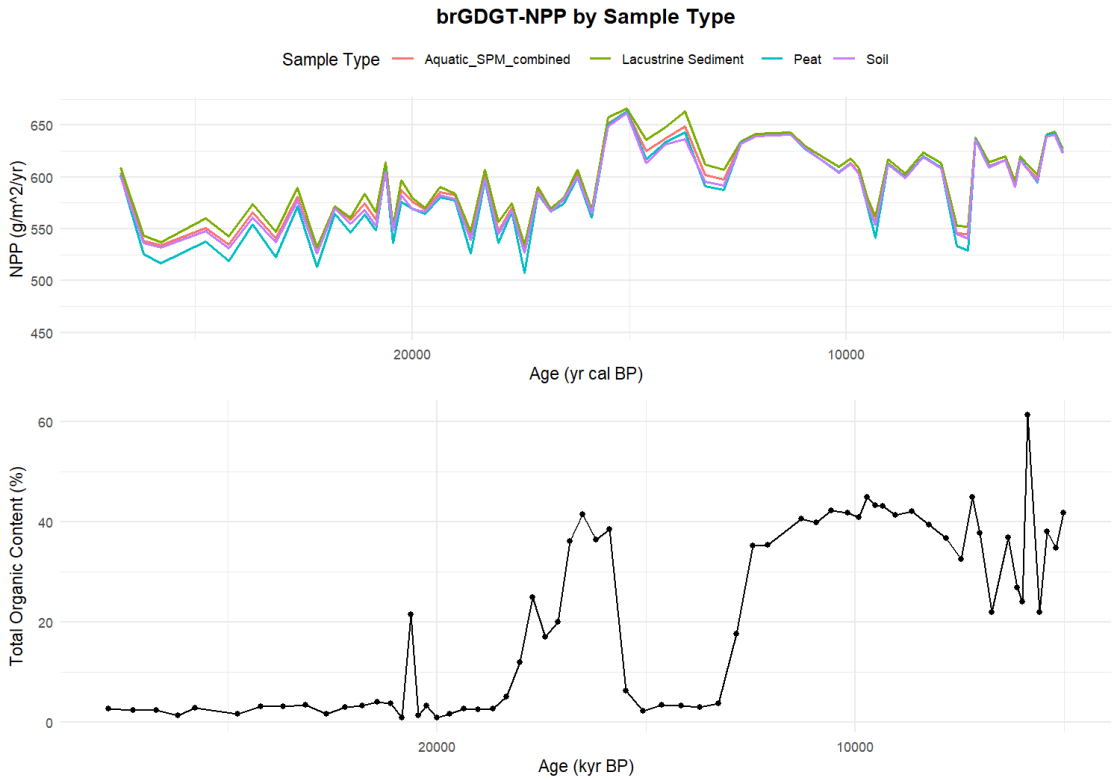


Figure S9. Padul NPP sensitivity to sample-type assignment and total organic carbon.

Figure 6D–E: NPP reconstruction

Sample MAT and MAP within the supplied 95% bounds, calculate Miami-model NPP and interpolate to the brGDGT chronology. Calculate the NPP deficit as potential NPP minus brGDGT-derived NPP, propagating MAT, MAP and Random Forest prediction uncertainty.

```
# Potential NPP and NPP deficit
Padul_Miami_climate <- read.xlsx("Dataset_1.xlsx", sheet = "Padul_climate")

n_iter <- 1000
n_samples <- nrow(Padul_df)

npp_miami_mc <- matrix(NA, nrow = n_samples, ncol = n_iter)
```

```r
for(i in 1:n_iter){

  mat_sim <- rtruncnorm(
    n = nrow(Padul_Miami_climate),
    a = Padul_Miami_climate$Lower95_MAT,
    b = Padul_Miami_climate$Upper95_MAT,
    mean = Padul_Miami_climate$Predicted_MAT,
    sd = (Padul_Miami_climate$Upper95_MAT - Padul_Miami_climate$Predicted_MAT
)/1.96
  )

  map_sim <- rtruncnorm(
    n = nrow(Padul_Miami_climate),
    a = Padul_Miami_climate$Lower95_MAP,
    b = Padul_Miami_climate$Upper95_MAP,
    mean = Padul_Miami_climate$Predicted_MAP,
    sd = (Padul_Miami_climate$Upper95_MAP - Padul_Miami_climate$Predicted_MAP
)/1.96
  )

  npp_miami_mat <- 3000 / (1 + exp(1.315 - 0.119 * mat_sim))
  npp_miami_map <- 3000 * (1 - exp(-0.000664 * map_sim))
  npp_miami_full <- pmin(npp_miami_mat, npp_miami_map) * 0.45

  npp_miami_interp <- approx(
    x = Padul_Miami_climate$Age,
    y = npp_miami_full,
    xout = Padul_df$`Age.(yr.cal.BP)`,
    rule = 2
  )$y

  npp_miami_mc[, i] <- npp_miami_interp
}



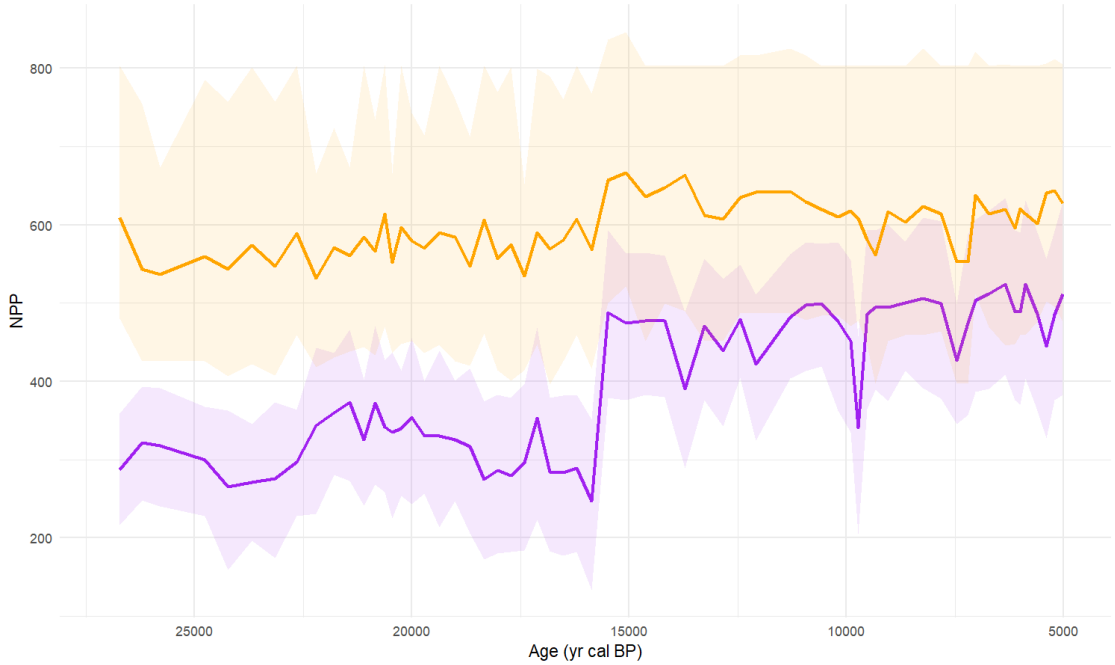
npp_miami_mean <- apply(npp_miami_mc, 1, mean, na.rm = TRUE)
npp_miami_low  <- apply(npp_miami_mc, 1, quantile, probs = 0.025, na.rm = TRU
E)
npp_miami_high <- apply(npp_miami_mc, 1, quantile, probs = 0.975, na.rm = TRU
E)

df_plot <- data.frame(
  Age = Padul_df$`Age.(yr.cal.BP)`,
  NPP_mean = npp_miami_mean,
  NPP_low  = npp_miami_low,
  NPP_high = npp_miami_high
)

ggplot(df_plot, aes(x = Age)) +
```

```
  geom_line(aes(y = NPP_mean), color = "purple", size = 1) +
  geom_ribbon(aes(ymin = NPP_low, ymax = NPP_high), fill = "purple", alpha =
0.1) +
  geom_line(data = results_complete,
            aes(x = Age_calBP, y = NPP_pred_mean),
            color = "orange", linewidth = 1) +
  geom_ribbon(data = results_complete,
              aes(x = Age_calBP, ymin = NPP_CI95_lower, ymax = NPP_CI95_upper
), fill = "orange", alpha = 0.1) +

  scale_x_reverse(limits = c(27000, 5000), breaks = seq(35000, 2000, by = -50
00)) +
  labs(
    x = "Age (yr cal BP)",
    y = "NPP"
  ) +
  theme_minimal()
```



Climate-derived and brGDGT-derived NPP in Padul.

```
# Potential NPP uncertainty from MAT and MAP RMSE
Padul_Miami_climate <- read.xlsx("Dataset_1.xlsx", sheet = "Padul_climate")
set.seed(123)

n_iter <- 5000
n_clim <- nrow(Padul_Miami_climate)
NPPpot_mc <- matrix(NA_real_, nrow = n_clim, ncol = n_iter)

for (i in seq_len(n_clim)) {
  mat_low <- Padul_Miami_climate$Predicted_MAT[i] - Padul_Miami_climate$MAT_R
```

```r
MSE[i]
  mat_high <- Padul_Miami_climate$Predicted_MAT[i] + Padul_Miami_climate$MAT_
RMSE[i]
  mat_sim <- rtruncnorm(
    n = n_iter, a = mat_low, b = mat_high,
    mean = Padul_Miami_climate$Predicted_MAT[i],
    sd = Padul_Miami_climate$MAT_RMSE[i]
  )

  map_low <- max(0, Padul_Miami_climate$Predicted_MAP[i] - Padul_Miami_climat
e$RMSE_MAP[i])
  map_high <- Padul_Miami_climate$Predicted_MAP[i] + Padul_Miami_climate$RMSE
_MAP[i]
  map_sim <- rtruncnorm(
    n = n_iter, a = map_low, b = map_high,
    mean = Padul_Miami_climate$Predicted_MAP[i],
    sd = Padul_Miami_climate$RMSE_MAP[i]
  )

  npp_miami_mat <- 3000 / (1 + exp(1.315 - 0.119 * mat_sim))
  npp_miami_map <- 3000 * (1 - exp(-0.000664 * map_sim))
  NPPpot_mc[i, ] <- pmin(npp_miami_mat, npp_miami_map) * 0.45
}

NPPpot_low <- apply(NPPpot_mc, 1, quantile, probs = 0.025, na.rm = TRUE)
NPPpot_high <- apply(NPPpot_mc, 1, quantile, probs = 0.975, na.rm = TRUE)
NPPpot_median <- apply(NPPpot_mc, 1, median, na.rm = TRUE)

df_NPPpot <- data.frame(
  Age = Padul_Miami_climate$Age,
  NPP_mean = Padul_Miami_climate$NPP_Miami,
  NPP_low = NPPpot_low,
  NPP_high = NPPpot_high,
  NPP_MC_median = NPPpot_median
)

# Prediction uncertainty from individual RF trees
rf_wide <- tree_long_main %>%
  dplyr::select(Age_calBP, Tree, NPP) %>%
  tidyr::pivot_wider(names_from = Tree, values_from = NPP) %>%
  dplyr::arrange(Age_calBP)

rf_age <- rf_wide$Age_calBP
RF_matrix <- as.matrix(rf_wide[, -1])
NPPact_central <- rowMeans(RF_matrix, na.rm = TRUE)
NPPpot_central_rf <- approx(
  x = Padul_Miami_climate$Age,
  y = Padul_Miami_climate$NPP_Miami,
  xout = rf_age, rule = 2
```

```
)$y
DeltaNPP_central <- NPPpot_central_rf - NPPact_central

set.seed(123)
n_delta <- 5000
DeltaNPP_mc <- matrix(NA_real_, nrow = length(rf_age), ncol = n_delta)

for (m in seq_len(n_delta)) {
  pot_id <- sample(seq_len(ncol(NPPpot_mc)), size = 1)
  NPPpot_draw <- approx(
    x = Padul_Miami_climate$Age,
    y = NPPpot_mc[, pot_id],
    xout = rf_age, rule = 2
  )$y
  rf_id <- sample(seq_len(ncol(RF_matrix)), size = 1)
  NPPact_draw <- RF_matrix[, rf_id]
  DeltaNPP_mc[, m] <- NPPpot_draw - NPPact_draw
}

hanpp_mean <- data.frame(
  Age_calBP = rf_age,
  HANPP_mean = DeltaNPP_central,
  HANPP_low = apply(DeltaNPP_mc, 1, quantile, probs = 0.025, na.rm = TRUE),
  HANPP_high = apply(DeltaNPP_mc, 1, quantile, probs = 0.975, na.rm = TRUE),
  HANPP_MC_median = apply(DeltaNPP_mc, 1, median, na.rm = TRUE)
)

hanpp_mean_main <- hanpp_mean
knitr::kable(head(hanpp_mean), digits = 2,
             col.names = gsub("HANPP", "DNPP", names(hanpp_mean)))
```

Age_calBP	DNPP_mean	DNPP_low	DNPP_high	DNPP_MC_median
-62.0	39.17	-197.56	252.10	34.90
-1.0	59.08	-169.90	263.78	57.15
123.0	45.00	-188.46	265.49	44.72
5015.0	-115.18	-352.45	88.64	-113.91
5203.5	-156.36	-388.06	36.86	-155.82
5404.5	-190.96	-415.70	-4.15	-187.86

```
DNPP_plot2 <- ggplot(hanpp_mean, aes(x = Age_calBP)) +
  geom_ribbon(aes(ymin = HANPP_low, ymax = HANPP_high),
              fill = "darkgreen", alpha = 0.15) +
  geom_line(aes(y = HANPP_mean), color = "darkgreen", linewidth = 0.7) +
  scale_x_reverse(limits = c(27000, 5000), breaks = seq(25000, 5000, -5000))
+
  ylim(c(-600, 200)) +
  labs(x = "Age (cal yr BP)",
       y = expression(Delta*NPP~(g~C~m^{-2}~yr^{-1}))) +
```

```
  theme_minimal()

DNPP_plot2
```

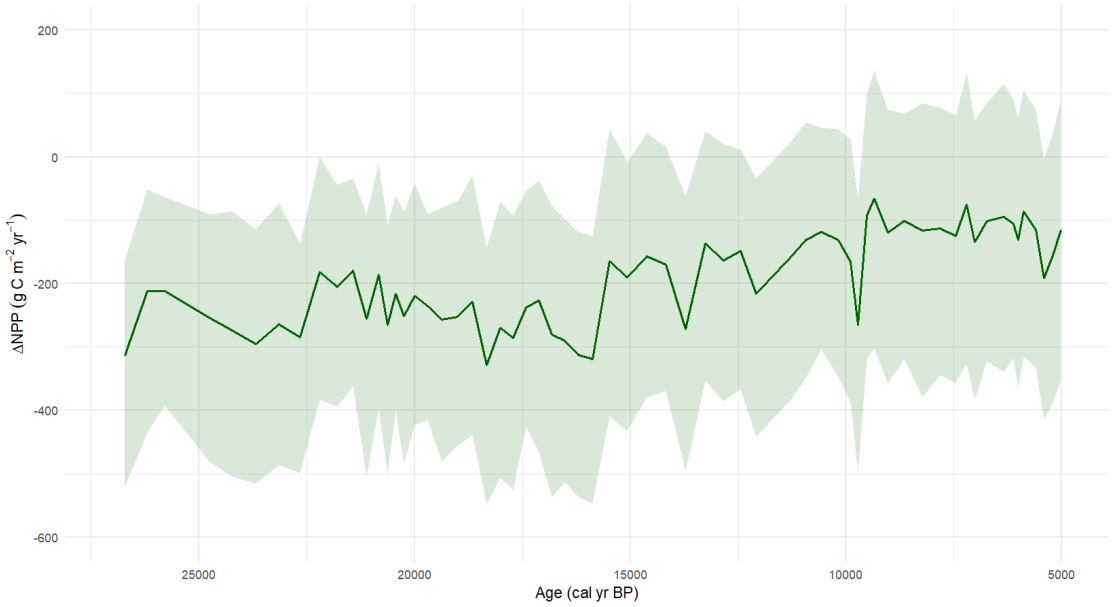


Figure 6E. DNPP reconstruction with uncertainty.

Figure 6B–C: Palaeoenvironmental context

Plot NGRIP $\delta^{18}O$ alongside Mediterranean pollen taxa and the pollen precipitation index over the Padul reconstruction interval.

```
NGRIP_df <- read.xlsx("Dataset_1.xlsx", sheet = "NGRIP", detectDates = FALSE)
head(NGRIP_df)

##    Age   d18O
## 1 5.00 -34.76
## 2 5.00 -34.65
## 3 5.05 -34.65
## 4 5.05 -34.98
## 5 5.10 -34.98
## 6 5.10 -34.41

events <- data.frame(
  Event = c( "H2", "LGM", "H1",  "Younger Dryas", "8.2"),
  start = c( 24500, 21500, 17000,  12900,  8400),
  end   = c( 23000, 19000, 15000, 11700,  8100)
)

NGRIP_plot <- ggplot() +
  geom_rect(data = events, aes(xmin = start, xmax = end, ymin = -Inf, ymax =
Inf),
            fill = "grey80", alpha = 0.4) +
```

```
  geom_line(data = NGRIP_df, aes(x = Age * 1000, y = d18O),
            color = "blue", size = 1) +
  scale_x_reverse(limits = c(27000, 5000),
                  breaks = seq(35000, 5000, by = -5000)) +
  labs(
    x = " ",
    y = expression(delta^{18}*O~("\u2030")),
    title = " "
  ) +
  theme_minimal(base_size = 14)

NGRIP_plot
```

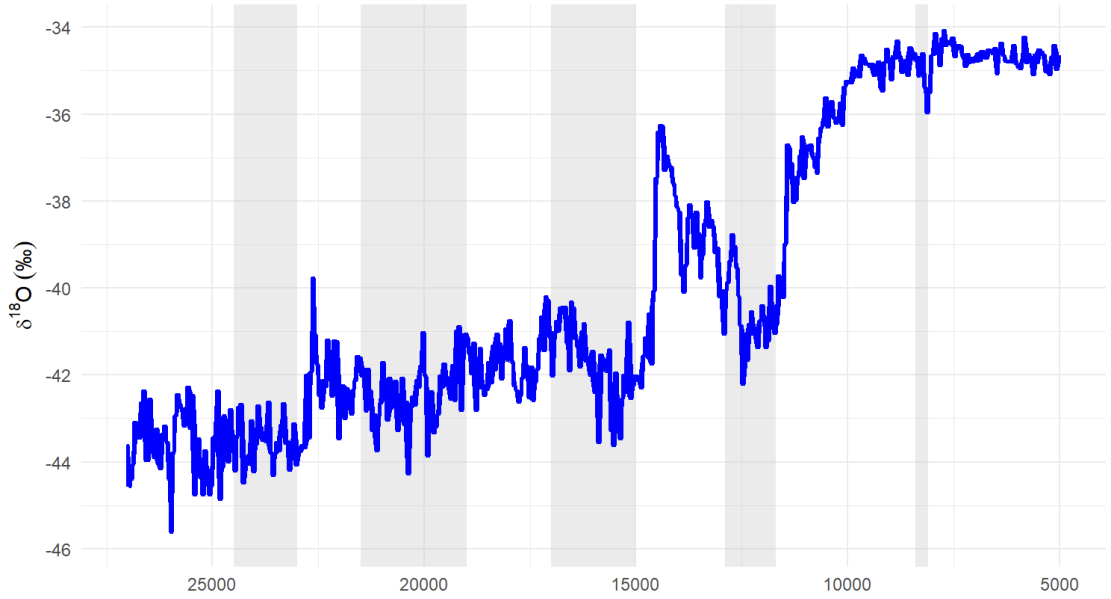


Figure 6B-C. NGRIP and pollen records.

```
pollen_df <- read.xlsx("Dataset_1.xlsx", sheet = "Pollen", detectDates = FALS
E)

lp_combined <- ggplot() +
  geom_line(data = pollen_df, aes(x = Age, y = Mediterranean), color = "darkg
reen") +
  geom_line(data = pollen_df, aes(x = Age, y = P.Index), color = "black") +
  scale_y_continuous(
    name = "Mediterranean",
    sec.axis = sec_axis(~., name = "P.Index")
  ) +
  scale_x_reverse(limits = c(27000, 5000),
                  breaks = seq(35000, 5000, by = -5000)) +
  labs(x = "Age (years BP)") +
  theme_minimal()
```

```
lp_combined
```

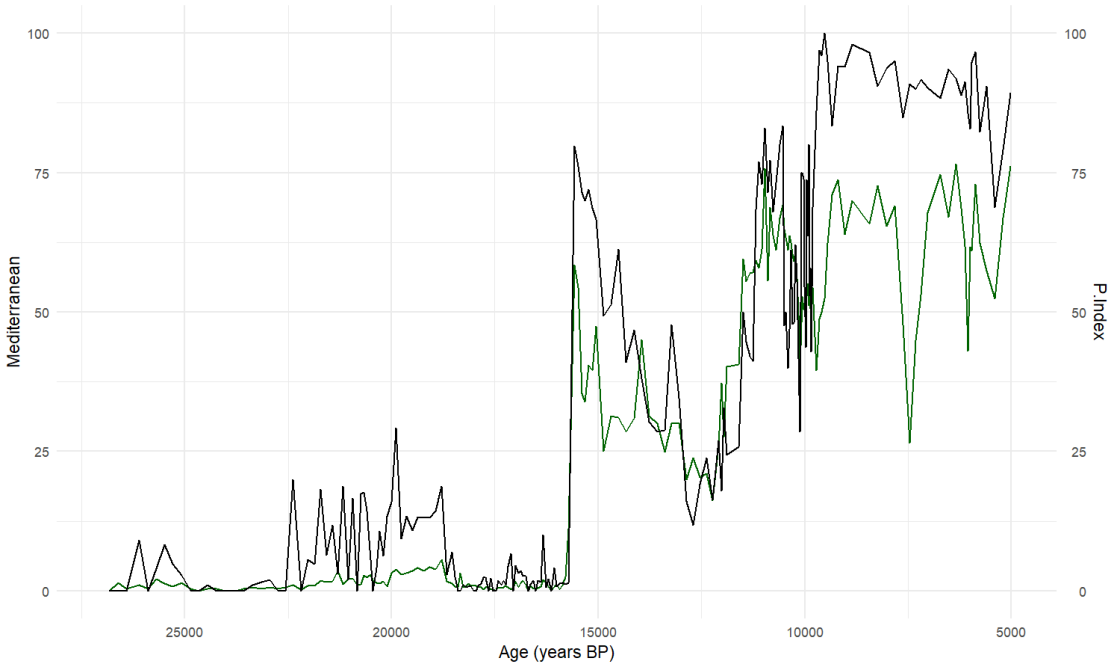


Figure 6B-C. NGRIP and pollen records.

Figure S13 and Table S5: Temporal-offset sensitivity

Shift the potential-NPP chronology by −1000 to +1000 years. Recalculate the NPP deficit and compare each offset with the zero-offset reconstruction using Spearman correlation and percentage sign agreement.

```
Padul_Miami_climate <- read.xlsx(
  "Dataset_1.xlsx",
  sheet = "Padul_climate"
)

rf_mean <- tree_long_main %>%
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(
    NPP_rf = mean(NPP, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  dplyr::arrange(Age_calBP)

lags <- c(-1000, -500, -250, -100, 0, 100, 250, 500, 1000)

lag_test <- bind_rows(lapply(lags, function(L){

  tmp <- rf_mean %>%
```

```
    mutate(
      Lag = L,
      NPP_Miami = approx(
        Padul_Miami_climate$Age + L,
        Padul_Miami_climate$NPP_Miami,
        xout = Age_calBP,
        rule = 1
      )$y,
      HANPP = NPP_Miami - NPP_rf
    )

  tmp
}))

base <- lag_test %>%
  filter(Lag == 0) %>%
  dplyr::select(Age_calBP, HANPP_0 = HANPP)

lag_summary <- lag_test %>%
  left_join(base, by = "Age_calBP") %>%
  group_by(Lag) %>%
  summarise(
    rho = cor(HANPP, HANPP_0,
              method = "spearman",
              use = "complete.obs"),
    sign_agreement = mean(
      sign(HANPP) == sign(HANPP_0),
      na.rm = TRUE
    ) * 100
  )


ggplot(
  lag_test,
  aes(
    x = Age_calBP,
    y = HANPP,
    color = Lag,
    group = Lag
  )
) +
  geom_line(linewidth = 0.8, alpha = 0.7) +

  geom_line(
    data = subset(lag_test, Lag == 0),
    aes(x = Age_calBP, y = HANPP),
    inherit.aes = FALSE,
    color = "black",
    linewidth = 1.3
```

```
) +

geom_hline(
  yintercept = 0,
  linetype = "dashed",
  color = "black"
) +

scale_color_gradient2(
  low = "blue",
  mid = "grey70",
  high = "red",
  midpoint = 0,
  breaks = c(-1000, -500, 0, 500, 1000),
  name = "Lag (yr)"
) +

scale_x_reverse(
  limits = c(27000, 5000),
  breaks = seq(25000, 5000, by = -5000)
) +

labs(
  x = "Age (cal yr BP)",
  y = expression("NPP Deficit (g C m"^{-2}*" yr"^{-1}*")")
) +

theme_minimal(base_size = 12)
```

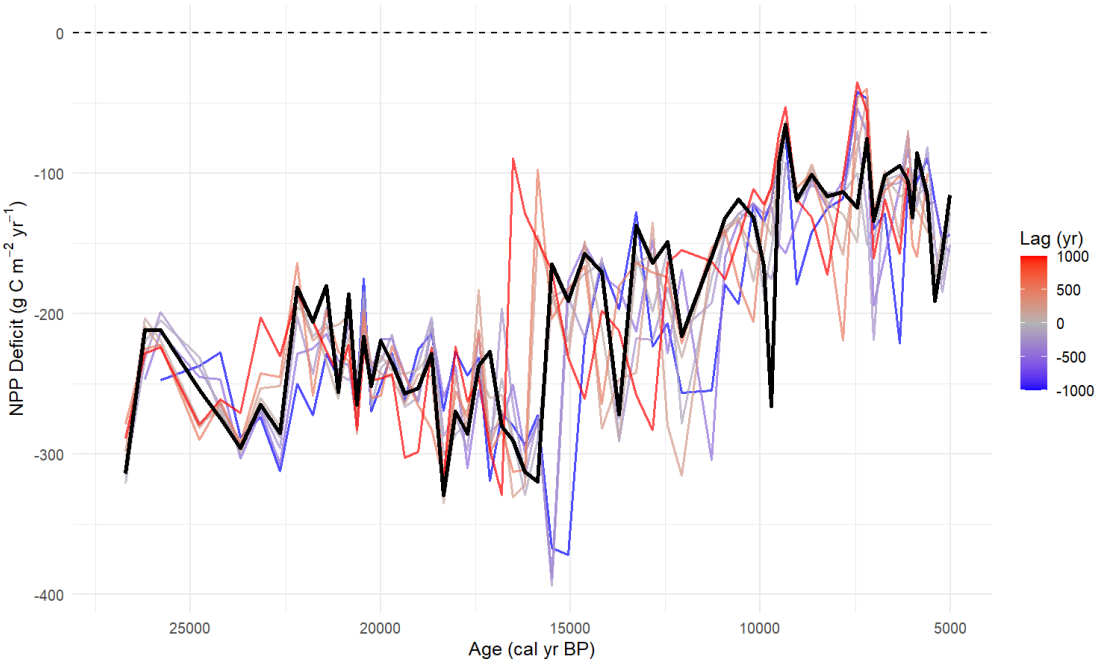


Figure S13. NPP-deficit sensitivity to temporal offsets.

```
base <- lag_test %>%
  dplyr::filter(Lag == 0) %>%
  dplyr::select(
    Age_calBP,
    HANPP_0 = HANPP
  )

lag_sensitivity <- lag_test %>%
  dplyr::left_join(base, by = "Age_calBP") %>%
  dplyr::group_by(Lag) %>%
  dplyr::summarise(

    n = sum(complete.cases(HANPP, HANPP_0)),

    Spearman_rho = cor(
      HANPP,
      HANPP_0,
      method = "spearman",
      use = "complete.obs"
    ),

    Sign_agreement = mean(
      sign(HANPP) == sign(HANPP_0),
      na.rm = TRUE
    ) * 100,

    .groups = "drop"
```

```
  ) %>%
  dplyr::arrange(Lag)


knitr::kable(lag_sensitivity, digits = 3, caption = "Table S5. Temporal-offse
t sensitivity statistics.")
```

Table S5. Temporal-offset sensitivity statistics.

Lag	n	Spearman_rho	Sign_agreement
-1000	61	0.734	100
-500	62	0.769	100
-250	63	0.838	100
-100	63	0.929	100
0	63	1.000	100
100	62	0.945	100
250	61	0.784	100
500	60	0.754	100
1000	57	0.584	100

Figure S12: Alternative palaeoclimate sources

Recalculate the potential-NPP minus brGDGT-NPP difference using the pollen, HadCM3B-M2.1 and CHELSA-TraCE21k potential-NPP records. Compare the three reconstructions using the same brGDGT tree predictions.

```
local({
  pollen_climate <- read.xlsx("Dataset_1.xlsx", sheet = "Padul_climate")
  require_columns(pollen_climate, c("Age", "NPP_Miami"), "Padul_climate")

  Padul_Miami_climate <- read.xlsx("Dataset_1.xlsx", sheet = "Regional_Paleoc
limate")
  head(Padul_Miami_climate)

  rf_mean_s <- tree_long %>%
    group_by(Age_calBP) %>%
    summarise(NPP_rf = mean(NPP, na.rm = TRUE)) %>%
    arrange(desc(Age_calBP))

  miami_interp_s <- approx(x = pollen_climate$Age,
                           y = pollen_climate$NPP_Miami,
                           xout = rf_mean_s$Age_calBP)$y

  hanpp_data_s <- tree_long %>%
    mutate(NPP_Miami = approx(pollen_climate$Age,
```

```
                              pollen_climate$NPP_Miami,
                              xout = Age_calBP)$y) %>%
  mutate(HANPP =   NPP_Miami -NPP  ) %>%
  arrange(desc(Age_calBP))

hanpp_mean_s <- hanpp_data_s %>%
  group_by(Age_calBP) %>%
  summarise(hanpp_mean_s = mean(HANPP, na.rm = TRUE),
            HANPP_sd = sd(HANPP, na.rm = TRUE))

DNPP_plot1 <- ggplot() +
  geom_line(data = hanpp_data_s,
            aes(x = Age_calBP, y = HANPP, group = Tree),
            alpha = 0.1, color = "darkgrey", linewidth = 0.3) +
  geom_line(data = hanpp_mean_s,
            aes(x = Age_calBP, y = hanpp_mean_s),
            color = "darkgreen", linewidth = 0.5) +
  scale_x_reverse(limits = c(27000, 5000),
                  breaks = seq(35000, 5000, by = -5000)) +
  labs(x = "Age (cal yr BP)",
       y = expression(paste("NPP Deficit (g C m"^{-2}, " yr"^{-1}, ")")),
       title = " ") +
  theme_minimal()

rf_mean_s <- tree_long %>%
  group_by(Age_calBP) %>%
  summarise(NPP_rf = mean(NPP, na.rm = TRUE)) %>%
  arrange(desc(Age_calBP))

miami_interp_s <- approx(x = Padul_Miami_climate$Age_GCM,
                         y = Padul_Miami_climate$NPP_Miami_GCM ,
                         xout = rf_mean_s$Age_calBP)$y

hanpp_data_s <- tree_long %>%
  mutate(NPP_Miami = approx(Padul_Miami_climate$Age_GCM,
                              Padul_Miami_climate$NPP_Miami_GCM ,
                              xout = Age_calBP)$y) %>%
  mutate(HANPP =   NPP_Miami -NPP  ) %>%
  arrange(desc(Age_calBP))

hanpp_mean_s <- hanpp_data_s %>%
  group_by(Age_calBP) %>%
  summarise(hanpp_mean_s = mean(HANPP, na.rm = TRUE),
            HANPP_sd = sd(HANPP, na.rm = TRUE))

DNPP_plot2 <- ggplot() +
  geom_line(data = hanpp_data_s,
            aes(x = Age_calBP, y = HANPP, group = Tree),
            alpha = 0.1, color = "darkgrey", linewidth = 0.3) +
```

```
    geom_line(data = hanpp_mean_s,
              aes(x = Age_calBP, y = hanpp_mean_s),
              color = "darkgreen", linewidth = 0.5) +
    scale_x_reverse(limits = c(27000, 5000),
                    breaks = seq(35000, 5000, by = -5000)) +
    labs(x = "Age (cal yr BP)",
         y = expression(paste("NPP Deficit (g C m"^{-2}, " yr"^{-1}, ")")),
         title = " ") +
    theme_minimal()

  miami_interp_s <- approx(x = Padul_Miami_climate$Age_CHELSA ,
                           y = Padul_Miami_climate$NPP_Miami_CHELSA ,
                           xout = rf_mean_s$Age_calBP)$y

  hanpp_data_s <- tree_long %>%
    mutate(NPP_Miami = approx(Padul_Miami_climate$Age_CHELSA ,
                              Padul_Miami_climate$NPP_Miami_CHELSA ,
                              xout = Age_calBP)$y) %>%
    mutate(HANPP =   NPP_Miami -NPP  ) %>%
    arrange(desc(Age_calBP))

  hanpp_mean_s <- hanpp_data_s %>%
    group_by(Age_calBP) %>%
    summarise(hanpp_mean_s = mean(HANPP, na.rm = TRUE),
              HANPP_sd = sd(HANPP, na.rm = TRUE))

  DNPP_plot3 <- ggplot() +
    geom_line(data = hanpp_data_s,
              aes(x = Age_calBP, y = HANPP, group = Tree),
              alpha = 0.1, color = "darkgrey", linewidth = 0.3) +
    geom_line(data = hanpp_mean_s,
              aes(x = Age_calBP, y = hanpp_mean_s),
              color = "darkgreen", linewidth = 0.5) +
    scale_x_reverse(limits = c(27000, 5000),
                    breaks = seq(35000, 5000, by = -5000)) +
    labs(x = "Age (cal yr BP)",
         y = expression(paste("NPP Deficit (g C m"^{-2}, " yr"^{-1}, ")")),
         title = " ") +
    theme_minimal()

  grid.arrange(DNPP_plot1,
               DNPP_plot2,
               DNPP_plot3)
})
```

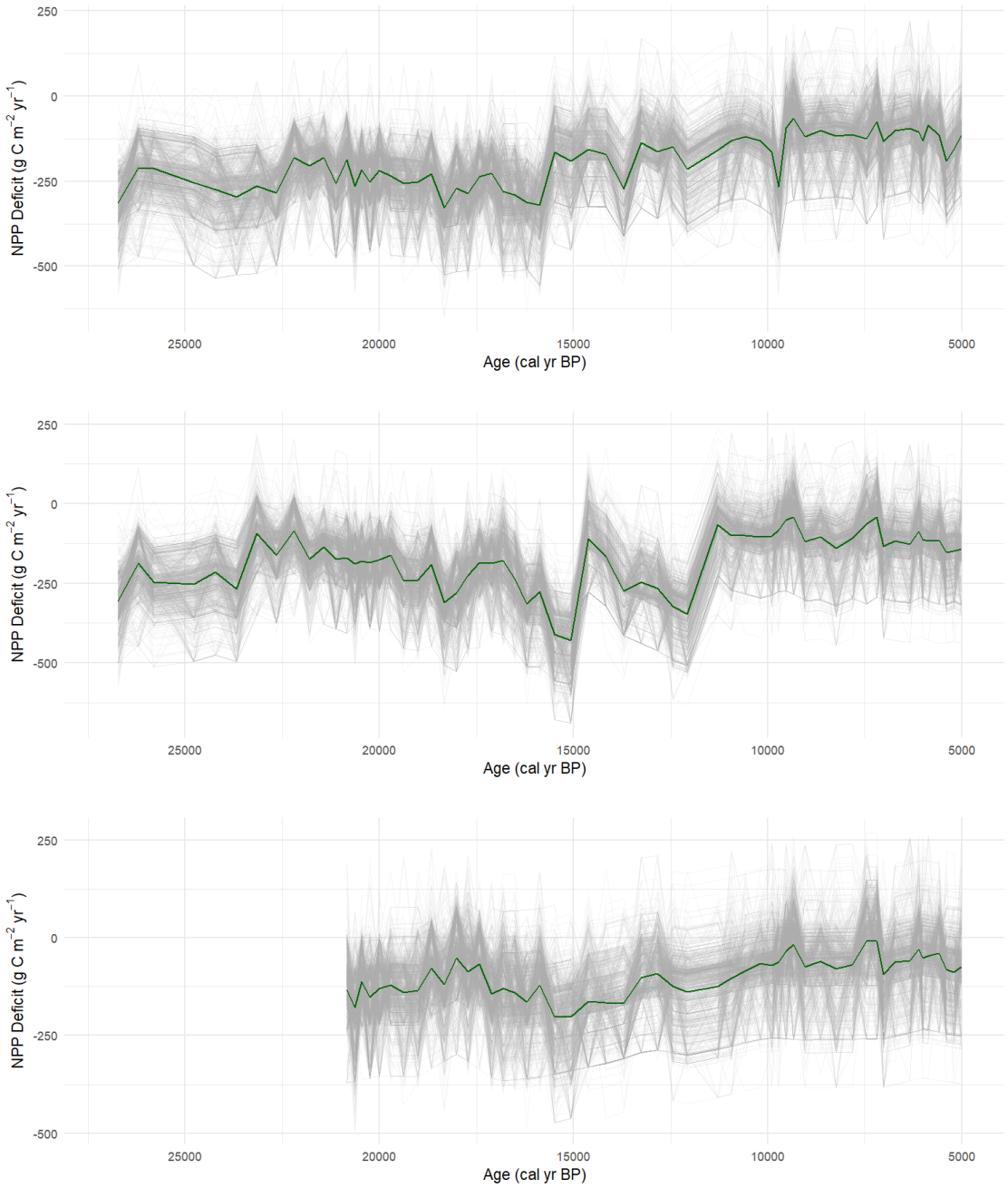


Figure S12. NPP deficit using alternative palaeoclimate records.

Summed probability distributions

Select included radiocarbon dates by culture, calibrate them with the specified terrestrial or marine curves, group dates within archaeological layers using 300-year bins and calculate summed probability distributions.

```
deltaR_val <- 94
deltaR_sd <- 61
```

```r
cultures <- c("Gravettian", "Solutrean", "Magdalenian", "Epipalaeolithic", "N
eolithic")
SPD_df <- read.xlsx("Dataset_1.xlsx", sheet = "SPD", detectDates = FALSE)
nrow(SPD_df)
```

```
## [1] 326
```

```r
SPD_df <- subset(SPD_df, Included == "y")
nrow(SPD_df)
```

```
## [1] 265
```

```r
# Script_1(6).R: bin individual calibrated dates by unique site-layer IDs.
# Only the outputs originally displayed in the guide remain visible.
counts_by_culture <- SPD_df %>%
  dplyr::filter(Included == "y", Culture %in% cultures,
                !is.na(Age), !is.na(s.dev.), !is.na(Curve),
                !is.na(Labref), !is.na(layer), !is.na(Sitename)) %>%
  dplyr::mutate(SiteLayer = paste(Sitename, layer, sep = "_")) %>%
  dplyr::group_by(Culture) %>%
  dplyr::summarise(n_sites = dplyr::n_distinct(Sitename),
                   n_layers = dplyr::n_distinct(SiteLayer),
                   n_dates = dplyr::n(), .groups = "drop") %>%
  dplyr::arrange(match(Culture, cultures))

Regions_SPD <- data.frame()
invisible(capture.output(suppressMessages(suppressWarnings({
  for (cult in cultures) {
    cult_df <- SPD_df %>%
      dplyr::filter(Included == "y", Culture == cult) %>%
      dplyr::filter(!is.na(Age), !is.na(s.dev.), !is.na(Curve),
                    !is.na(Labref), !is.na(layer))
    if (nrow(cult_df) == 0) next
    curves <- ifelse(cult_df$Curve == "Terrestrial", "intcal20", "marine20")
    deltaR_vals <- ifelse(cult_df$Curve == "Marine", deltaR_val, 0)
    deltaR_sds <- ifelse(cult_df$Curve == "Marine", deltaR_sd, 0)
    caldates_ind <- calibrate(
      x = cult_df$Age, errors = cult_df$s.dev., calCurves = curves,
      ids = paste0(cult_df$Labref, "_", seq_len(nrow(cult_df))),
      delta.R = deltaR_vals, delta.STD = deltaR_sds, calMatrix = TRUE)
    cult_df$SiteLayer <- paste(cult_df$Sitename, cult_df$layer, sep = "_")
    bins <- binPrep(sites = as.integer(factor(cult_df$SiteLayer)),
                    ages = cult_df$Age, h = 300)
    spd_cult <- spd(caldates_ind, timeRange = c(35000, 2000), bins = bins)
    Regions_SPD <- rbind(Regions_SPD, data.frame(
      PrDens = spd_cult$grid$PrDens, Age = spd_cult$grid$calBP,
      Culture = cult))
  }
})))
```

Figure 6F: SPD and charcoal

Plot cultural SPD curves with Padul charcoal concentration. Scale charcoal to the SPD plotting range and use the corresponding inverse transformation for the secondary axis.

```
scale_factor <- max(Regions_SPD$PrDens, na.rm = TRUE) /
  max(pollen_df$Charcoals_cm3, na.rm = TRUE)

SPD_l3_charcoal <- ggplot() +

  geom_area(
    data = Regions_SPD,
    aes(x = Age, y = PrDens, fill = Culture),
    alpha = 0.5,
    position = "identity",
    color = "black"
  ) +

  geom_point(
    data = pollen_df,
    aes(x = Age, y = Charcoals_cm3 * scale_factor),
    color = "black",
    size = 1.5
  ) +

  scale_x_reverse(
    limits = c(27000, 5000),
    breaks = seq(25000, 5000, by = -5000)
  ) +

  scale_y_continuous(
    name = "Probability Density",
    sec.axis = sec_axis(
      ~ . / scale_factor,
      name = expression("Charcoal (particles cm"^{-3}*")")
    )
  ) +

  scale_fill_brewer(palette = "Set2") +

  labs(
    x = "cal BP",
    title = "Summed Probability Distribution and Charcoal"
  ) +

  theme_classic() +

  theme(
```

```
    legend.position = "none",
    axis.title.y.right = element_text(),
    axis.text.y.right = element_text()
  )

SPD_13_charcoal
```

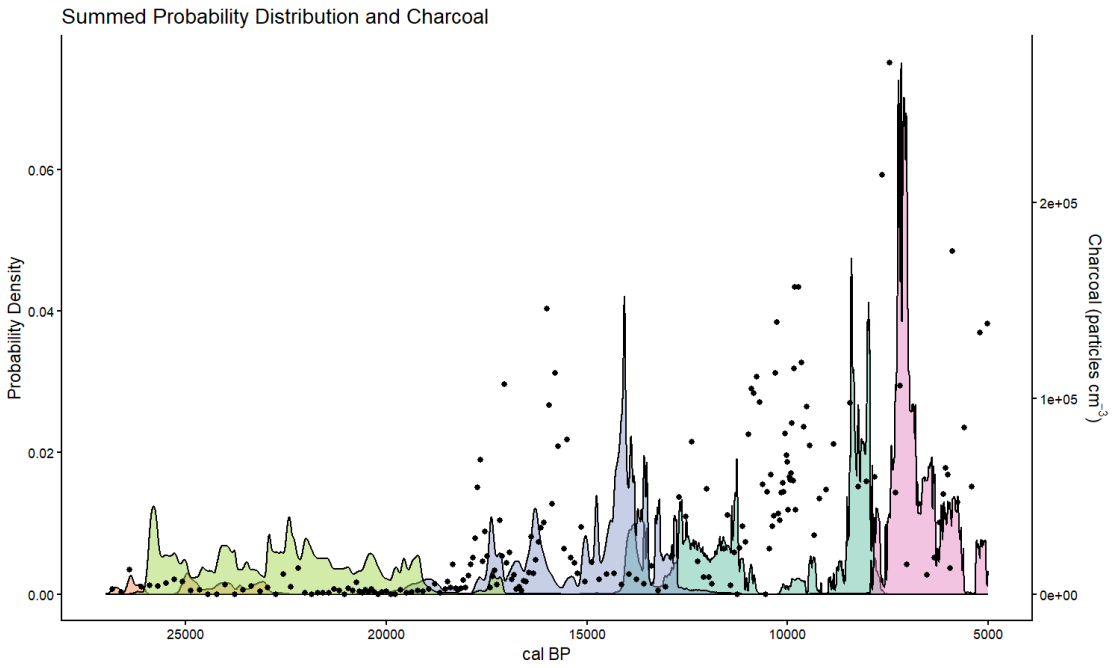


Figure 6F. Cultural SPD curves and charcoal concentration.

Figure 7A–C: NPP distributions by culture

Select ages with SPD density ≥0.001, interpolate brGDGT-derived and climate-derived NPP to those ages and calculate their difference. Plot boxplots of brGDGT-derived NPP, potential NPP and NPP deficit.

```
Pollen_df <- read.xlsx("Dataset_1.xlsx", sheet = "Pollen")

spd_ages <- Regions_SPD %>%
  filter(PrDens >= 0.001) %>%
  dplyr::select(Age, Culture, PrDens) %>%
  distinct()

npp_by_age <- results_complete %>%
  dplyr::select(Age_calBP, NPP_pred_mean) %>%
  arrange(Age_calBP)

npp_interp <- approx(
  x = npp_by_age$Age_calBP,
  y = npp_by_age$NPP_pred_mean,
```

```r
  xout = spd_ages$Age,
  rule = 1
)$y

miami_interp <- approx(
  x = Pollen_df$Age,
  y = Pollen_df$NPP_Miami,
  xout = spd_ages$Age,
  rule = 1
)$y

all_data_complete <- spd_ages %>%
  mutate(
    NPP_brGDGT = npp_interp,
    NPP_Miami = miami_interp,
    HANPP = NPP_Miami - NPP_brGDGT
  )

summary(dplyr::rename(all_data_complete, DNPP = HANPP))
```

```
##       Age          Culture              PrDens           NPP_brGDGT
##  Min.   : 3078   Length:29114       Min.   :0.001000   Min.   :531.8
##  1st Qu.:12121   Class :character   1st Qu.:0.002159   1st Qu.:561.7
##  Median :17696   Mode  :character   Median :0.004386   Median :587.2
##  Mean   :18080                      Mean   :0.006306   Mean   :592.6
##  3rd Qu.:24321                      3rd Qu.:0.007020   3rd Qu.:617.8
##  Max.   :33238                      Max.   :0.075100   Max.   :666.3
##                                                        NA's   :4087
##    NPP_Miami          DNPP
##  Min.   :236.3   Min.   :-354.35
##  1st Qu.:313.7   1st Qu.:-265.97
##  Median :352.3   Median :-228.59
##  Mean   :380.1   Mean   :-213.33
##  3rd Qu.:471.3   3rd Qu.:-163.65
##  Max.   :528.9   Max.   : -65.43
##  NA's   :6093    NA's   :6093
```

```r
stats_by_culture <- all_data_complete %>%
  group_by(Culture) %>%
  summarise(

    NPP_brGDGT_mean = mean(NPP_brGDGT, na.rm = TRUE),
    NPP_brGDGT_sd = sd(NPP_brGDGT, na.rm = TRUE),
    NPP_brGDGT_min = min(NPP_brGDGT, na.rm = TRUE),
    NPP_brGDGT_max = max(NPP_brGDGT, na.rm = TRUE),

    NPP_Miami_mean = mean(NPP_Miami, na.rm = TRUE),
    NPP_Miami_sd = sd(NPP_Miami, na.rm = TRUE),
    NPP_Miami_min = min(NPP_Miami, na.rm = TRUE),
    NPP_Miami_max = max(NPP_Miami, na.rm = TRUE),
```

```
    HANPP_mean = mean(HANPP, na.rm = TRUE),
    HANPP_sd = sd(HANPP, na.rm = TRUE),
    HANPP_min = min(HANPP, na.rm = TRUE),
    HANPP_max = max(HANPP, na.rm = TRUE),

    PrDens_mean = mean(PrDens, na.rm = TRUE),
    PrDens_sd = sd(PrDens, na.rm = TRUE),

    Age_min = min(Age, na.rm = TRUE),
    Age_max = max(Age, na.rm = TRUE),
    Age_span = Age_max - Age_min,
    n_ages = n()
  ) %>%
  arrange(desc(Age_min))

p1 <- ggplot(all_data_complete, aes(x = reorder(Culture, -Age), y = NPP_brGDG
T, fill = Culture)) +
  geom_boxplot(alpha = 0.7,   outlier.color = "gray",
               outlier.alpha = 0.5, size = 0.7) +
  stat_summary(fun = mean, geom = "point",  color = "black") +
  labs(
    title = "",
    x = "",
    y = "NPP brGDGT (g C m⁻² yr⁻¹)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

p2 <- ggplot(all_data_complete, aes(x = reorder(Culture, -Age), y = NPP_Miami
, fill = Culture)) +
  geom_boxplot(alpha = 0.7,   outlier.color = "gray",
               outlier.alpha = 0.5, size = 0.7) +
  stat_summary(fun = mean, geom = "point",  color = "black") +
  labs(
    title = "",
    x = " ",
    y = "NPP climate (g C m⁻² yr⁻¹)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

p3 <- ggplot(all_data_complete, aes(x = reorder(Culture, -Age), y = HANPP, fi
```

```
ll = Culture)) +
  geom_boxplot(alpha = 0.7,   outlier.color = "gray",
               outlier.alpha = 0.5, size = 0.7) +
  stat_summary(fun = mean, geom = "point",  color = "black") +
  labs(
    title = " ",
    x = " ",
    y = "Deficit NPP (g C m⁻² yr⁻¹)"
  ) +
  geom_hline(yintercept = 139, linetype = "dashed", color = "darkred", linewi
dth = 0.8) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

grid.arrange(p1, p2,p3 ,ncol = 3, nrow = 1)
```

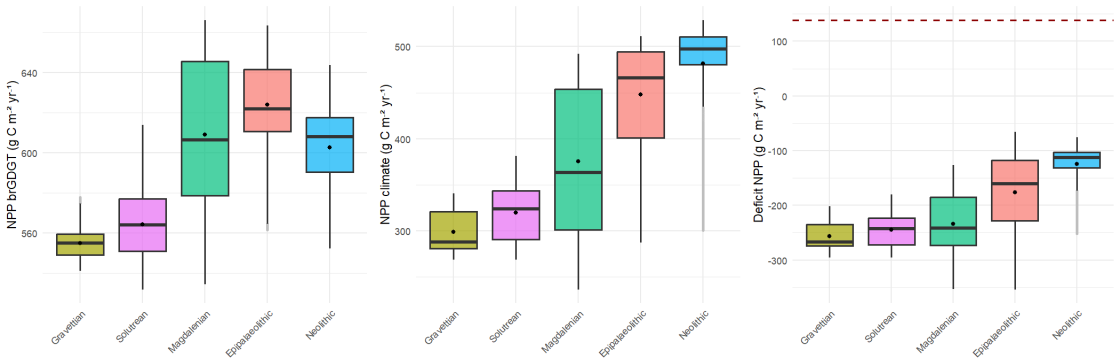


Figure 7A-C. NPP and NPP-deficit distributions by culture.

```
knitr::kable(stats_by_culture, digits = 3, caption = "Table S4. NPP and SPD s
tatistics by culture.",
             col.names = gsub("HANPP", "DNPP", names(stats_by_culture)))
```

Table S4. NPP and SPD statistics by culture.

Culture	NPP_brGDGT_mean	NPP_brGDGT_sd	NPP_brGDGT_min	NPP_brGDGT_max	NPP_Miami_mean	NPP_Miami_sd	NPP_Miami_min	NPP_Miami_max	DNPP_mean	DNPP_sd	DNPP_min	DNPP_max	PrDens_mean	PrDens_sd	Age_min	Age_max	Age_span	n_ages
Gravettian	555.161	8.090	541.260	577.932	299.159	22.254	268.760	340.623	-256.401	24.691	-295.748	-201.288	0.003	0.002	22963	33238	10275	6542
Solutrean	564.337	16.532	531.817	614.022	320.095	29.405	268.760	381.053	-244.242	28.027	-295.748	-179.989	0.005	0.002	17084	26024	8940	7451
Magdalenian	609.065	36.689	534.553	666.337	375.352	76.591	236.253	492.356	-233.713	52.544	-353.615	-126.603	0.007	0.006	12721	19491	6770	5840
Epipaleolithic	624.175	20.487	561.698	663.471	448.102	53.463	287.286	511.301	-176.072	65.809	-354.347	-65.426	0.007	0.007	7673	14123	6450	5274

Culture	NPP_brGDGT_mean	NPP_brGDGT_sd	NPP_brGDGT_min	NPP_brGDGT_max	NPP_Miami_mean	NPP_Miami_sd	NPP_Miami_min	NPP_Miami_max	DNPP_mean	DNPP_sd	DNPP_min	DNPP_max	Pr_Dens_mean	Pr_Dens_sd	Age_min	Age_max	Age_span	Age_n
Neolithic	602.762	23.440	552.455	643.954	481.748	49.204	300.603	528.914	-124.2107	33.587	-251.9707	-75.370	0.013	0.014	3078	7930	4852	407

Figure 8D: SPD density and NPP deficit

Fit a linear relationship between log10 SPD density and NPP deficit.

```
# Reproduce exactly the deterministic HANPP calculation in Script1.R.
# Unlike hanpp_mean_main (Monte Carlo), this uses NPP_Miami - NPP
# for each tree, then averages the deficit at each age.
fig08_climate <- openxlsx::read.xlsx("Dataset_1.xlsx", sheet = "Padul_climate
")

fig08_hanpp_data <- tree_long_main %>%
  dplyr::mutate(
    NPP_Miami = stats::approx(
      x = fig08_climate$Age,
      y = fig08_climate$NPP_Miami,
      xout = Age_calBP
    )$y,
    HANPP = NPP_Miami - NPP
  ) %>%
  dplyr::arrange(dplyr::desc(Age_calBP))

DNPP_data <- fig08_hanpp_data %>%
  dplyr::mutate(DNPP = HANPP)

fig08_hanpp_by_age <- fig08_hanpp_data %>%
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(HANPP = mean(HANPP, na.rm = TRUE), .groups = "drop")

fig08_npp_by_age <- tree_long_main %>%
```

```r
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(NPP_brGDGT = mean(NPP, na.rm = TRUE), .groups = "drop")

# As in Script1.R, use an EXACT age join (not interpolation) for 8D.
all_data_fig08d <- Regions_SPD %>%
  dplyr::left_join(fig08_npp_by_age, by = c("Age" = "Age_calBP")) %>%
  dplyr::left_join(fig08_hanpp_by_age, by = c("Age" = "Age_calBP"))

all_data_fixed_thresh <- all_data_fig08d %>%
  dplyr::filter(PrDens >= 0.001)

DNPP_Dens_plot <- ggplot(all_data_fixed_thresh,
                         aes(x = log10(PrDens), y = HANPP)) +
  geom_point(aes(color = Culture), size = 2, alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2, color = "darkred") +
  stat_cor(method = "pearson",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 3) +
  labs(
    x = "PrDens",
    y = "DNPP",
    title = " "
  ) +
  theme_minimal()

DNPP_Dens_plot
```

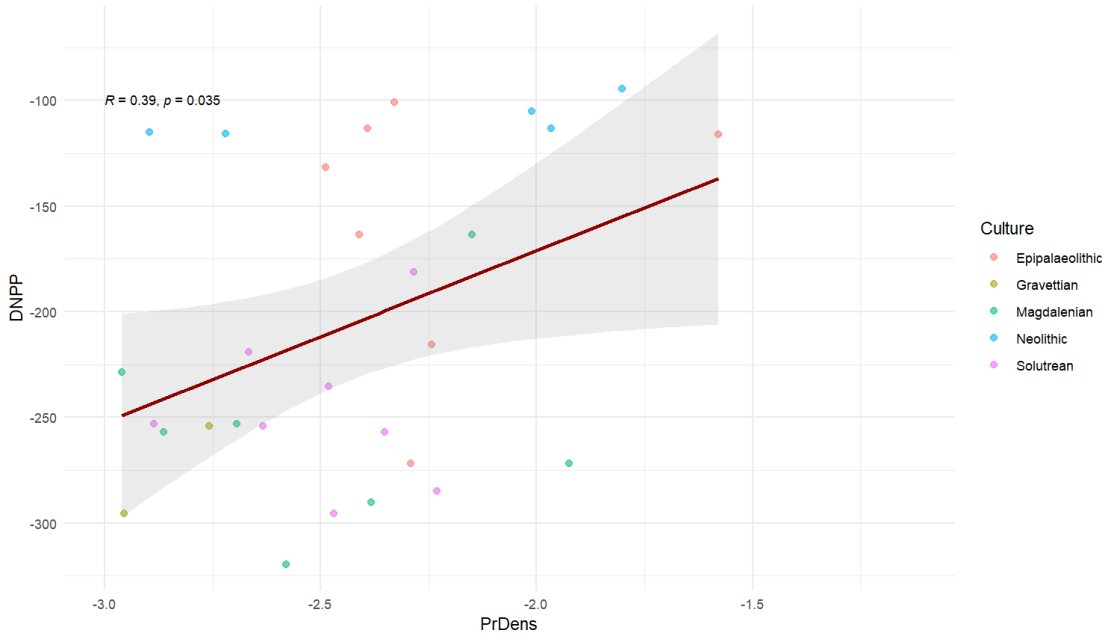


Figure 8D. SPD density and NPP deficit in logarithmic coordinates.

Figure S10: Pollen and NPP associations

Interpolate the pollen and charcoal records to brGDGT ages and retain 5000–27000 cal BP. Plot brGDGT-derived NPP against potential NPP, total pollen and pollen accumulation rate, colouring points by age and reporting correlations.

```
Pollen_df<- read.xlsx("Dataset_1.xlsx", sheet="Pollen")

rf_mean_s10 <- tree_long_main %>%
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(NPP_rf = mean(NPP, na.rm = TRUE), .groups = "drop") %>%
  dplyr::arrange(dplyr::desc(Age_calBP))

hanpp_interp <- approx(
  x = hanpp_mean_main$Age_calBP,
  y = hanpp_mean_main$HANPP_mean,
  xout = rf_mean_s10$Age_calBP,
  rule = 2
)$y

pollen_interp <- data.frame(
  Charcoal_cm3  = approx(Pollen_df$Age, Pollen_df$Charcoals_cm3, xout = rf_me
an_s10$Age_calBP, rule = 2)$y,
  PAR           = approx(Pollen_df$Age, Pollen_df$PAR, xout = rf_mean_s10$Age
_calBP, rule = 2)$y,
  Total_f           = approx(Pollen_df$Age, Pollen_df$Total_f, xout = rf_mean
_s10$Age_calBP, rule = 2)$y,
  NPP_climate    = approx(Pollen_df$Age, Pollen_df$NPP_Miami, xout = rf_mean_
s10$Age_calBP, rule = 2)$y,
  CAR           = approx(Pollen_df$Age, Pollen_df$CAR, xout = rf_mean_s10$Age
_calBP, rule = 2)$y,
  Charcoals           = approx(Pollen_df$Age, Pollen_df$Charcoals, xout = rf_
mean_s10$Age_calBP, rule = 2)$y
  )

combined_df <- rf_mean_s10 %>%
  mutate(
    HANPP = hanpp_interp
  ) %>%
  bind_cols(pollen_interp)


combined_df<- subset(combined_df, combined_df$Age_calBP> 4999 & combined_df$A
ge_calBP<27000)

NPP_actvs_pot<-ggplot(combined_df, aes(x = NPP_rf, y = NPP_climate)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +
```

```
  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "spearman",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "NPP brGDGT",
    y = "NPP Climate"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )


NPP_PAR<-ggplot(combined_df, aes(x = log10(NPP_rf), y = log10(PAR))) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "spearman",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "NPP brGDGT",
    y = "PAR"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )
```

```r
NPP_PollenTotal<-ggplot(combined_df, aes(x = log10(NPP_rf), y = log10(Total_f
))) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "spearman",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "NPP brGDGT",
    y = "Pollen count"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )


grid.arrange(NPP_actvs_pot, NPP_PollenTotal, NPP_PAR, ncol =1 )
```

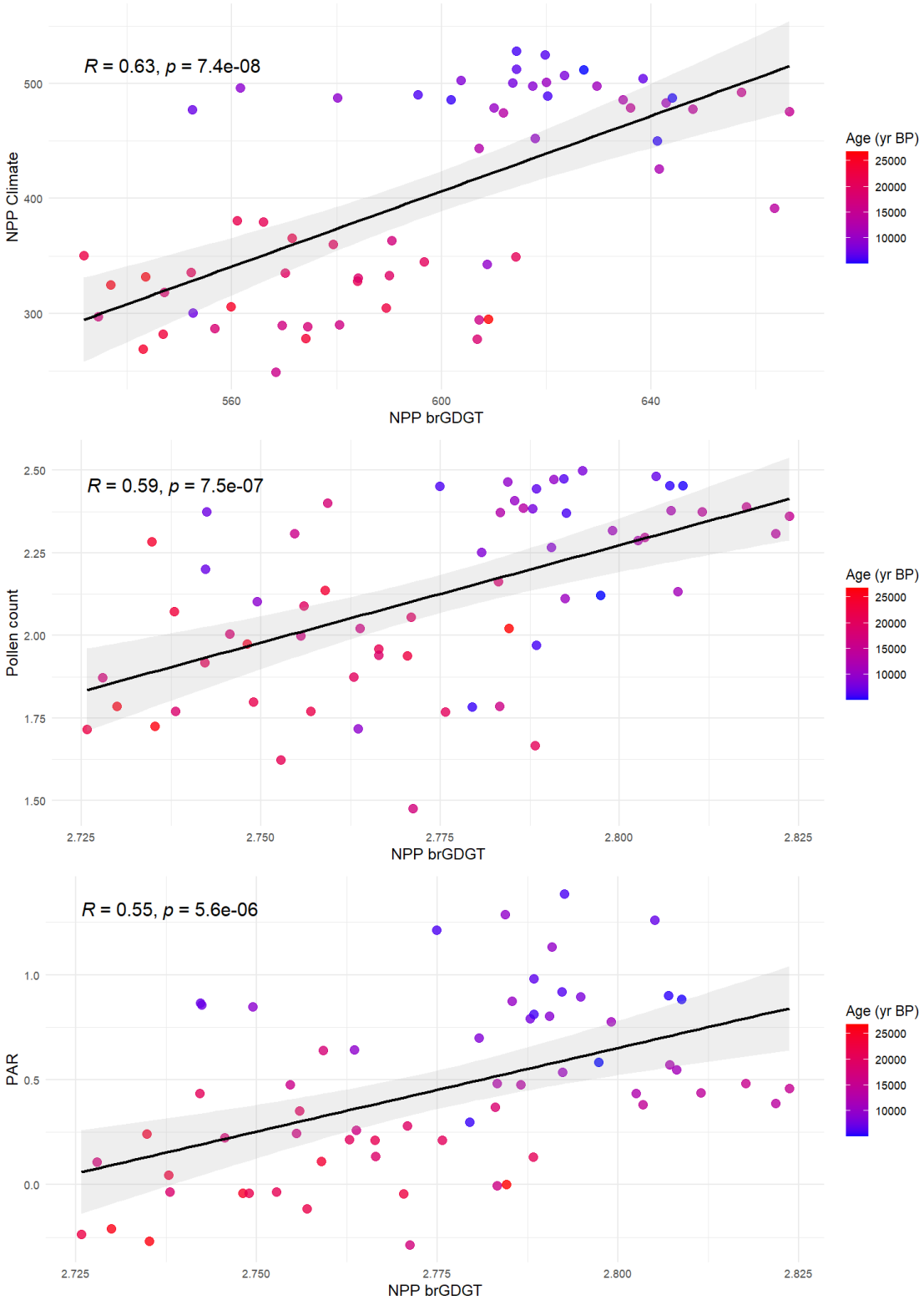


Figure S10. NPP associations with potential NPP, total pollen and pollen accumulation rate.

Figure S11 and Figure 8E: Charcoal associations

Plot charcoal concentration against potential and brGDGT-derived NPP and the charcoal–NPP-deficit relationship..

```
combined_df_P<- subset(combined_df, combined_df$Age_calBP<16000) # < Bolling Allerod

Pot_Carchoal_All<-ggplot(combined_df, aes(x = combined_df$Charcoal_cm3, y = NPP_climate)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "Charcoal",
    y = "DNPP"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )

Act_Carchoal_All<-ggplot(combined_df, aes(x = combined_df$Charcoal_cm3, y = NPP_rf)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "Charcoal",
    y = "DNPP"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )


Act_Carchoal_Magd<-ggplot(combined_df_P, aes(x = combined_df_P$Charcoal_cm3,
```

```
y = NPP_rf)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "pearson",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "Charcoal",
    y = "DNPP"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )


Pot_Carchoal_Magd<-ggplot(combined_df_P, aes(x = combined_df_P$Charcoal_cm3,
y = NPP_climate)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "pearson",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "Charcoal",
    y = "DNPP"
  ) +

  theme_minimal() +
```

```
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )

grid.arrange(Pot_Carchoal_All, Pot_Carchoal_Magd, Act_Carchoal_All, Act_Carch
oal_Magd, ncol=2)
```

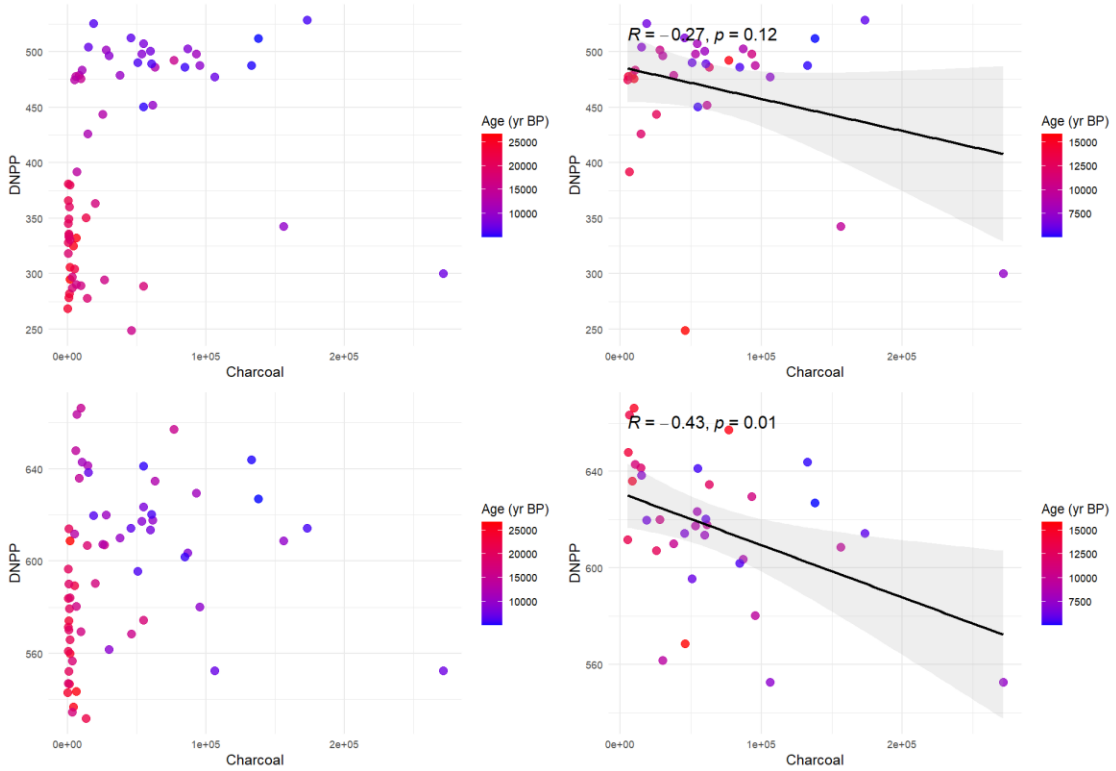


Figure S11. Charcoal-NPP associations

```
# Figure 8E: use the *same* deterministic HANPP as Script1.R.
# Script1 interpolates HANPP (with rule=2) to the RF-age grid, and
# charcoal concentration from the Pollen sheet to that grid.
fig08_rf_mean <- tree_long_main %>%
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(NPP_rf = mean(NPP, na.rm = TRUE), .groups = "drop") %>%
  dplyr::arrange(dplyr::desc(Age_calBP))

fig08_pollen <- openxlsx::read.xlsx("Dataset_1.xlsx", sheet = "Pollen")
combined_df_fig08e <- fig08_rf_mean %>%
  dplyr::mutate(
    HANPP = stats::approx(fig08_hanpp_by_age$Age_calBP,
                          fig08_hanpp_by_age$HANPP,
                          xout = Age_calBP, rule = 2)$y,
    Charcoal_cm3 = stats::approx(fig08_pollen$Age,
                                 fig08_pollen$Charcoals_cm3,
```

```
                                  xout = Age_calBP, rule = 2)$y
  ) %>%
  dplyr::filter(Age_calBP > 4999, Age_calBP < 27000)

DNPP_Charcoal_cm3 <- ggplot(combined_df_fig08e,
                            aes(x = log10(Charcoal_cm3), y = HANPP)) +

  geom_point(aes(color = Age_calBP ), size = 3, alpha = 0.8) +

  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "darkgrey", a
lpha = 0.2) +

  stat_cor(method = "pearson",
           label.x.npc = "left",
           label.y.npc = "top",
           size = 5) +

  scale_color_gradient(low = "blue", high = "red", name = "Age (yr BP)") +

  labs(
    x = "DNPP",
    y = "Total Charcoal"
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(size = 12),
    legend.position = "right"
  )

DNPP_Charcoal_cm3
```

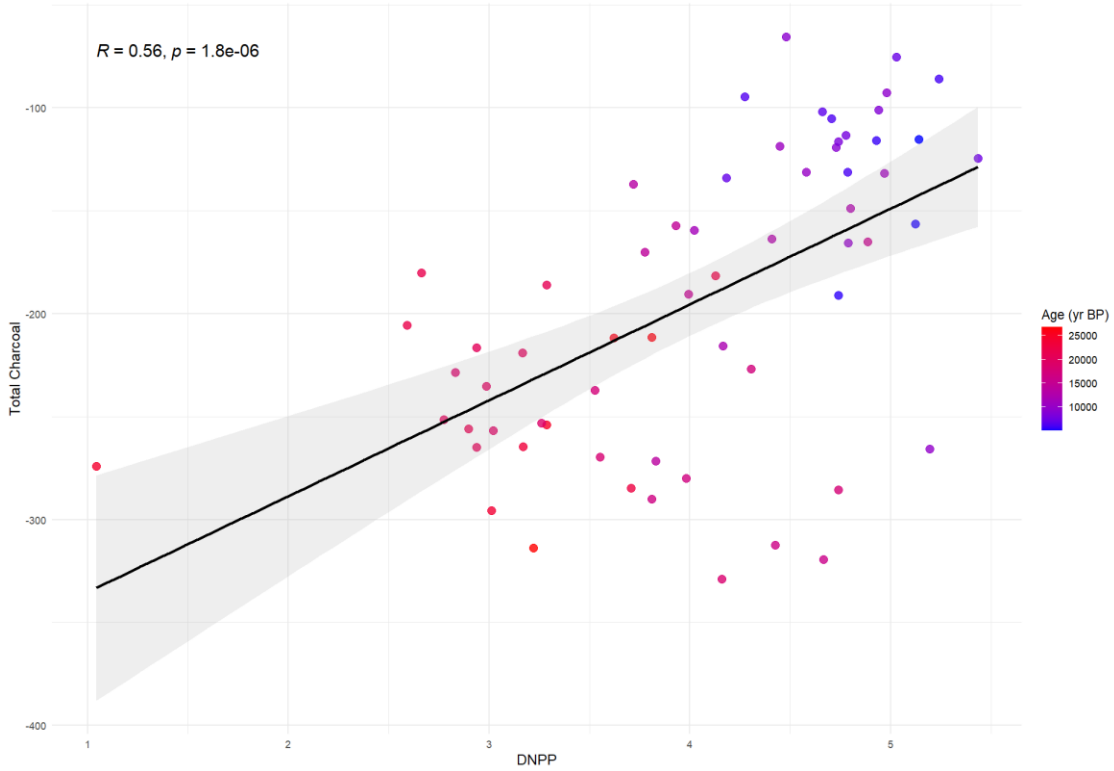


Figure S11. Charcoal-NPP associations

Table S3: Between-culture comparisons

Fit ANOVA models for brGDGT-derived NPP, potential NPP and NPP deficit by culture. Calculate Tukey pairwise mean differences, 95% intervals and adjusted p-values.

```
anova_brgdgt <- aov(
  NPP_brGDGT ~ Culture,
  data = all_data_complete
)

anova_miami <- aov(
  NPP_Miami ~ Culture,
  data = all_data_complete
)

anova_hanpp <- aov(
  HANPP ~ Culture,
  data = all_data_complete
)

tukey_brgdgt <- TukeyHSD(anova_brgdgt)
tukey_miami  <- TukeyHSD(anova_miami)
tukey_hanpp  <- TukeyHSD(anova_hanpp)
```

```r
tidy_tukey <- function(tukey_object, variable_name) {

  out <- as.data.frame(tukey_object$Culture)

  out$Comparison <- rownames(out)
  rownames(out) <- NULL

  out %>%
    transmute(
      Variable = variable_name,
      Comparison = Comparison,
      Mean_difference = diff,
      CI_95_lower = lwr,
      CI_95_upper = upr,
      p_adjusted = `p adj`
    )
}

table_brgdgt <- tidy_tukey(
  tukey_brgdgt,
  "brGDGT-derived NPP"
)

table_miami <- tidy_tukey(
  tukey_miami,
  "Climate-derived NPP"
)

table_hanpp <- tidy_tukey(
  tukey_hanpp,
  "NPP deficit"
)

tukey_table <- bind_rows(
  table_brgdgt,
  table_miami,
  table_hanpp
)


summary(anova_brgdgt)
```

```
##                Df   Sum Sq Mean Sq F value Pr(>F)
## Culture         4 16647924 4161981    7197 <2e-16 ***
## Residuals   25022 14470590     578
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 4087 observations deleted due to missingness
```

```r
summary(anova_miami)
```

```
##                Df   Sum Sq  Mean Sq F value Pr(>F)
## Culture         4 90405897 22601474    8258 <2e-16 ***
## Residuals   23016 62994156     2737
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 6093 observations deleted due to missingness
```

```r
summary(anova_hanpp)
```

```
##                Df   Sum Sq  Mean Sq F value Pr(>F)
## Culture         4 41015384 10253846    4829 <2e-16 ***
## Residuals   23016 48876708     2124
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 6093 observations deleted due to missingness
```

```r
knitr::kable(tukey_table, digits = 4, caption = "Table S3. Tukey comparisons between cultures.")
```

Table S3. Tukey comparisons between cultures.

Variable	Comparison	Mean_difference	CI_95_lower	CI_95_upper	p_adjusted
brGDGT-derived NPP	Gravettian-Epipalaeolithic	-69.0137	-70.6164	-67.4110	0
brGDGT-derived NPP	Magdalenian-Epipalaeolithic	-15.1094	-16.3555	-13.8633	0
brGDGT-derived NPP	Neolithic-Epipalaeolithic	-21.4128	-22.7875	-20.0381	0
brGDGT-derived NPP	Solutrean-Epipalaeolithic	-59.8377	-61.0181	-58.6572	0
brGDGT-derived NPP	Magdalenian-Gravettian	53.9043	52.3265	55.4822	0
brGDGT-derived NPP	Neolithic-Gravettian	47.6009	45.9196	49.2821	0
brGDGT-derived NPP	Solutrean-Gravettian	9.1760	7.6495	10.7026	0
brGDGT-derived NPP	Neolithic-Magdalenian	-6.3035	-7.6491	-4.9578	0
brGDGT-derived NPP	Solutrean-Magdalenian	-44.7283	-45.8747	-43.5818	0
brGDGT-derived NPP	Solutrean-Neolithic	-38.4248	-39.7099	-37.1397	0

Variable	Comparison	Mean_difference	CI_95_lower	CI_95_upper	p_adjusted
Climate-derived NPP	Gravettian-Epipalaeolithic	-148.9438	-152.8025	-145.0850	0
Climate-derived NPP	Magdalenian-Epipalaeolithic	-72.7504	-75.4614	-70.0393	0
Climate-derived NPP	Neolithic-Epipalaeolithic	33.6452	30.2292	37.0611	0
Climate-derived NPP	Solutrean-Epipalaeolithic	-128.0075	-130.5757	-125.4393	0
Climate-derived NPP	Magdalenian-Gravettian	76.1934	72.3835	80.0033	0
Climate-derived NPP	Neolithic-Gravettian	182.5889	178.2490	186.9288	0
Climate-derived NPP	Solutrean-Gravettian	20.9363	17.2266	24.6459	0
Climate-derived NPP	Neolithic-Magdalenian	106.3955	103.0348	109.7563	0
Climate-derived NPP	Solutrean-Magdalenian	-55.2571	-57.7514	-52.7629	0
Climate-derived NPP	Solutrean-Neolithic	-161.6526	-164.8993	-158.4060	0
NPP deficit	Gravettian-Epipalaeolithic	-80.3286	-83.7275	-76.9296	0
NPP deficit	Magdalenian-Epipalaeolithic	-57.6410	-60.0290	-55.2530	0
NPP deficit	Neolithic-Epipalaeolithic	51.8624	48.8535	54.8714	0
NPP deficit	Solutrean-Epipalaeolithic	-68.1698	-70.4320	-65.9076	0
NPP deficit	Magdalenian-Gravettian	22.6876	19.3317	26.0436	0
NPP deficit	Neolithic-Gravettian	132.1910	128.3682	136.0138	0
NPP deficit	Solutrean-Gravettian	12.1587	8.8911	15.4264	0

Variable	Comparison	Mean_difference	CI_95_lower	CI_95_upper	p_adjusted
NPP deficit	Neolithic-Magdalenian	109.5034	106.5431	112.4637	0
NPP deficit	Solutrean-Magdalenian	-10.5289	-12.7259	-8.3318	0
NPP deficit	Solutrean-Neolithic	-120.0323	-122.8920	-117.1725	0

Figure 8A–E: Archaeological associations

Calculate cultural mean SPD density and NPP deficit. Compare mean deficit with mean SPD density, layer counts and date counts using Kendall correlation, and assemble these plots with the SPD-density and charcoal-deficit panels.

```
# The cultural statistics are calculated below from the individual-date SPD
# and age-averaged DNPP values, following Script_1(6).R.

cultures_corr <- c(
  "Gravettian",
  "Solutrean",
  "Magdalenian",
  "Epipalaeolithic",
  "Neolithic"
)

Regions_SPD_corr <- data.frame()
invisible(capture.output(suppressMessages(suppressWarnings({
  for (cult in cultures_corr) {
    cult_df <- SPD_df %>%
      dplyr::filter(Included == "y", Culture == cult) %>%
      dplyr::filter(!is.na(Age), !is.na(s.dev.), !is.na(Curve),
                    !is.na(Labref), !is.na(layer))
    if (nrow(cult_df) == 0) next
    curves <- ifelse(cult_df$Curve == "Terrestrial", "intcal20", "marine20")
    deltaR_vals <- ifelse(cult_df$Curve == "Marine", deltaR_val, 0)
    deltaR_sds <- ifelse(cult_df$Curve == "Marine", deltaR_sd, 0)
    caldates_ind <- calibrate(
      x = cult_df$Age, errors = cult_df$s.dev., calCurves = curves,
      ids = paste0(cult_df$Labref, "_", seq_len(nrow(cult_df))),
      delta.R = deltaR_vals, delta.STD = deltaR_sds, calMatrix = TRUE)
    cult_df$SiteLayer <- paste(cult_df$Sitename, cult_df$layer, sep = "_")
    bins <- binPrep(sites = as.integer(factor(cult_df$SiteLayer)),
                    ages = cult_df$Age, h = 300)
    spd_cult <- spd(caldates_ind, timeRange = c(35000, 2000), bins = bins)
    Regions_SPD_corr <- rbind(Regions_SPD_corr, data.frame(
      PrDens = spd_cult$grid$PrDens, Age = spd_cult$grid$calBP,
```

```
      Culture = cult))
  }
})))

if (!"Culture" %in% names(Regions_SPD_corr)) {
  stop("Figure 8 SPD produced no cultural results. Check the SPD input sheet.
")
}
missing_spd_cultures <- setdiff(cultures_corr, unique(as.character(Regions_SP
D_corr$Culture)))
if (length(missing_spd_cultures)) {
  stop("Figure 8 SPD is missing: ", paste(missing_spd_cultures, collapse = ",
"),
       ". Check spelling and Included/required fields in Dataset_1.xlsx (SPD)
.")
}

spd_ages <- Regions_SPD_corr %>%
  filter(PrDens >= 0.001) %>%
  dplyr::select(Age, Culture, PrDens) %>%
  distinct()

DNPP_by_age <- DNPP_data %>%
  dplyr::group_by(Age_calBP) %>%
  dplyr::summarise(DNPP_mean = mean(DNPP, na.rm = TRUE), .groups = "drop")

DNPP_interp <- approx(
  x = DNPP_by_age$Age_calBP,
  y = DNPP_by_age$DNPP_mean,
  xout = spd_ages$Age,
  rule = 2
)

all_data_complete <- spd_ages %>%
  mutate(
    DNPP = DNPP_interp$y
  ) %>%
  filter(
    !is.na(DNPP),
    !is.na(PrDens)
  )

culture_stats <- all_data_complete %>%
  group_by(Culture) %>%
  summarise(
    mean_PrDens = mean(PrDens, na.rm = TRUE),
    sd_PrDens   = sd(PrDens, na.rm = TRUE),
    mean_DNPP   = mean(DNPP, na.rm = TRUE),
    sd_DNPP     = sd(DNPP, na.rm = TRUE),
```

```
    n = n(),
    .groups = "drop"
  )

missing_culture_stats <- setdiff(cultures_corr, as.character(culture_stats$Culture))
if (length(missing_culture_stats)) {
  stop("Figure 8 cultural summary lacks: ", paste(missing_culture_stats, collapse = ", "),
       ". Check PrDens >= 0.001 and DNPP interpolation for those cultures.")
}

culture_colors <- c(
  "Gravettian" = "#3b528b",
  "Solutrean" = "#21918c",
  "Magdalenian" = "#5ec962",
  "Epipalaeolithic" = "#fde725",
  "Neolithic" = "#f98e09"
)

culture_stats <- culture_stats %>%
  dplyr::left_join(counts_by_culture %>%
                     dplyr::select(Culture, n_layers, n_dates), by = "Culture") %>%
  dplyr::rename(Layers = n_layers, Dates = n_dates)
if (anyNA(culture_stats$Layers) || anyNA(culture_stats$Dates))
  stop("Missing layers or dates for a culture in Figure 8.")

cor_test <- cor.test(culture_stats$mean_PrDens, culture_stats$mean_DNPP,
                     method = "pearson")
r2 <- round(unname(cor_test$estimate), 3)
p_value <- round(cor_test$p.value, 4)

p <- ggplot(culture_stats, aes(x = mean_PrDens, y = mean_DNPP, color = Culture)) +
  geom_errorbar(aes(ymin = mean_DNPP - sd_DNPP, ymax = mean_DNPP + sd_DNPP),
                width = 0.05 * max(culture_stats$mean_PrDens, na.rm = TRUE),
                linewidth = 0.8, alpha = 0.7) +
  geom_point(size = 5) +
  geom_smooth(method = "lm", se = TRUE, color = "black", linetype = "dashed", alpha = 0.2) +
  annotate("text", x = min(culture_stats$mean_PrDens, na.rm = TRUE),
           y = max(culture_stats$mean_DNPP, na.rm = TRUE) * 0.95,
           label = paste0("Pearson r = ", r2, "\np = ", p_value),
           size = 3.5, hjust = 0, vjust = 1) +
  scale_color_manual(values = culture_colors) +
  labs(x = "Probability Density", y = "DNPP (g C m⁻² yr⁻¹)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none", panel.grid.minor = element_blank(),
```

```
        panel.border = element_rect(fill = NA, color = "grey70"),
        axis.text = element_text(size = 12), axis.title = element_text(size =
14))

cor_test <- cor.test(culture_stats$Layers, culture_stats$mean_DNPP,
                     method = "pearson")
r2 <- round(unname(cor_test$estimate), 3)
p_value <- round(cor_test$p.value, 4)

p_l <- ggplot(culture_stats, aes(x = Layers, y = mean_DNPP, color = Culture))
+
  geom_errorbar(aes(ymin = mean_DNPP - sd_DNPP, ymax = mean_DNPP + sd_DNPP),
                width = 0.5, linewidth = 0.8, alpha = 0.7) +
  geom_point(size = 5) +
  geom_smooth(method = "lm", se = TRUE, color = "black", linetype = "dashed",
alpha = 0.2) +
  annotate("text", x = min(culture_stats$Layers, na.rm = TRUE),
           y = max(culture_stats$mean_DNPP, na.rm = TRUE) * 0.95,
           label = paste0("Pearson r = ", r2, "\np = ", p_value),
           size = 3.5, hjust = 0, vjust = 1) +
  scale_color_manual(values = culture_colors) +
  scale_x_continuous(breaks = seq(10, 45, 5)) +
  labs(x = "Number of layers", y = "DNPP (g C m⁻² yr⁻¹)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none", panel.grid.minor = element_blank(),
        panel.border = element_rect(fill = NA, color = "grey70"),
        axis.text = element_text(size = 12), axis.title = element_text(size =
14))

knitr::kable(culture_stats, digits = 4, caption = "Cultural statistics used i
n Figure 8.",
             col.names = names(culture_stats))
```

Cultural statistics used in Figure 8.

Culture	mean_PrDens	sd_PrDens	mean_DNPP	sd_DNPP	n	Layers	Dates
Epipalaeolithic	0.0071	0.0075	-159.2405	45.4574	5274	37	41
Gravettian	0.0030	0.0021	-298.4836	27.2781	6542	11	28
Magdalenian	0.0066	0.0062	-225.6445	52.6314	5840	32	50
Neolithic	0.0134	0.0142	-112.9723	14.8027	4007	45	96
Solutrean	0.0046	0.0020	-241.0153	29.4139	7451	21	43

```
grid.arrange(p, p_l, DNPP_Dens_plot, DNPP_Charcoal_cm3, ncol = 2)
```

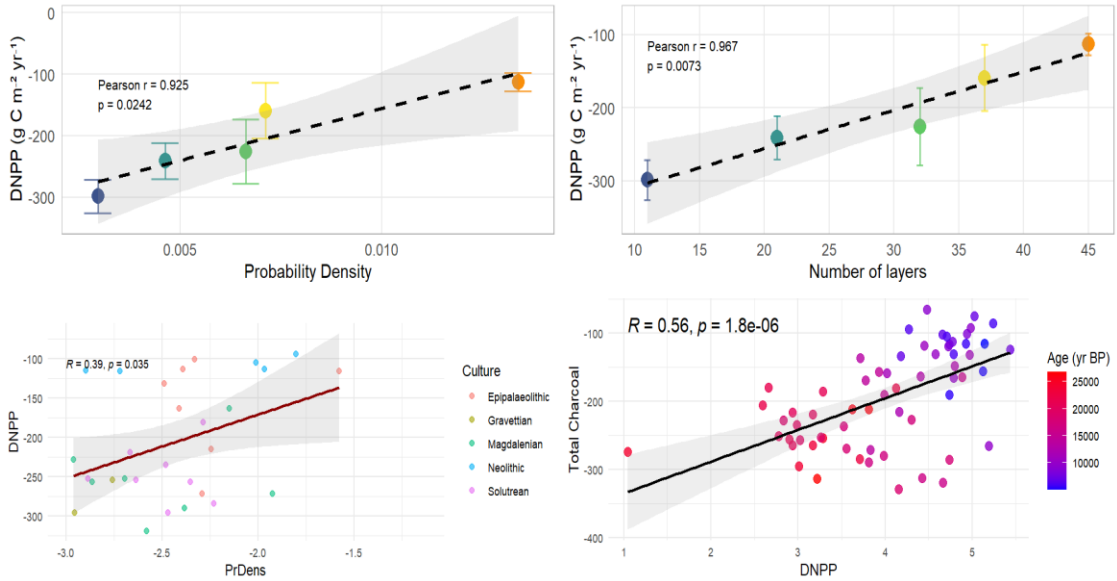


Figure 8. Archaeological indicators and charcoal associations with NPP deficit.

Script 2: Herbivore biomass and consumer carrying capacity

Packages for consumer analyses

```
library(openxlsx)
library(gridExtra)
library(beanplot)
library(ggplot2)
library(patchwork)
library(ggpubr)
library(dplyr)
library(tidyverse)
library(magick)
library(robustbase)
library(RColorBrewer)
library(tidyr)
library(igraph)
```

Herbivore density estimation

Read herbivore body masses and NPP estimates for each cultural period from Dataset 2. Calculate total herbivore biomass from NPP and then distribute density among species using model described in Methods. Calculate actual densities for each culture using brGDGT-derived NPP. It is used the mammal taxonomic composition observed in the

Padul region during the Late Pleistocene and the Early Holocene. The selected input is `Fauna_Padul_Holocene`.

```
df <- read.xlsx("Dataset_2.xlsx", rowNames = FALSE, colNames = TRUE,
                sheet = "Fauna_Padul_Holocene")

df$THB_act <- 10^(1.401 * log10(df$NPP_act) - 0.642)

outputs <- data.frame(
  Primary_Consumers = df$Primary_Consumers,
  BM_pc = df$BM_pc
)

cultures <- unique(df$Culture)
cultures <- cultures[!is.na(cultures)]

for (i in seq_along(cultures)) {
  culture_name <- cultures[i]
  df_cult <- df[df$Culture == culture_name, ]
  THB_act_value <- df_cult$THB_act[1]
  sum_BM25 <- sum(df$BM_pc^0.25, na.rm = TRUE)
  c_act <- THB_act_value / sum_BM25
  Density_act <- c_act * (df$BM_pc^-0.75)
  outputs[[paste0(culture_name, "_act")]] <- Density_act
}

write.csv(outputs, "Outputs_Density_Pleistocene.csv", row.names = FALSE)
knitr::kable(outputs, digits = 5, caption = "Herbivore densities by cultural
period.")
```

Herbivore densities by cultural period.

Primary_Consumers	BM_pc	Gravettian_act	Solutrean_act	Magdalenian_act	Epipaleolithic_act	Neolithic_act
Equus ferus	200.00000	1.18198	1.18723	1.32106	1.36644	1.30954
Bos primigenius	900.00000	0.38256	0.38426	0.42758	0.44226	0.42385
Cervus elaphus	131.25000	1.62110	1.62829	1.81184	1.87409	1.79604
Capreolus capreolus	22.50000	6.08480	6.11182	6.80077	7.03440	6.74146
Capra pyrenaica	63.38883	2.79817	2.81059	3.12742	3.23485	3.10014
Rupicapra rupicapra	28.05000	5.15743	5.18033	5.76429	5.96231	5.71401

Primary_Consumers	BM_pc	Gravettian_act	Solutrean_act	Magdalenian_act	Epipaleolithic_act	Neolithic_act
Sus scrofa	101.05210	1.97230	1.98106	2.20438	2.28011	2.18515
Oryctolagus	1.96000	37.94818	38.11668	42.41341	43.87045	42.04347
Lepus granatensis	2.30000	33.65791	33.80736	37.61832	38.91063	37.29020

Population and biomass inputs

Define the stationary-population equation and select the species and period-specific density inputs. The calculation below uses the `Actual_Pleistocene` density worksheet and actual-NPP period columns.

```
stability_function <- function(params, a, beta) {
  alpha <- params[1]
  sum(a * exp(-((1:length(a) - 1) / alpha)^beta)) - 1
}

species_list<- c("Mammuthus.primigenius",
                 "Equus.hydruntinus",
                 "Equus.ferus",
                 "Bison.priscus",
                 "Bos.primigenius",
                 "Cervus.elaphus",
                 "Capreolus.capreolus",
                 "Capra.pyrenaica",
                 "Rupicapra.rupicapra",
                 "Sus.scrofa",
                 "Oryctolagus",
                 "Lepus.granatensis")

df_PC <- read.xlsx("Dataset_2.xlsx",
                   rowNames = FALSE,
                   colNames = TRUE,
                   sheet = "Actual_Pleistocene")

chrons <- c("Gravettian_act", "Solutrean_act", "Magdalenian_act", "Epipaleolithic_act")

outputs_bm <- data.frame(
  Species = character(),
  Chron = character(),
  Small_Biomass = numeric(),
  Medium_Biomass = numeric(),
  MediumLarge_Biomass = numeric(),
```

```
  Large_Biomass = numeric(),
  stringsAsFactors = FALSE
)
```

Figure S15: Stationary survival curves

Generate 100 candidate β values between 0 and 1. For each candidate, solve the stationary-population equation for α, calculate the survival curve and plot the accepted curves with their mean. The example uses Mammuthus primigenius.

```
df<-read.xlsx("Dataset_2.xlsx", rowNames=FALSE,
              colNames=TRUE, sheet="Mammuthus.primigenius")

Species <- na.omit(df$Species[1])
a<- na.omit(df$Fecundity)

beta_values <- runif(100, min = 0, max = 1)
survival_curves <- list()

for (beta in beta_values) {

  test_interval <- function(lower, upper, stability_function, a, beta) {
    tryCatch({
      alpha_estimate <- uniroot(stability_function, interval = c(lower, upper
), a = a, beta = beta)$root
      return(alpha_estimate)
    }, error = function(e) {
      return(NA)
    })
  }

  search_range <- seq(0, 10, by = 0.1)
  valid_intervals <- lapply(1:(length(search_range) - 1), function(i) {
    lower <- search_range[i]
    upper <- search_range[i + 1]
    alpha_estimate <- test_interval(lower, upper, stability_function, a, beta
)
    if (!is.na(alpha_estimate)) {
      return(list(lower = lower, upper = upper, alpha_estimate = alpha_estima
te))
    } else {
      return(NULL)
    }
  })

  valid_intervals <- Filter(Negate(is.null), valid_intervals)

  if (length(valid_intervals) == 0) next
```

```
  alpha_estimate <- valid_intervals[[1]]$alpha_estimate

  proportion_alive <- exp(-(((1:length(a)) - 1) / alpha_estimate) ^ beta))
  survival_curves[[length(survival_curves) + 1]] <- list(proportion_alive = p
roportion_alive, beta = beta, alpha = alpha_estimate)
}

proportion_matrix <- do.call(cbind, lapply(survival_curves, function(curve) c
urve$proportion_alive))

mean_proportion_alive <- rowMeans(proportion_matrix)

plot(1, type = "n", xlab = "Age", ylab = "Alive (%)", xlim = c(1, length(a)),
ylim = c(0, 1), main = paste("", Species))
for (curve in survival_curves) {
  lines(curve$proportion_alive, col = "grey")
  lines(mean_proportion_alive, col = "red", lwd = 3)
}
```

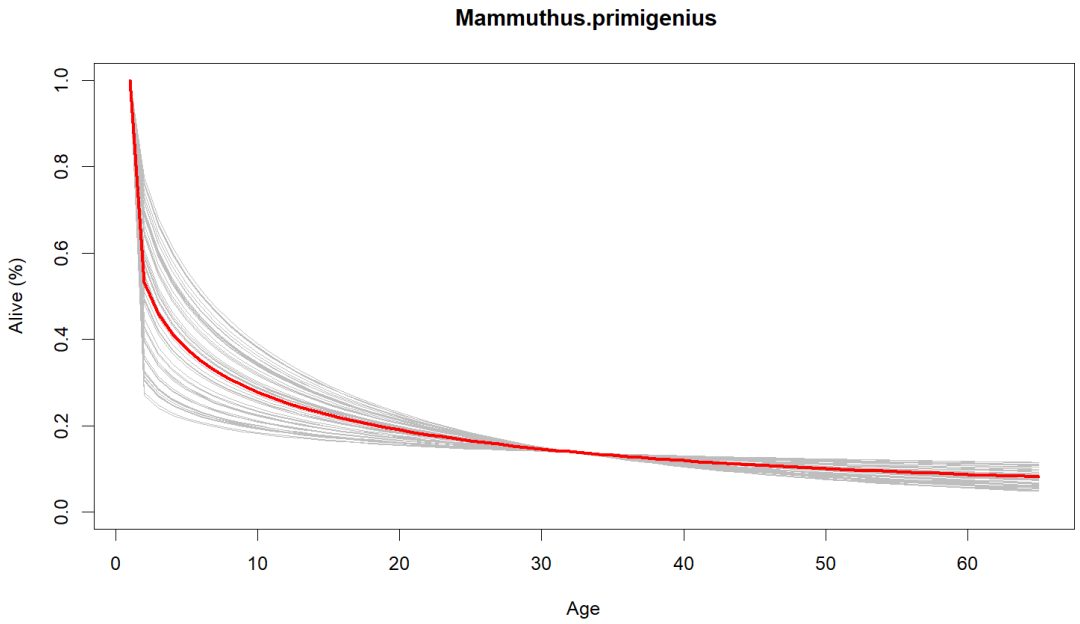


Figure S15. Simulated stationary survival curves for Mammuthus primigenius.

Biomass by species, period and body-size class

Repeat the survival calculation for each herbivore species. Normalise the mean survival curve, combine it with age-specific weights and densities, and calculate biomass in small (<10 kg), medium (10–<100 kg), medium-large (100–<500 kg) and large (≥500 kg) classes.

```
for (j in species_list){
```

```r
  df<-read.xlsx("Dataset_2.xlsx", rowNames=FALSE,
                colNames=TRUE, sheet=j)
  Species <- na.omit(df$Species[1])
  a<- na.omit(df$Fecundity)

  beta_values <- runif(100, min = 0, max = 1)
  survival_curves <- list()

  for (beta in beta_values) {

    test_interval <- function(lower, upper, stability_function, a, beta) {
      tryCatch({
        alpha_estimate <- uniroot(stability_function, interval = c(lower, upp
er), a = a, beta = beta)$root
        return(alpha_estimate)
      }, error = function(e) {
        return(NA)
      })
    }

    search_range <- seq(0, 10, by = 0.1)
    valid_intervals <- lapply(1:(length(search_range) - 1), function(i) {
      lower <- search_range[i]
      upper <- search_range[i + 1]
      alpha_estimate <- test_interval(lower, upper, stability_function, a, be
ta)
      if (!is.na(alpha_estimate)) {
        return(list(lower = lower, upper = upper, alpha_estimate = alpha_esti
mate))
      } else {
        return(NULL)
      }
    })

    valid_intervals <- Filter(Negate(is.null), valid_intervals)

    if (length(valid_intervals) == 0) next

    alpha_estimate <- valid_intervals[[1]]$alpha_estimate

    proportion_alive <- exp(-(((1:length(a)) - 1) / alpha_estimate) ^ beta))
    survival_curves[[length(survival_curves) + 1]] <- list(proportion_alive =
proportion_alive, beta = beta, alpha = alpha_estimate)
  }

  proportion_matrix <- do.call(cbind, lapply(survival_curves, function(curve)
curve$proportion_alive))
```

```r
  mean_proportion_alive <- rowMeans(proportion_matrix)


  proportion_alive <- mean_proportion_alive

  alpha_values <- sapply(survival_curves, function(curve) curve$alpha)
  beta_values <- sapply(survival_curves, function(curve) curve$beta)

  mean_alpha <- mean(alpha_values, na.rm = TRUE)
  mean_beta <- mean(beta_values, na.rm = TRUE)

  alpha_estimate <- mean_alpha
  beta <- mean_beta

  Fec <- a
  e <- proportion_alive * a
  print(round(sum(e)))

  for (chron_value in chrons) {

    row_index <- which(df_PC$Primary_Consumers == j)

    if (length(row_index) == 0) {
      cat("Check species' name: species not found in df_PC:", j, "\n")
      next
    }

    Density <- df_PC[[chron_value]][row_index]

    if (is.na(Density)) {
      cat("Check species' name: no density for", j, "in", chron_value, "\n")
      next
    }

    df$factor_death <- abs(c(proportion_alive[1], diff(proportion_alive)))
    df$factor_death<- proportion_alive / sum(proportion_alive)

    df$Category[df$Weight < 10] <- "Small"
    df$Category[df$Weight >= 10 & df$Weight < 100] <- "Medium"
    df$Category[df$Weight >= 100 & df$Weight < 500] <- "MediumLarge"
    df$Category[df$Weight >= 500] <- "Large"

Small_f <- sum(subset(df, Category=="Small")$factor_death * subset(df, Catego
ry=="Small")$Weight)
Medium_f <- sum(subset(df, Category=="Medium")$factor_death * subset(df, Cate
gory=="Medium")$Weight)
MediumLarge_f <- sum(subset(df, Category=="MediumLarge")$factor_death * subse
t(df, Category=="MediumLarge")$Weight)
```

```
Large_f <- sum(subset(df, Category=="Large")$factor_death * subset(df, Catego
ry=="Large")$Weight)

Small_Biomass <- Small_f * Density
Medium_Biomass <- Medium_f * Density
MediumLarge_Biomass <- MediumLarge_f * Density
Large_Biomass <- Large_f * Density

outputs_bm <- rbind(outputs_bm, data.frame(
  Species = j,
  Chron = chron_value,
  Small_Biomass = Small_Biomass,
  Medium_Biomass = Medium_Biomass,
  MediumLarge_Biomass = MediumLarge_Biomass,
  Large_Biomass = Large_Biomass,
  stringsAsFactors = FALSE
))

  }
}


write.csv(outputs_bm, "outputs__Biomass_Actual_Pleisto.csv")

knitr::kable(outputs_bm, digits = 4, caption = "Species-level biomass by peri
od and body-size class.")
```

Tables S6–S8: Biomass summaries

Read the biomass results from Dataset 2 and calculate mean and standard deviation by cultural period and body-size class for the selected community and NPP source. The selected subset is Holocene composition and actual NPP.

```
Biomass_df<- read.xlsx("Dataset_2.xlsx", sheet = "Biomass")

Biomass_df<- subset(Biomass_df, Biomass_df$Composition=="Holocene")
Biomass_df<- subset(Biomass_df, Biomass_df$NPP=="ACT")
outputs_bm <- as.data.frame(Biomass_df)
outputs_bm$Chron <- as.character(outputs_bm$Chron)

summary_biomass <- aggregate(
  cbind(Small_Biomass, Medium_Biomass, MediumLarge_Biomass, Large_Biomass) ~
Chron,
  data = outputs_bm,
  FUN = function(x) c(mean = mean(x, na.rm = TRUE), sd = sd(x, na.rm = TRUE))
)

summary_biomass <- do.call(data.frame, summary_biomass)
```

```r
colnames(summary_biomass) <- c(
  "Chron",
  "Small_mean", "Small_sd",
  "Medium_mean", "Medium_sd",
  "MediumLarge_mean", "MediumLarge_sd",
  "Large_mean", "Large_sd"
)

summary_biomass <- summary_biomass[order(as.character(summary_biomass$Chron)), , drop = FALSE]


write.csv(summary_biomass, "summary_biomass.csv")

knitr::kable(summary_biomass, digits = 4, caption = "Biomass summary by cultural period and body-size class.")
```

Biomass summary by cultural period and body-size class.

Chron	Small_mean	Small_sd	Medium_mean	Medium_sd	MediumLarge_mean	MediumLarge_sd	Large_mean	Large_sd
Epipaleolithic_act	12.3189	19.1078	51.0271	52.3986	46.3395	83.9883	24.3724	73.1172
Gravettian_act	10.6559	16.5283	44.1387	45.3251	40.0839	72.6504	21.0823	63.2468
Magdalenian_act	11.9098	18.4732	49.3323	50.6583	44.8004	81.1989	23.5629	70.6888
Solutrean_act	10.7032	16.6017	44.3347	45.5263	40.2619	72.9730	21.1759	63.5276

Secondary-consumer carrying capacity

Read community biomass and prey preferences. Apply wastage factors of 0.8, 0.75, 0.65 and 0.6 to the four body-size classes. Allocate edible biomass among predators according to their normalised prey preferences and divide available biomass by annual biomass requirements. The selected calculation uses Pleistocene composition and the 60% human-meat diet worksheet.

```r
df_biomass<- read.xlsx("Dataset_2.xlsx", sheet = "Summary_Biomass")


df_biomass<- subset(df_biomass, df_biomass$Composition=="Pleistocene")

Prey_preferences<- read.xlsx("Prey_preferences.xlsx", rowNames=FALSE,
```

```
                              colNames=TRUE, sheet="Diet_60_Pleistocene")
head(Prey_preferences)
```

```
##   Weight_Category Ursus.arctos Panthera.pardus Crocuta.crocuta Canis.lupus
## 1              BM     180.5200         54.9900      62.9900000     32.1800
## 2              db     212.1944        926.7993    1003.5251218    638.0108
## 3   %Meat_in_Diet      10.0000        100.0000      98.5000000    100.0000
## 4           Small       0.2500          0.0200       0.0000000      0.0300
## 5          Medium       0.2500          0.8000       0.4517766      0.3000
## 6        MedLarge       0.2500          0.1800       0.4010152      0.6400
##   Meles.meles Vulpes.vulpes Felis.silvestris Lynx.pardinus Cuon.alpinus
## 1    13.00000       5.31000         5.500000        9.4000   12.7600000
## 2    16.96228      18.17779         9.314311      270.6359  301.3819342
## 3     5.00000      10.00000         5.000000      100.0000   90.0000000
## 4     1.00000       1.00000         1.000000        0.7500    0.2222222
## 5     0.00000       0.00000         0.000000        0.2500    0.4444444
## 6     0.00000       0.00000         0.000000        0.0000    0.2777778
##   Martes.martes Homo.sapiens
## 1     1.3000000      59.5000
## 2    40.9077337     328.4202
## 3    60.0000000      60.0000
## 4     0.8333333       0.1400
## 5     0.1666667       0.4600
## 6     0.0000000       0.3000
```

```
db_values <- data.frame(
  Predator = colnames(Prey_preferences)[-1],
  AnnualBiomass = as.numeric(Prey_preferences[2, -1])
)

print(db_values)
```

```
##            Predator AnnualBiomass
## 1      Ursus.arctos    212.194363
## 2   Panthera.pardus    926.799345
## 3   Crocuta.crocuta   1003.525122
## 4       Canis.lupus    638.010775
## 5       Meles.meles     16.962278
## 6     Vulpes.vulpes     18.177785
## 7  Felis.silvestris      9.314311
## 8     Lynx.pardinus    270.635947
## 9      Cuon.alpinus    301.381934
## 10    Martes.martes     40.907734
## 11     Homo.sapiens    328.420214
```

```
WastageFactor <- c(Small = 0.8, Medium = 0.75, MedLarge = 0.65, Large = 0.6)

df_biomass_corrected <- df_biomass
df_biomass_corrected$Small_mean      <- df_biomass$Small_mean      * WastageF
actor["Small"]
```

```r
df_biomass_corrected$Medium_mean     <- df_biomass$Medium_mean     * WastageF
actor["Medium"]
df_biomass_corrected$MediumLarge_mean <- df_biomass$MediumLarge_mean* Wastage
Factor["MedLarge"]
df_biomass_corrected$Large_mean      <- df_biomass$Large_mean      * WastageF
actor["Large"]

prey_subset <- Prey_preferences[Prey_preferences$Weight_Category %in% c("Smal
l","Medium","MedLarge","Large"), ]

prey_numeric <- prey_subset
for (col in colnames(prey_numeric)[-1]) {
  prey_numeric[[col]] <- as.numeric(as.character(prey_numeric[[col]]))
}

predators <- colnames(prey_numeric)[-1]
prey_long <- data.frame(
  SizeCategory = rep(prey_numeric$Weight_Category, times = length(predators))
,
  Predator = rep(predators, each = nrow(prey_numeric)),
  DietFraction = unlist(prey_numeric[, predators]),
  stringsAsFactors = FALSE
)

Dataset_CC_list <- list()

for (culture in df_biomass_corrected$Culture) {

  EB_row <- df_biomass_corrected[df_biomass_corrected$Culture == culture,
                                 c("Small_mean","Medium_mean","MediumLarge_me
an","Large_mean")]
  EB <- data.frame(
    SizeCategory = c("Small","Medium","MedLarge","Large"),
    EdibleBiomass = as.numeric(EB_row[1,])
  )

  B_table <- data.frame()

  for (size in EB$SizeCategory) {
    edible <- EB$EdibleBiomass[EB$SizeCategory == size]

    diet_row <- prey_long[prey_long$SizeCategory == size, ]

    total_demand <- sum(diet_row$DietFraction, na.rm = TRUE)

    diet_row$BiomassAvailable <- edible * diet_row$DietFraction / total_deman
d
```

```
    B_table <- rbind(B_table, diet_row)
  }

  Dataset_CC_culture <- aggregate(BiomassAvailable ~ Predator, data = B_table
, sum)
  Dataset_CC_culture$AnnualBiomass <- db_values$AnnualBiomass[match(Dataset_C
C_culture$Predator, db_values$Predator)]
  Dataset_CC_culture$Dataset_CC <- Dataset_CC_culture$BiomassAvailable / Data
set_CC_culture$AnnualBiomass
  Dataset_CC_culture$Culture <- culture

  Dataset_CC_list[[culture]] <- Dataset_CC_culture[, c("Culture","Predator","
BiomassAvailable","AnnualBiomass","Dataset_CC")]
}

Dataset_CC_final <- do.call(rbind, Dataset_CC_list)


write.csv(Dataset_CC_final, "Output_CC_Pleistocene_60.csv")

knitr::kable(Dataset_CC_final, digits = 5, caption = "Calculated carrying cap
acity by consumer and culture.")
```

Calculated carrying capacity by consumer and culture.

	Culture	Predator	BiomassAvailable	AnnualBiomass	Dataset_CC
Gravettian.1	Gravettian	Canis.lupus	8.33685	638.01077	0.01307
Gravettian.2	Gravettian	Crocuta.crocuta	11.66812	1003.52512	0.01163
Gravettian.3	Gravettian	Cuon.alpinus	6.99660	301.38193	0.02322
Gravettian.4	Gravettian	Felis.silvestris	0.71017	9.31431	0.07624
Gravettian.5	Gravettian	Homo.sapiens	9.00571	328.42021	0.02742
Gravettian.6	Gravettian	Lynx.pardinus	1.73444	270.63595	0.00641
Gravettian.7	Gravettian	Martes.martes	1.39302	40.90773	0.03405
Gravettian.8	Gravettian	Meles.meles	0.71017	16.96228	0.04187
Gravettian.9	Gravettian	Panthera.pardus	5.45216	926.79935	0.00588
Gravettian.10	Gravettian	Ursus.arctos	13.69410	212.19436	0.06454
Gravettian.11	Gravettian	Vulpes.vulpes	0.71017	18.17779	0.03907
Solutrean.1	Solutrean	Canis.lupus	8.37387	638.01077	0.01312

	Culture	Predator	BiomassAvailable	AnnualBiomass	Dataset_CC
Solutrean.2	Solutrean	Crocuta.crocuta	11.71993	1003.52512	0.01168
Solutrean.3	Solutrean	Cuon.alpinus	7.02767	301.38193	0.02332
Solutrean.4	Solutrean	Felis.silvestris	0.71332	9.31431	0.07658
Solutrean.5	Solutrean	Homo.sapiens	9.04570	328.42021	0.02754
Solutrean.6	Solutrean	Lynx.pardinus	1.74214	270.63595	0.00644
Solutrean.7	Solutrean	Martes.martes	1.39920	40.90773	0.03420
Solutrean.8	Solutrean	Meles.meles	0.71332	16.96228	0.04205
Solutrean.9	Solutrean	Panthera.pardus	5.47637	926.79935	0.00591
Solutrean.10	Solutrean	Ursus.arctos	13.75491	212.19436	0.06482
Solutrean.11	Solutrean	Vulpes.vulpes	0.71332	18.17779	0.03924
Magdalenian.1	Magdalenian	Canis.lupus	9.31782	638.01077	0.01460
Magdalenian.2	Magdalenian	Crocuta.crocuta	13.04106	1003.52512	0.01300
Magdalenian.3	Magdalenian	Cuon.alpinus	7.81986	301.38193	0.02595
Magdalenian.4	Magdalenian	Felis.silvestris	0.79373	9.31431	0.08522
Magdalenian.5	Magdalenian	Homo.sapiens	10.06538	328.42021	0.03065
Magdalenian.6	Magdalenian	Lynx.pardinus	1.93853	270.63595	0.00716
Magdalenian.7	Magdalenian	Martes.martes	1.55693	40.90773	0.03806
Magdalenian.8	Magdalenian	Meles.meles	0.79373	16.96228	0.04679
Magdalenian.9	Magdalenian	Panthera.pardus	6.09370	926.79935	0.00657
Magdalenian.10	Magdalenian	Ursus.arctos	15.30544	212.19436	0.07213
Magdalenian.11	Magdalenian	Vulpes.vulpes	0.79373	18.17779	0.04366

	Culture	Predator	BiomassAvailable	AnnualBiomass	Dataset_CC
Epipaleolithic.1	Epipaleolithic	Canis.lupus	9.63792	638.01077	0.01511
Epipaleolithic.2	Epipaleolithic	Crocuta.crocuta	13.48907	1003.52512	0.01344
Epipaleolithic.3	Epipaleolithic	Cuon.alpinus	8.08850	301.38193	0.02684
Epipaleolithic.4	Epipaleolithic	Felis.silvestris	0.82100	9.31431	0.08814
Epipaleolithic.5	Epipaleolithic	Homo.sapiens	10.41116	328.42021	0.03170
Epipaleolithic.6	Epipaleolithic	Lynx.pardinus	2.00512	270.63595	0.00741
Epipaleolithic.7	Epipaleolithic	Martes.martes	1.61042	40.90773	0.03937
Epipaleolithic.8	Epipaleolithic	Meles.meles	0.82100	16.96228	0.04840
Epipaleolithic.9	Epipaleolithic	Panthera.pardus	6.30304	926.79935	0.00680
Epipaleolithic.10	Epipaleolithic	Ursus.arctos	15.83123	212.19436	0.07461
Epipaleolithic.11	Epipaleolithic	Vulpes.vulpes	0.82100	18.17779	0.04516

Carrying-capacity summaries

Read the stored carrying-capacity scenarios and average results by palaeocommunity, NPP source, culture and species.

```
CC_df<-read.xlsx("Dataset_2.xlsx", sheet= "Carnivores_CC")

head(CC_df)
```

```
##      NPP Paleocommunity Human_Diet            Culture          Species
## 1 brGDGT              H         30 Epipaleolithic_act      Canis.lupus
## 2 brGDGT              H         30 Epipaleolithic_act Felis.silvestris
## 3 brGDGT              H         30 Epipaleolithic_act     Homo.sapiens
## 4 brGDGT              H         30 Epipaleolithic_act    Lynx.pardinus
## 5 brGDGT              H         30 Epipaleolithic_act    Martes.martes
## 6 brGDGT              H         30 Epipaleolithic_act      Meles.meles
##   BiomassAvailable         CC
## 1        25.460407 0.03990592
## 2         1.969711 0.21147151
```

```
## 3        24.056967 0.14650113
## 4         8.183530 0.03023815
## 5         6.112257 0.14941569
## 6         1.969711 0.11612305
```

```
summary_table_df <- CC_df %>%
  group_by(Paleocommunity, NPP, Culture, Species) %>%
  summarise(mean_CC = mean(CC, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = Species,
    values_from = mean_CC
  )

write.xlsx(summary_table_df, "summary_stats_cc.xlsx")

knitr::kable(summary_table_df, digits = 5, caption = "Carrying-capacity means across the stored scenarios.")
```

Carrying-capacity means across the stored scenarios.

Paleocommunity	NPP	Culture	Canis.lupus	Felis.silvestris	Homo.sapiens	Lynx.pardinus	Martes.martes	Meles.meles	Ursus.arctos	Vulpes.vulpes	Crocuta.crocuta	Cuon.alpinus	Panthera.pardus
H	brGDGT	Epipaleolithic_act	0.03991	0.21147	0.10438	0.03024	0.14942	0.11612	0.10909	0.10836	NA	NA	NA
H	brGDGT	Gravettian_act	0.03452	0.18292	0.09029	0.02616	0.12925	0.10045	0.09436	0.09373	NA	NA	NA
H	brGDGT	Magdalenian_act	0.03858	0.20445	0.10092	0.02923	0.14445	0.11227	0.10546	0.10476	NA	NA	NA

Palaeocommunity	NPP	Culture	Canis.lupus	Felis.silvestris	Homo.sapiens	Lynx.pardinus	Martes.martes	Meles.meles	Ursus.arctos	Vulpes.vulpes	Crocuta.crocuta	Cuon.alpinus	Panthera.pardus
H	brGDGT	Solutrean_act	0.034667	0.18374	0.090669	0.02627	0.12982	0.10089	0.09478	0.09415	NA	NA	NA
P	brGDGT	Epipaleolithic_act	0.017797	0.13681	0.04965	0.01132	0.06030	0.07513	0.06986	0.07010	0.01414	0.03146	0.00966
P	brGDGT	Gravettian_act	0.015554	0.11834	0.04295	0.00979	0.05216	0.06499	0.06043	0.06064	0.01223	0.02722	0.00836
P	brGDGT	Magdalenian_act	0.017237	0.13227	0.04800	0.01094	0.05829	0.07263	0.06754	0.06778	0.01367	0.03042	0.00934
P	brGDGT	Solutrean_act	0.015561	0.11887	0.04314	0.00983	0.05239	0.06527	0.06070	0.06091	0.01229	0.02734	0.00839

Figure 9: Herbivore biomass and human carrying capacity

Compare herbivore biomass by culture, body-size class and community composition. Summarise Homo sapiens carrying capacity from the stored consumer scenarios and plot means and standard deviations.

```
getwd()
```

```
## [1] "C:/Users/mav54/OneDrive - University of Cambridge/Desktop/NatComm/Git
HUB/Final_GitHub"

Biomass_herbivores_df<- read.xlsx("Dataset_2.xlsx", sheet="Summary_Biomass")
head(Biomass_herbivores_df)

##       NPP Composition        Culture Small_mean  Small_sd Medium_mean Medi
um_sd
## 1  brGDGT Pleistocene     Gravettian   4.656528  8.682730    20.01675  24.
87552
## 2  brGDGT Pleistocene      Solutrean   4.677204  8.721283    20.10563  24.
98597
## 3  brGDGT Pleistocene    Magdalenian   5.204445  9.704395    22.37205  27.
80253
## 4  brGDGT Pleistocene Epipaleolithic   5.383235 10.037774    23.14061  28.
75764
## 5 Climate Pleistocene     Gravettian   9.076666 15.901473    38.22603  45.
59229
## 6 Climate Pleistocene      Solutrean  10.147004 17.776603    42.73371  50.
96862
##   MediumLarge_mean MediumLarge_sd Large_mean  Large_sd
## 1         27.88011       44.41641   39.25275  78.99832
## 2         28.00390       44.61363   39.42704  79.34909
## 3         31.16066       49.64272   43.87148  88.29377
## 4         32.23114       51.34812   45.37862  91.32696
## 5         41.80372       71.72499   39.49432  89.90005
## 6         46.73329       80.18294   44.15157 100.50123

df_plot <- Biomass_herbivores_df %>%
  filter(NPP == "brGDGT") %>%

  mutate(Culture = factor(Culture,
                          levels = c("Gravettian",
                                     "Solutrean",
                                     "Magdalenian",
                                     "Epipaleolithic")))

df_long <- df_plot %>%
  dplyr::select(Composition, Culture,
         Small_mean, Medium_mean, MediumLarge_mean, Large_mean) %>%

  pivot_longer(cols = c(Small_mean, Medium_mean, MediumLarge_mean, Large_mean
),
               names_to = "Size",
               values_to = "Biomass") %>%

  mutate(Size = recode(Size,
                       Small_mean = "Small",
                       Medium_mean = "Medium",
                       MediumLarge_mean = "Medium-large",
```

```
                        Large_mean="Large"))

ggplot(df_long, aes(x = Culture, y = Biomass, fill = Size)) +
  geom_col(position = position_dodge(width = 0.8)) +
  facet_wrap(~ Composition) +
  labs(
    x = "Culture",
    y = "Herbivore biomass (kg km⁻² yr⁻¹)",
    fill = "Herbivore size"
  ) +

  theme_bw() +

  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(face = "bold")
  )
```

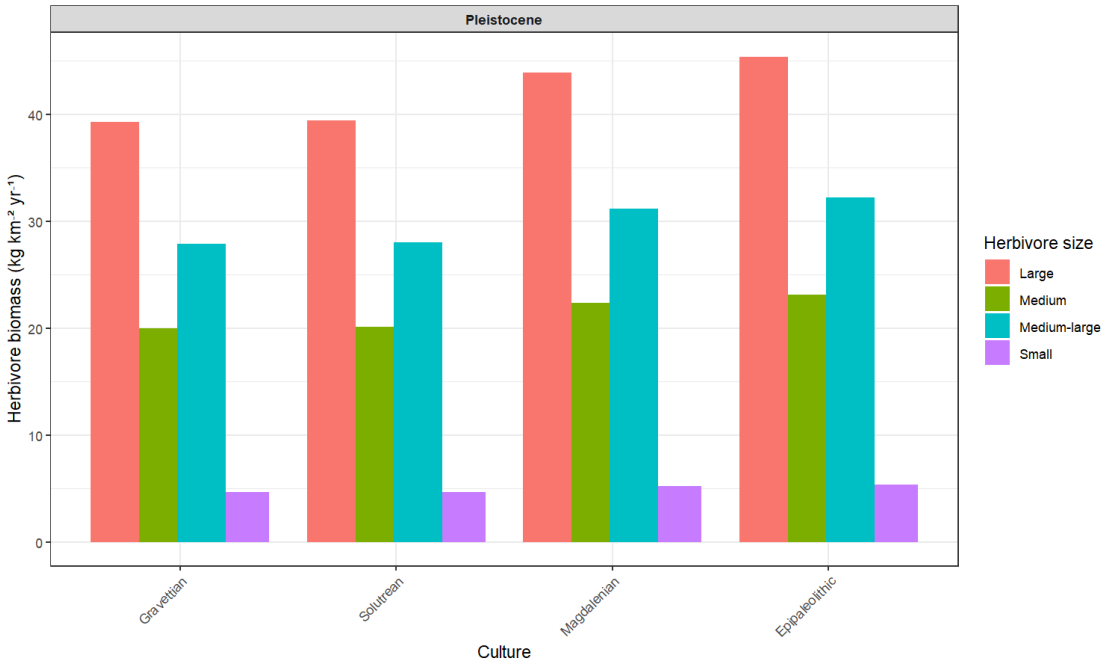


Figure 9. Herbivore biomass by culture, size class and community composition.

```
df_hs <- CC_df %>%
  filter(Species == "Homo.sapiens")

summary_df <- df_hs %>%
  group_by(Culture, NPP, Paleocommunity) %>%
  summarise(
    mean_CC = mean(CC, na.rm = TRUE),
    sd_CC   = sd(CC, na.rm = TRUE),
    .groups = "drop"
  )
```

```r
summary_df <- subset(summary_df, summary_df$NPP == "brGDGT")

ggplot(summary_df, aes(x = Culture, y = mean_CC, fill = Paleocommunity)) +
  geom_point() +
  geom_errorbar(aes(ymin = mean_CC - sd_CC, ymax = mean_CC + sd_CC), width =
0.2) +
  facet_wrap(~ NPP) +
  labs(x = "Culture", y = "Carrying Capacity (CC)", fill = "Paleocommunity")
+
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

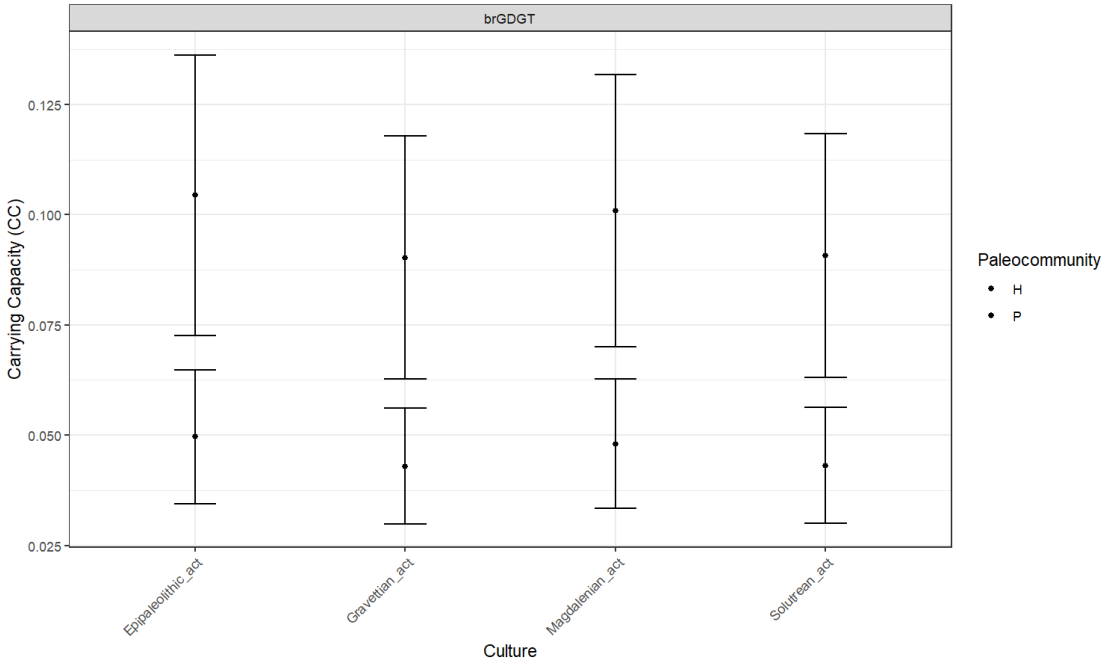


Figure 9. Human carrying-capacity means and standard deviations for brGDGT-derived NPP.

```r
homo_data <- CC_df %>%
  filter(Species == "Homo.sapiens") %>%

  mutate(Culture = gsub("_act", "", Culture)) %>%

  mutate(Culture = factor(Culture,
                          levels = c("Gravettian", "Solutrean",
                                     "Magdalenian", "Epipaleolithic")))

summary_homo <- homo_data %>%
  group_by(Paleocommunity, Culture) %>%
  summarise(
    Mean_CC = mean(CC, na.rm = TRUE),
```

```
    SD_CC = sd(CC, na.rm = TRUE),
    .groups = 'drop'
  )

ggplot(summary_homo, aes(x = Culture, y = Mean_CC, color = Paleocommunity, gr
oup = Paleocommunity)) +
  geom_point(size = 3, position = position_dodge(width = 0.3)) +
  geom_errorbar(aes(ymin = Mean_CC - SD_CC, ymax = Mean_CC + SD_CC),
                width = 0.2, position = position_dodge(width = 0.3)) +
  geom_line(position = position_dodge(width = 0.3)) +
  labs(
    title = "Homo sapiens CC across Cultures",
    subtitle = "Comparison between Paleocommunity A and B",
    x = "Culture",
    y = "Mean CC (± SD)",
    color = "Paleocommunity"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
```

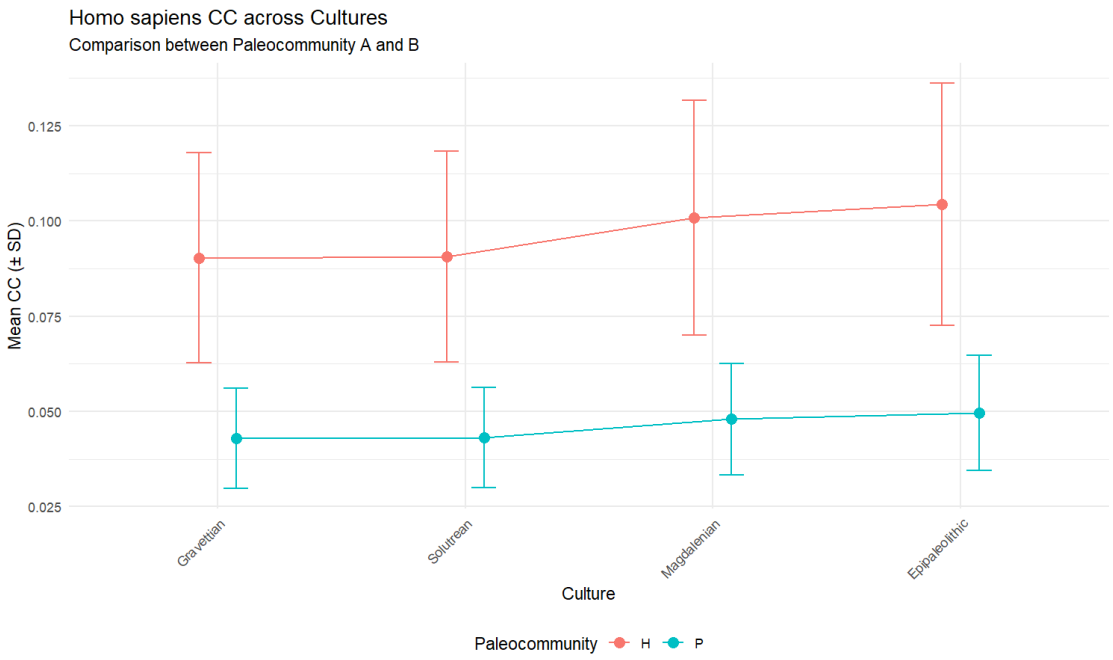


Figure 9. Human carrying capacity across cultural periods and palaeocommunities.

Consumer–body-size connections

Construct an undirected graph linking consumer species to prey body-size classes. Set edge width according to the biomass-demand values in dB and place body-size classes and consumers in separate columns.

```
df <- read.xlsx("Prey_preferences.xlsx", sheet = "dB")
names(df)[1] <- "X1"

species_order <- c("Ursus.arctos", "Crocuta.crocuta", "Panthera.pardus",
                   "Homo.sapiens", "Canis.lupus", "Cuon.alpinus",
                   "Lynx.pardinus", "Felis.silvestris", "Meles.meles",
                   "Vulpes.vulpes", "Martes.martes")

df_long <- df %>%
  pivot_longer(cols = -X1, names_to = "Species", values_to = "Value") %>%
  filter(Value > 0)

edges <- data.frame(from = df_long$X1, to = df_long$Species, weight = df_long
$Value)
size_categories <- unique(df$X1)
species <- colnames(df)[-1]
nodes <- data.frame(
  name = c(size_categories, species),
  type = c(rep("Category", length(size_categories)), rep("Species", length(sp
ecies)))
)
g <- graph_from_data_frame(edges, vertices = nodes, directed = FALSE)

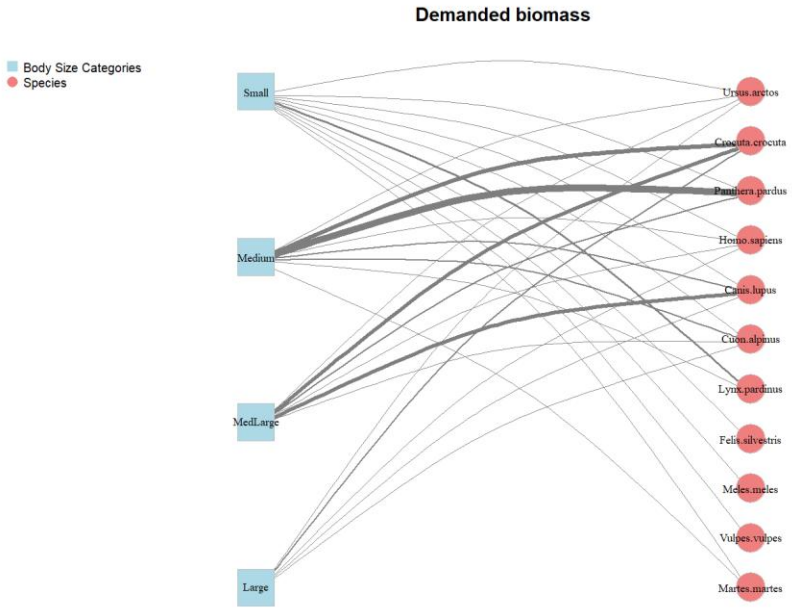
layout_matrix <- matrix(0, nrow = length(nodes$name), ncol = 2)
category_positions <- seq(1, 0, length.out = length(size_categories))
for (i in seq_along(size_categories)) {
  idx <- which(nodes$name == size_categories[i])
  layout_matrix[idx, ] <- c(0, category_positions[i])
}

species_positions <- seq(1, 0, length.out = length(species_order))
for (i in seq_along(species_order)) {
  idx <- which(nodes$name == species_order[i])
  if (length(idx) > 0) layout_matrix[idx, ] <- c(1, species_positions[i])
}

V(g)$color <- ifelse(V(g)$type == "Category", "lightblue", "lightcoral")
V(g)$shape <- ifelse(V(g)$type == "Category", "square", "circle")
V(g)$size <- ifelse(V(g)$type == "Category", 15, 12)
E(g)$width <- E(g)$weight / 100

plot(g, layout = layout_matrix,
     vertex.label = V(g)$name, vertex.label.cex = 0.7,
```

```
     vertex.label.color = "black", vertex.frame.color = "gray",
     edge.color = "gray50", edge.curved = 0.2,
     main = "Demanded biomass", margin = c(0, 0, 0, 0))
legend("topleft", legend = c("Body Size Categories", "Species"),
       pch = c(15, 19), col = c("lightblue", "lightcoral"),
       pt.cex = 1.5, cex = 0.8, bty = "n")
```



Consumer connections to prey body-size classes, weighted by biomass demand.

Species contributions to body-size biomass

Aggregate biomass across periods and community compositions, calculate species percentages within each body-size class and plot four pie charts. Also calculate summaries by composition and by period.

```
Biomass_df<- read.xlsx("Dataset_2.xlsx", sheet= "Biomass")
head(Biomass_df)
```

```
##   Composition NPP               Species              Chron Small_Biomass
## 1 Pleistocene ACT Mammuthus.primigenius     Gravettian_act             0
## 2 Pleistocene ACT Mammuthus.primigenius      Solutrean_act             0
## 3 Pleistocene ACT Mammuthus.primigenius    Magdalenian_act             0
## 4 Pleistocene ACT Mammuthus.primigenius Epipaleolithic_act             0
## 5 Pleistocene ACT     Equus.hydruntinus     Gravettian_act             0
## 6 Pleistocene ACT     Equus.hydruntinus      Solutrean_act             0
##   Medium_Biomass MediumLarge_Biomass Large_Biomass
## 1       0.000000            1.103408      250.7478
## 2       0.000000            1.108307      251.8612
```

```
## 3       0.000000            1.233242      280.2524
## 4       0.000000            1.275608      289.8800
## 5       8.515949          107.205681        0.0000
## 6       8.553762          107.681699        0.0000

biomass_long <- Biomass_df %>%

  dplyr::select(Composition, Species, Chron, Small_Biomass, Medium_Biomass,
         MediumLarge_Biomass, Large_Biomass) %>%

  pivot_longer(cols = c(Small_Biomass, Medium_Biomass, MediumLarge_Biomass, L
arge_Biomass),
               names_to = "Size_Class",
               values_to = "Biomass") %>%

  mutate(Size_Class = case_when(
    Size_Class == "Small_Biomass" ~ "Small",
    Size_Class == "Medium_Biomass" ~ "Medium",
    Size_Class == "MediumLarge_Biomass" ~ "MediumLarge",
    Size_Class == "Large_Biomass" ~ "Large"
  )) %>%

  filter(Biomass > 0)

biomass_aggregated <- biomass_long %>%
  group_by(Size_Class, Species) %>%
  summarise(Total_Biomass = sum(Biomass, na.rm = TRUE), .groups = 'drop') %>%
  group_by(Size_Class) %>%
  mutate(Percentage = Total_Biomass / sum(Total_Biomass) * 100) %>%
  arrange(Size_Class, desc(Percentage))

biomass_by_composition <- biomass_long %>%
  group_by(Composition, Size_Class, Species) %>%
  summarise(Total_Biomass = sum(Biomass, na.rm = TRUE), .groups = 'drop') %>%
  group_by(Composition, Size_Class) %>%
  mutate(Percentage = Total_Biomass / sum(Total_Biomass) * 100) %>%
  arrange(Composition, Size_Class, desc(Percentage))

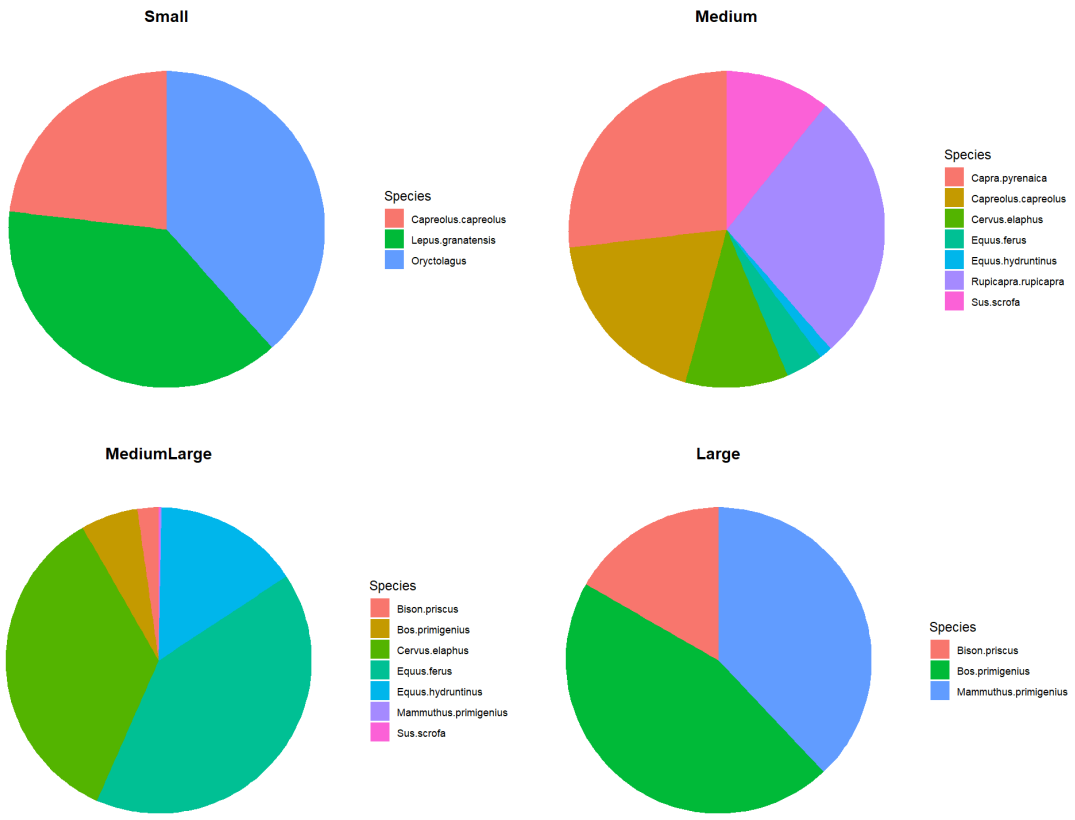
biomass_by_chron <- biomass_long %>%
  group_by(Chron, Size_Class, Species) %>%
  summarise(Total_Biomass = sum(Biomass, na.rm = TRUE), .groups = 'drop') %>%
  group_by(Chron, Size_Class) %>%
  mutate(Percentage = Total_Biomass / sum(Total_Biomass) * 100) %>%
  arrange(Chron, Size_Class, desc(Percentage))

create_pie <- function(data, size_class) {
  data %>%
    filter(Size_Class == size_class) %>%
    ggplot(aes(x = "", y = Percentage, fill = Species)) +
```

```r
    geom_bar(stat = "identity", width = 1) +
    coord_polar("y", start = 0) +
    labs(title = size_class) +
    theme_minimal() +
    theme(
      axis.title.x = element_blank(),
      axis.title.y = element_blank(),
      axis.text = element_blank(),
      axis.ticks = element_blank(),
      panel.grid = element_blank(),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      legend.position = "right"
    )
}

pie_small <- create_pie(biomass_aggregated, "Small")
pie_medium <- create_pie(biomass_aggregated, "Medium")
pie_mediumlarge <- create_pie(biomass_aggregated, "MediumLarge")
pie_large <- create_pie(biomass_aggregated, "Large")

grid.arrange(pie_small, pie_medium, pie_mediumlarge, pie_large)
```



Species contributions to biomass within the four body-size classes.

Software versions

```
sessionInfo()
```

```
## R version 4.5.3 (2026-03-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United Kingdom.utf8
## [2] LC_CTYPE=English_United Kingdom.utf8
## [3] LC_MONETARY=English_United Kingdom.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United Kingdom.utf8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] splines   grid      stats     graphics  grDevices utils     datasets
## [8] methods   base
##
## other attached packages:
##  [1] igraph_2.3.3         robustbase_0.99-7    magick_2.9.1
##  [4] beanplot_1.3.1       maps_3.4.3           RANN_2.6.2
##  [7] terra_1.9-11         ppcor_1.1            MASS_7.3-65
## [10] spdep_1.4-2          spData_2.3.4         blockCV_3.2-0
## [13] sf_1.1-0             mgcv_1.9-4           nlme_3.1-168
## [16] rcarbon_1.5.2        reshape2_1.4.5       randomForest_4.7-1.2
## [19] ggExtra_0.11.0       caret_7.0-1          lattice_0.22-9
## [22] cowplot_1.2.0        RColorBrewer_1.1-3   lubridate_1.9.5
## [25] forcats_1.0.1        stringr_1.6.0        readr_2.2.0
## [28] tibble_3.3.1         tidyverse_2.0.0      corrplot_0.95
## [31] analogue_0.18.1      gridExtra_2.3        ggpubr_0.6.3
## [34] patchwork_1.3.2      purrr_1.2.2          compositions_2.0-9
## [37] ggrepel_0.9.8        vegan_2.7-3          permute_0.9-10
## [40] truncnorm_1.0-9      openxlsx_4.2.8.1     scales_1.4.0
## [43] ggplot2_4.0.2        tidyr_1.3.2          dplyr_1.2.1
##
## loaded via a namespace (and not attached):
##   [1] later_1.4.8            polyclip_1.10-7        hardhat_1.4.3
##   [4] pROC_1.19.0.1          rpart_4.1.24           lifecycle_1.0.5
##   [7] rstatix_0.7.3          globals_0.19.1         backports_1.5.1
##  [10] magrittr_2.0.4         rmarkdown_2.31         yaml_2.3.12
##  [13] httpuv_1.6.17          otel_0.2.0             zip_2.3.3
##  [16] sp_2.2-1               spatstat.sparse_3.1-0  bayesm_3.1-7
##  [19] DBI_1.3.0              abind_1.4-8            spatstat.model_3.7-0
```

```
##  [22] nnet_7.3-20             tensorA_0.36.2.1       ipred_0.9-15
##  [25] lava_1.9.0              listenv_0.10.1         spatstat.utils_3.2-2
##  [28] units_1.0-1             goftest_1.2-3          spatstat.random_3.4-5
##  [31] parallelly_1.46.1       codetools_0.2-20       tidyselect_1.2.1
##  [34] farver_2.1.2            stats4_4.5.3           spatstat.explore_3.8-0
##  [37] e1071_1.7-17            Formula_1.2-5          survival_3.8-6
##  [40] iterators_1.0.14        foreach_1.5.2          tools_4.5.3
##  [43] snow_0.4-4              Rcpp_1.1.1             glue_1.8.0
##  [46] prodlim_2026.03.11      xfun_0.60              withr_3.0.2
##  [49] fastmap_1.2.0           boot_1.3-32            digest_0.6.39
##  [52] timechange_0.4.0        R6_2.6.1               mime_0.13
##  [55] wk_0.9.5                tensor_1.5.1           spatstat.data_3.1-9
##  [58] utf8_1.2.6              generics_0.1.4         data.table_1.18.2.1
##  [61] recipes_1.3.2           class_7.3-23           ModelMetrics_1.2.2.2
##  [64] pkgconfig_2.0.3         gtable_0.3.6           timeDate_4052.112
##  [67] S7_0.2.1                htmltools_0.5.9        carData_3.0-6
##  [70] doSNOW_1.0.20           gower_1.0.2            spatstat.univar_3.1-7
##  [73] knitr_1.51              rstudioapi_0.18.0      tzdb_0.5.0
##  [76] spatstat.linnet_3.5-0   proxy_0.4-29           KernSmooth_2.23-26
##  [79] parallel_4.5.3          miniUI_0.1.2           s2_1.1.9
##  [82] pillar_1.11.1           vctrs_0.7.3            promises_1.5.0
##  [85] car_3.1-5               xtable_1.8-8           cluster_2.1.8.2
##  [88] princurve_2.1.6         evaluate_1.0.5         cli_3.6.5
##  [91] compiler_4.5.3          rlang_1.1.7            future.apply_1.20.2
##  [94] ggsignif_0.6.4          labeling_0.4.3         classInt_0.4-11
##  [97] plyr_1.8.9              stringi_1.8.7          viridisLite_0.4.3
## [100] deldir_2.0-4            profileModel_0.6.2     spatstat.geom_3.7-3
## [103] Matrix_1.7-4            hms_1.1.4              future_1.70.0
## [106] shiny_1.13.0            spatstat_3.6-0         brglm_0.7.3
## [109] broom_1.0.12            DEoptimR_1.1-4
```